

Programa de Doctorado en Economía y Gobierno

Tesis Doctoral

Un Nuevo Enfoque para Viejos Debates:
Ensayos sobre Empresas en Mercados
Laborales y Financieros

Daniel Fernandez

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A New Approach to Old Debates:
Essays on Firms in Labor and
Financial Markets

by

Daniel Fernandez

Advisors:

Guillermo Caruana

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To Astrid.

*I once heard that Paul Dirac
gave his wife a comfortable living,
but she gave him a life worth living.
I still owe you the first one.*

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*The purpose of studying economics is not to
acquire a set of ready-made answers to economic
questions, but to learn how to avoid being
deceived by economists*

Joan Robinson

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All errors, omissions, and infelicities that remain are my own.

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Resumen

Esta tesis doctoral se ubica a mitad de camino entre la evidencia empírica reciente y debates profundamente arraigados en la literatura económica. Cada capítulo es una combinación de trabajo teórico y empírico que se propone revisar un supuesto canónico en la literatura, a la vez que se pregunta: ¿qué sucede cuando nuevas ideas se encuentran con preguntas de antaño? Las respuestas, en cada caso, sugieren que algunos debates de larga data han pasado por alto dimensiones lo suficientemente relevantes como para merecer un estudio más profundo.

El segundo capítulo analiza cómo las empresas reaccionan ante aumentos del salario mínimo. A diferencia de estudios previos, que se centran en la disyuntiva entre empleo y salario, esta sección desarrolla un marco teórico en el que las empresas no solo eligen sueldos, sino también ciertas comodidades laborales, todo ello sujeto a pisos salariales estipulados por ley. El modelo predice, bajo algunos supuestos mínimos en el caso de Brasil, que un aumento del salario mínimo comprime la distribución salarial a la vez que aumenta la desigualdad en las comodidades laborales entre empresas. A su vez, cuando las empresas ofrecen las mismas condiciones laborales a todos sus empleados, distintas compensaciones salariales pueden lograr que los aumentos del salario mínimo se propagen a lo largo y lo ancho de la distribución salarial.

El trabajo empírico, por su parte, se sustenta en datos administrativos pertenecientes a la seguridad social brasileña. De esta forma, estimo las condiciones laborales de cada establecimiento al analizar cuándo los trabajadores renuncian a una empresa para pasar a otra. A su vez, como el salario mínimo es más o menos vinculante en cada uno de los 27 estados de Brasil, puedo identificar efectos marginales de la política en distintos puntos de la distribución salarial. Mis resultados respaldan las predicciones del modelo y me ayudan a reconciliar la intuición económica, que rechaza que las empresas se queden de brazos cruzados

cuando aumenta el salario mínimo, con la evidencia empírica reciente, que no encuentra caídas significativas en el nivel de empleo.

El tercer capítulo investiga cómo la estructura de propiedad de una empresa puede afectar los incentivos de sus gerentes. En particular, cómo la presencia de accionistas comunes en dos empresas —es decir, quienes tienen participación en ambas— puede dar lugar a una relación comercial entre ellas. Esta sección de la tesis doctoral utiliza diversas fuentes de datos abiertos para Estados Unidos y describe una nueva estrategia de identificación basada en los ajustes de cartera que realizan los inversores cuando los componentes del S&P 500 cambian. Los resultados sugieren que entre potenciales clientes y proveedores, quienes tienen estructuras de propiedad de mayor similitud son más proclives a iniciar relaciones comerciales. Sin embargo, el marco teórico asociado también advierte de posibles efectos adversos. Así, se observa que la probabilidad de una relación comercial aumenta significativamente si uno de los directores tiene mayores incentivos para transferir activos fuera de la empresa. De esta manera, es posible que decisiones gerenciales que benefician principalmente a accionistas comunes sean subsidiadas directamente por los accionistas de la empresa cuyo control es más disperso.

El cuarto capítulo cuestiona abiertamente una presunción en la literatura, según la cual la correlación negativa entre la productividad de la firma y la tasa de desvinculaciones laborales se explica principalmente por finalizaciones de contrato que resultan eficientes. Esta sección utiliza datos administrativos de Brasil para mostrar que la tasa de despidos es la que disminuye de manera marcada con la productividad de la firma, mientras que la tasa de renuncias muestra diferencias preponderantemente moderadas. Ante lo visto, desarrollamos un modelo teórico de búsqueda de empleo en el que rigideces salariales e incertidumbre sobre la productividad dan lugar a un mayor número de despidos que, en última instancia, resultan ineficientes. El modelo describe cómo las empresas más productivas tienen mayores márgenes —pagan salarios por debajo de la productividad laboral—. En consecuencia, dichas empresas pueden afrontar situaciones adversas con mayor facilidad, sin necesidad de recurrir a despidos. Seguidamente, recurrimos a los datos con el fin de validar los mecanismos clave del modelo. En primer lugar, encontramos que las empresas que enfrentan una mayor rigidez salarial despiden más y exhiben una relación entre productividad y despidos que es más pronunciada. En segundo lugar, mostramos que los márgenes crecen con la productividad de la firma y que éstos se correlacionan negativamente con la tasa de despidos.

Summary

The following dissertation sits at the intersection of the most recent empirical findings and debates deeply rooted in the Economics literature. By combining theoretical and empirical work, each chapter revisits a canonical assumption and wonders: what changes when new ideas meet old questions? The answer, in each case, suggests that overlooked margins in long-standing debates are consequential enough to warrant further study.

The second chapter studies how firms respond to minimum wage increases. While previous studies focus on the employment-wage tradeoff, I develop a framework in which firms choose wages and job amenities, subject to binding wage floors. Under minimal assumptions, the model predicts that wage floors compress the wage distribution while increasing dispersion in job amenities. Furthermore, when firms offer the same amenities to all employees, compensating differentials can propagate increases in the minimum wage across the wage distribution. The empirical work relies on matched employer-employee administrative data from Brazil. I recover firm-level job amenities using worker flows, and trace wage spillovers by exploiting cross-state variation in the effective minimum wage. My empirical findings support the predictions and help reconcile the economic intuition that firms respond to higher wage floors with recent evidence of limited employment declines.

The third chapter investigates how ownership structures can affect managerial incentives. In particular, how supply chain relationships can emerge from overlapping institutional shareholders. Using several publicly available US data sources, I present a novel identification strategy that leverages changes in portfolio holdings following index entries and exits from the S&P 500. The evidence suggests that more similar customer-supplier ownership structures are more likely to become active trading partnerships. Furthermore, combining my findings with theoretical insights uncovers potential adverse effects. I observe a significant increase in the likelihood of an active trading partnership when one of the managers has

incentives to transfer resources out of the company. The results flag that overlapping shareholders may privately benefit from certain vertical relations at the expense of shareholders in firms with more diluted ownership structures.

The fourth chapter challenges the view that the negative correlation between firm productivity and separation rates reflects efficient separations. Using Brazilian administrative data, we show that layoff rates decline sharply with productivity, whereas quit rates show more modest differences. We develop a job search model in which wage rigidity and productivity uncertainty lead to inefficient layoffs. The model describes how more productive firms set larger markdowns and, consequently, can absorb negative shocks more easily without resorting to layoffs. Empirically, we validate the two key mechanisms in the model. First, we show that firms more tightly constrained by wage rigidity experience more layoffs and a steeper productivity-layoff relationship. Second, we document that markdowns are more prevalent among more productive firms and that they negatively correlate with layoffs.

Chapter 1

Introduction

Whenever a theory appears to you as the only possible one, take this as a sign that you have neither understood the theory nor the problem which it was intended to solve

Karl Popper

Markets do not exist. In Economics, we coined the term to organize the countless interactions and exchanges that shape economic life. The word became the discipline's signature concept and the baseline for studying, analyzing, and even discussing both individual choices and aggregate outcomes. Yet the concept itself remains thorny. Both empirical work and competition courts require drawing market boundaries, as conclusions often change depending on the definitions used. Markets are not fixed entities, which is why many fields have devoted several pages to unraveling them, including Industrial Organization and Labor Economics.

Although firms seem less abstract, we make them appear as monolithic agents maximizing profits. In practice, firms consist of people with different roles, incentives, and agendas; they are collections of daily interactions between owners, managers, and employees. It comes as no surprise that firms have therefore been a core object of study in Economics, with fields contributing perspectives as diverse as Corporate Finance, Contract Theory, and Organizational Economics.

It is no coincidence that my doctoral thesis opens with this digression. Firms and markets do not immediately appear to me as empirical objects, nor do individual choices as mechanisms. Drawn to what lies beneath these abstractions, throughout my Ph.D. I revisited simplifying assumptions that never quite sat right, either in theory or empirically. By questioning them, I kept returning to the same question: what does our analysis overlook when it abstracts from the everyday economic exchanges?

The three chapters that follow engage with long-standing debates in Economics and draw on recent literature to explore overlooked margins. Some of these insights remain contested, and not all economists would agree they belong in the conversation. However, my dissertation combines theoretical and empirical work that bridges these more recent ideas with the older questions. In each case, the innovation turns out to matter, not as a refinement of previous results, but as a dimension worth continuing to explore.

The first chapter acknowledges that individuals not only care about wages but also about non-wage attributes in their employment contracts. I lean on a growing body of empirical evidence showing that workers are willing to give up a fraction of their wages for certain job amenities (Mas and Pallais, 2017; Maestas et al., 2023), such as better working conditions or specific employee benefits. With these findings in mind, my work tackles a puzzle at the heart of the minimum wage debate.

While canonical models predict that higher wage floors lead to layoffs (Stigler, 1946; Brown et al., 1982; Neumark and Wascher, 2000, 2008; Neumark et al., 2014), recent empirical work finds little evidence of employment declines (Card and Krueger, 1994, 2000; Addison et al., 2009, 2012; Dube et al., 2010; Allegretto et al., 2017; Cengiz et al., 2019). My work aims to bridge the gap between economic intuition and empirical findings by arguing that firms respond to minimum wage increases by reducing job amenities for low-paid workers rather than resorting to dismissals.

I propose a theoretical model that illustrates the tradeoff between wages and job amenities and highlights how exposure to the minimum wage depends on the cost of providing non-wage benefits to employees. Although predictions are empirically challenging when firms can freely set job amenities, the framework provides a testable implication when employers choose a single level of job amenities for all employees. In such cases, firms employing workers at the statutory minimum and above reduce amenities for everyone. As a result, high-paid workers require compensating differentials that propagate the minimum wage increases throughout the wage distribution.

For the empirical analysis, Brazil provides an ideal setting. Not only does the country have annual minimum wage adjustments, but its labor institutions support the assumption that firms cannot tailor job amenities to individual workers. In this way, I recover firm-level job amenities using worker flows from matched employer-employee administrative records. Then, I trace wage spillovers by exploiting cross-state variation in the bindingness of the minimum wage, enabling me to calculate marginal effects of higher wage floors across different percentiles of the hourly wage distribution.

Although firm-specific job amenities and wage spillovers make the conclusion about wage inequality and dispersion in job amenities less obvious, taken together, my empirical findings uncover patterns consistent with the theoretical predictions. The increase in the real hourly minimum wage between 2012 and 2019 led to a decline in wage distribution variance and an increase in the job amenity dispersion. Furthermore, my findings corroborate that the higher wage floor rippled well beyond the lower tail of the wage distribution, pushing wages upward up to the 95th percentile.

The second chapter moves to financial markets. In recent decades, the United States has witnessed a rapid expansion of institutional financial intermediaries and a proliferation of passive indexing strategies. As a result, a few large asset management institutions, such as Vanguard, BlackRock, and State Street, have entered the top shareholder list of most publicly listed companies in the country (Schmalz, 2018, 2021).

In the meantime, a growing body of literature on common ownership has shown that overlapping shareholders can soften competition by influencing managerial decisions (Antón et al., 2023). But, if that is the case, what would stop overlapping shareholders from influencing executives when they sit on both sides of a supply chain?

The common ownership literature revisits the separation of ownership and control from a different angle. These studies depart from the traditional standpoint in corporate finance that managers may pursue private benefits rather than maximizing firm profits (Jensen and Meckling, 1976). Instead, they focus on managers maximizing the joint returns of shareholders' portfolios (Rotemberg, 1984). Compared with horizontal settings, where firms and shareholders reap the benefits of softer competition, supply chains may have winners and losers—shifting the tension between ownership and control once more.

My work builds on the theoretical framework in Rotemberg (1984) and combines several open data sources for publicly listed companies in the United States and Canada. First, I employ the model's insights to construct overlapping ownership measures using 13D, 13F,

and 13G filings (Backus et al., 2021; Schwartz-Ziv and Volkova, 2025) submitted to the Securities and Exchange Commission (SEC). Similarly, I identify potential suppliers from vertical-relatedness measures in Frésard et al. (2020) and retrieve active trading partnerships from 10-K filings (Barrot and Sauvagnat, 2016).

The identification strategy relies on an instrumental variables approach. Using panel data on quarterly shareholdings over a decade, I predict ownership ties, focusing on companies and investors with more stable ownership structures and portfolios, and leveraging additions and deletions from the S&P 500 index. Using predicted rather than actual shareholdings, I estimate two-way fixed-effects regressions, weighting each observation by its customer-supplier vertical-relatedness.

The empirical analysis supports prior findings in the literature. Higher degrees of overlapping ownership make active trading partnerships more likely, allowing firms to circumvent contractual and informational frictions (Riva, 2022; Gao et al., 2022; Deng et al., 2025; Freeman, 2025). Nonetheless, my findings uncover potential adverse effects. The likelihood of active trading partnerships increases if they are one-sided. In other words, when one of the firms has a more diluted ownership structure, its manager has incentives to tunnel resources out of the company, privately benefiting overlapping owners at the expense of non-overlapping shareholders at the firm that appointed the manager.

The third chapter is based on a joint paper with Cauê Dobbin and Tom Zohar, in which we revisit a well-documented fact in the labor economics literature. That is, more productive firms have lower separation rates (Topel and Ward, 1992; Davis et al., 2013; Haltiwanger et al., 2018*a,b*). As a recent strand of the literature finds evidence of an increasing number of inefficient, involuntary separations (Schmieder and von Wachter, 2010; Jäger et al., 2023; Davis and Krolikowski, 2025), we reconsider previous explanations about the negative correlation between productivity and separations.

Canonical models often attribute separations to better job offers at another firm or declines in the value of employment relationships. According to these frameworks, both the employer and the employee agree to the separation, rendering the distinction between quits and layoffs irrelevant (Burdett and Mortensen, 1998; Postel-Vinay and Robin, 2002; Postel-Vinay and Turon, 2010). The immediate assessment from the literature has therefore been that workers at more productive firms are less likely to receive better poaching offers, thereby interpreting the negative relationship between productivity and separations as mostly driven by quits.

Our empirical analysis combines matched employer-employee administrative records from Brazil with aggregate reports from annual surveys of manufacturing firms. The Brazilian government mandates employers to file the exact date and reason for every separation. Leveraging that, we decompose separations into quits and layoffs, which reveals that the pattern is almost entirely explained by a declining layoff rate. While quits also decrease with firm productivity, we find a significantly less pronounced relationship.

As the pattern could emerge from worker segregation and sorting across firms, we further control for worker heterogeneity using a rich set of covariates. Although the negative correlation becomes less steep, almost half of the variation still persists.

Based on these observations, we develop a job search model with ex-ante homogeneous workers, in which wage rigidity and productivity uncertainty are enough to replicate the empirical patterns. In our framework, firms set wages by weighing the tradeoff between average markdowns (the gap between average productivity and wages) and worker retention.

The model shows that, in equilibrium, more productive firms choose higher wages but still keep larger markdowns. These markdowns act as a cushion that can absorb negative productivity realizations, shielding the firm from worker-specific losses. On the contrary, less productive firms are more likely to observe worker productivity falling short of the posted wage, resorting to inefficient layoffs to preclude the loss.

We then turn to the data to empirically validate the two mechanisms present in the model. First, we show that firms with less access to flexible pay, and therefore more constrained by wage rigidity, exhibit higher layoff rates and a steeper correlation between productivity and layoffs. Second, we document that more productive firms exhibit larger markdowns, which in turn lead to a decline in the layoff rate.

Chapter 2

The wage-amenities tradeoff: Do firms offset minimum wage increases by cutting job amenities?

*Work is about a search for daily meaning as well
as daily bread, for recognition as well as cash,
for astonishment rather than torpor*

Studs Terkel

2.1 Introduction

What happens when governments increase minimum wages? Economists have long disagreed over whether wage floor policies actually benefit low-paid workers.¹ Whereas supporters of the policy emphasize earnings gains at the bottom of the wage distribution, critics warn about unintended consequences for employment among low earners. Despite the varied opinions, concerns about poverty and earnings inequality have made these policies a standard

¹. The debate over the effects of higher minimum wages dates back to Stigler (1946), who laid the foundations for a strong, though not universal, disdain for wage floor policies. Consensus began to break down in the early 1990s, with the work of David Card and Alan Krueger. Since then, the profession has witnessed numerous meta-analyses (e.g., Card and Krueger, 1995; Neumark and Wascher, 2008; Neumark, 2017; Allegretto and Reich, 2018; Clemens, 2019; Wolfson and Belman, 2019) and academic back-and-forth, such as Card and Krueger (1994, 2000) versus Neumark and Wascher (2000), or Neumark et al. (2014) versus Allegretto et al. (2017).

recommendation among international organizations and poverty experts, leading to their implementation worldwide.²

In Economics, the academic debate centers on a tension between economic intuition and recent empirical evidence regarding the employment-wage tradeoff. Canonical models predict that firms employing low-paid workers respond to a higher minimum wage by cutting labor, leading to layoffs and fewer vacancies (Brown et al., 1982). In contrast, empirical studies find no significant decline in employment following minimum wage increases (Addison et al., 2009, 2012; Dube et al., 2010; Allegretto et al., 2017; Cengiz et al., 2019). Critics of these findings, nonetheless, continue to dispute the results and question the approaches employed (Neumark et al., 2014; Clemens, 2019; Neumark, 2019).

In this chapter, I focus on a feature often overlooked in the debate about the minimum wage: job amenities.³ I argue that firms compensate for higher wage floors by reducing job amenities among low-paid workers. By incorporating a wage-amenities tradeoff, my approach bridges the gap between the economic intuition that firms respond to minimum wage increases and the empirical findings of limited employment declines.

I propose a framework that explicitly depicts a tradeoff between wages and job amenities, yielding two intuitive empirical predictions. A higher minimum wage compresses the wage distribution; however, it worsens working conditions by reducing job amenities at low-paying firms.

Empirically, separating responses to minimum wages from unrelated shifts in firm compensation policies can be challenging. In fact, the model shows that if firms can freely adjust amenities across employees, then taking these predictions to the data would require comprehensive worker-level data. That is, all tangible and intangible non-wage attributes related to employment relationships.

The framework, nevertheless, offers an additional, testable implication when firms choose a single level of amenities for all employees. In such cases, firms that employ workers both at the minimum wage and above must reduce job amenities for everyone. As a result, highly paid workers require a compensating differential to remain indifferent, which ripples throughout the whole wage distribution.

² Over the last two decades, wage inequality has declined in roughly two-thirds of countries worldwide. Nonetheless, the global trend cannot mask the fact that wage gaps remain large, and some countries now experience a reversal of the decline (International Labour Organization, 2024).

³ In previous studies, amenities are often used interchangeably with benefits or perks (Mas and Pallais, 2017). In this chapter, I define amenities as any non-wage feature of labor contracts, including workplace conditions, career advancement, task characteristics, or the effort required for the job.

Brazil provides a particularly suitable setting for testing the implications of the framework. First, the Brazilian government makes reasonable adjustments to the federal minimum wage each January. Compared to other settings where the statutory minimum remains fixed for long periods and changes abruptly, this feature addresses concerns in the literature that prior findings reflect short-run distortions rather than effective minimum wage increases (Card and Krueger, 2016; Clemens and Strain, 2021; Hurst et al., 2022). Second, I leverage institutional features of the Brazilian labor market that support the presumption that firms choose a single level of job amenities for all employees. Most notably, the consolidated labor laws and the widespread use of municipal-level collective agreements impose rigid limits on tailoring amenities to individual workers within the same establishment. Third, Brazil provides high-quality matched employer-employee data that are central to the empirical exercises in this study.

The two main empirical challenges the chapter faces are recovering unobserved job amenities and tracing spillovers across the entire wage distribution in response to higher wage floors. For the first challenge, I use worker flows across firms and recover job amenities at the establishment level. Building on Sorkin (2018), I leverage the fact that job-to-job transitions initiated by the worker, namely quits, reveal preferences over firms. By incorporating information on wage changes across such transitions, following Lehmann (2025), I separate wage and non-wage attributes of compensation. For the second challenge, I replicate Engbom and Moser (2022) by exploiting cross-state variation in the bindingness of the statutory minimum, thereby enabling me to compute marginal effects of minimum wage increases across different percentiles of the hourly wage distribution.

Taken together, my empirical findings uncover patterns consistent with the framework's predictions. Between 2012 and 2019, the real hourly minimum wage increased by around 9%. In the same period, my analysis indicates a decline in wage variance of around 6 log points and an increase in job amenity variance of around 35 log points. Although these findings do not constitute causal evidence for the wage-amenities tradeoff, I draw on the model's third prediction to strengthen the evidence of the mechanism. Consistent with the prediction, minimum wage increases between 2012 and 2019 propagate well beyond the lower tail of the wage distribution, pushing wages upward up to the 95th percentile.

The chapter contributes to the literature on the minimum wage and its effects on labor market outcomes. A growing body of literature argues that employers can accommodate higher wages by adjusting several margins beyond employment (Clemens, 2019, 2021), with

recent empirical work emphasizing two potential explanations: monopsony power (Azar et al., 2024; Berger et al., 2025; Faia et al., 2026) and passthrough to prices (Allegretto and Reich, 2018; Harasztosi and Lindner, 2019; Renkin et al., 2022). I complement this literature by showing suggestive evidence of the wage-amenities tradeoff as an additional adjustment margin, whereby firms respond to higher minimum wages by reducing job amenities.

Contrary to my approach, prior studies examine the effects of minimum wages on specific employee benefits. Clemens et al. (2018), for instance, find that workers become less likely to report employer-sponsored health insurance, while Kanayama et al. (2025) find almost no decline in flexible non-wage benefits such as transport reimbursements. The contrast in their conclusions points to a key limitation in their strategy: findings for specific benefits cannot rule out simultaneous adjustments to other non-wage attributes—especially since specific adjustments to job amenities can vary considerably across firms. My work, instead, relies on empirical predictions when employers cannot differentiate amenities across employees. This approach applies to a wide range of preferences over job amenities and compensation schemes, providing suggestive evidence that reconciles the economic intuition that firms adjust compensation in response to higher minimum wages with the recent empirical findings.⁴

Furthermore, the chapter contributes to the literature on how minimum wage increases spread across the wage distribution. Previous work in the United States (Lee, 1999; Autor et al., 2016; Cengiz et al., 2019; Derenoncourt and Weil, 2025) and the United Kingdom (Stewart, 2011) indicates small, often negligible spillover effects. In contrast, Engbom and Moser (2022) demonstrate that minimum wage increases in Brazil, between 1996 and 2018, affected wages up to the 90th percentile. My framework offers an explanation for these disparate results. When employers cannot tailor job amenities to different employees, wages at the top of the distribution rise due to compensating differentials.

The remainder of the chapter proceeds as follows. Section 2.2 introduces the framework and its empirical predictions on wages, job amenities, and wage floors. Section 2.3 describes the Brazilian institutional background and data sources, while Section 2.4 presents the empirical strategies used and the corresponding results.

⁴ Beyond research on the wage-employment tradeoff, recent studies also document negative effects on vacancies (Horton, 2025; Kudlyak et al., 2025) and separations (Dube et al., 2016), as well as worker flows toward larger, higher-paying, and more productive firms (Dustmann et al., 2021; Rao and Risch, 2026), which aligns with predictions in Board and Meyer-Ter-Vehn (2015) and Wang (2025b).

2.2 Framework: Wages, job amenities, and wage floors

The following section develops a simple framework that formalizes the tradeoff between wages and job amenities and enumerates the three main predictions I take to the data in Section 2.4.

2.2.1 Model

The basic framework consists of a static, random, posting, partial equilibrium model in which firms choose wages and job amenities, subject to worker outside options and binding minimum wages. In this subsection, I describe the model's setting and assumptions, and characterize two distinct equilibria: one in which firms have access to flexible amenities and another in which firms allocate the same amenities to all employees. For a complete derivation of the solution, see Section A.1 in the Appendix.

Setting and assumptions. Consider a firm $j \in \mathcal{J}$ that employs two workers indexed by $i \in \{L, H\}$, where L and H denote a low-paid and a high-paid worker, respectively. Workers differ in productivity ($\alpha_L < \alpha_H$) and outside options ($u_L < u_H$), but share common preferences over wages w and amenities a , as given by $U_i = w_i^\gamma a_i$.

The firm posts worker-specific contracts $\langle w_i, a_i \rangle$. Since this is a static model, firms cannot substitute labor.⁵ Therefore, if any worker rejects the contract, the firm exits and both workers receive their outside option u_i . Otherwise, the firm produces output equal to the workers' joint productivity, pays wages, and provides amenities at a unit cost p_a , earning profits equal to

$$\pi_j = f(\alpha_L, \alpha_H) - (w_L + w_H) - p_a(a_L + a_H)$$

for some generic production function $f(\alpha)$.

Total labor costs rise with wage floors. To keep the model simple, I assume that firms earn strictly positive profits, even when paying the minimum wage to all workers. An alternative interpretation is that new entrants, with a p_a low enough to break even, replace the exiting firms. Since optimal wages and job amenities do not depend on p_a for constrained firms, both interpretations are equivalent.

⁵ This implicitly rules out firms changing their worker composition. The simplification not only focuses attention on compensation rather than workforce adjustments, but also appears empirically plausible. Workplace heterogeneity (Handwerker, 2023) among Brazilian firms remained broadly stable between 2012 and 2019. If anything, worker composition became more heterogeneous over time.

Under these assumptions, I can set up the firm's cost-minimization problem, subject to each worker's participation and minimum-wage constraints. Namely,

$$\begin{aligned} \min_{\{(w_i, a_i)\}_{i=L}^H} \quad & (w_L + w_H) + p_a(a_L + a_H) \\ \text{s.t.} \quad & w_i^\gamma a_i \geq u_i, \quad i \in \{L, H\} \quad (PC) \\ & w_i \geq \underline{w}, \quad i \in \{L, H\} \quad (MW) \end{aligned}$$

Flexible amenities. When firms can differentiate job amenities among workers, each participation constraint pins down the worker's utility separately. As a result, individual optimal wages and job amenities depend on the bindingness of the corresponding minimum-wage constraint.

When the minimum wage is non-binding for worker i , the firm posts a contract comprising the following wage and non-wage components

$$w_i^* = u_i^{\frac{1}{1+\gamma}} (p_a \gamma)^{\frac{1}{1+\gamma}} \quad \text{and} \quad a_i^* = u_i^{\frac{1}{1+\gamma}} (p_a \gamma)^{-\frac{\gamma}{1+\gamma}}$$

Otherwise, the firm offers the minimum wage $\underline{w}$ and reduces job amenities to alleviate the mandatory wage increase, so that $a_i^* = u_i \underline{w}^{-\gamma}$.

Higher wage floors raise labor costs for workers with a binding minimum wage, potentially leading to small employment declines that I have assumed away for simplicity. Nevertheless, note that job amenities absorb some of the upward pressure on labor costs, implying that firms will resort to layoffs less often than in the absence of job amenities.

Furthermore, the solution highlights an important aspect of the framework. As long as the firm has access to flexible amenities, variation in contracts arising from minimum wage increases and other sources—such as changes in the firm's compensation policy—are indistinguishable.

Firm-level amenities. When firms rely on a single level of amenities a for all employees, posted contracts become tightly interconnected. Even if workers earn wages above the statutory minimum, optimal contracts depend on the outside options of both low- and high-paid employees. A salient case arises when the minimum wage binds both of them. Then, the firm posts contracts paying the minimum wage and sets job amenities based on the outside option of the high-paid worker, i.e., $a^* = u_H \underline{w}^{-\gamma}$.

As a result, the low-paid worker benefits from the higher wage floor and pockets a differential equal to $u_H - u_L$. Notice that further increases in the minimum wage do not affect the differential paid to the low earner, so that the firm can fully offset further upward pressure on labor costs by reducing job amenities.

Next, consider the case in which the low-paid worker is the only one with a binding wage floor. Then, the firm must offer the statutory minimum to the low-paid employee, and faces two options when setting job amenities.

On the one hand, the employer can use job amenities to satisfy the participation constraint of the high-paid worker and absorb the higher wage floor. In this case, the firm refuses to use job amenities as an offsetting mechanism. Although the differential paid to the low-paid worker represents less than the gap in outside options, firms taking this direction must fully absorb the increase in labor costs.

On the other hand, the firm can set jobs amenities based on the low-paid worker's outside option, i.e., $a^* = u_L \underline{w}^{-\gamma}$. However, by doing so, the firm reduces the high-paid worker's total compensation, who now requires a compensating differential to remain indifferent. As a result, the firm has to post a higher wage equal to $w_H^* = \underline{w} (u_H/u_L)^{1/\gamma}$, which originates spillovers across the wage distribution. In this case, the low-paid worker does not pocket a differential, and both employees achieve utilities equal to their respective outside options.

2.2.2 Empirical predictions

One notable implication of the model is that, when firms cannot tailor job amenities to individual workers, a minimum wage increase need not compress the wage distribution nor widen the dispersion of job amenities. This contradicts the conventional wisdom that higher wage floors trim the lower tail of the wage distribution, as well as the predictions in the case of flexible amenities. However, what stands out is that without further assumptions, the model predicts spillovers among workers who earn above the statutory minimum.

Firms react to a minimum wage increase depending on their exposure to wage floors, which is roughly captured by the number of workers earning exactly the statutory minimum. For that reason, I define three categories. All employees at the high-paying firm earn wages above the minimum wage. Rather, low-paid workers at the medium-paying firm and all employees at the low-paying firm earn the statutory minimum.

Medium-paying firms, however, may differ in their optimal contracts depending on the unit cost of amenities and workplace heterogeneity—the gap in employees' outside options.

Firms with a more homogeneous workforce (henceforth specialized employers) prefer to pass along minimum wage increases to the high-paid worker. Conversely, firms with higher workplace heterogeneity (hereafter vertical employers) favor leaving rents to low-paid employees.

Without further assumptions about the distribution of firms across categories, the model delivers empirical predictions I can map to aggregate moments observed in the data. This feature eliminates the need to track firms as they move across categories. And, more importantly, it addresses a previous challenge in the literature: insufficient exogenous variation and limited detail on workers' contracts.

Prediction 1: Wage compression. While intuition suggests that wage floors compress the wage distribution, spillovers to higher earners make the conclusion less obvious. A significant passthrough to the average wage can compress the upper tail of the distribution while increasing dispersion at the lower tail. Therefore, a higher minimum wage can actually increase dispersion in the wage distribution by shifting its skewness upward.

The framework accounts for this possibility and concludes that, everything else equal, if specialized firms have limited workplace heterogeneity, minimum wage increases compress the wage distribution, $\downarrow \mathbb{V}(w_i)$. While not necessary in other contexts, both the assumption and the prediction align with empirical patterns in Brazilian labor markets.

Prediction 2: Increased dispersion in job amenities. While low-paying firms and specialized employers reduce job amenities to offset wage increases, the effect on the overall dispersion depends on the interplay between two opposing forces. First, the gap in job amenities compared to high-paying employers and vertical firms widens, increasing the variance. Second, the within-variance in job amenities among specialized employers and low-paying firms decreases.

The framework, however, establishes that overall dispersion increases with the minimum wage, $\uparrow \mathbb{V}(a_j)$, provided the initial dispersion in job amenities at the bottom of the distributions is limited. In other words, if workers paid the minimum wage have similar outside options or earn compensation mostly in wages. Notoriously, in a context of continuous adjustments to the federal minimum wage—and the corresponding downward adjustments in job amenities—both circumstances become extremely plausible.

Prediction 3: Wage increases above the minimum wage. As previously shown, reducing job amenities in specialized firms negatively affects high-paid workers, who, as a result, demand a compensating differential to remain indifferent. As a result, an increase

in the minimum wage propagates throughout the wage distribution, leading to a significant passthrough to the average wage. In particular, the model anticipates positive wage spillovers up to the percentile in the distribution occupied by the highest earner in a specialized firm. Interestingly, nothing prevents this worker from locating near the top of the distribution.

2.3 Brazil: Institutional background and data

The following section describes the institutional setting and data used in the empirical analysis. Brazil provides a particularly suitable environment to study how firms adjust wages and job amenities in response to minimum wage increases. More precisely, the Brazilian labor market aligns with the context described in Section 2.2, and the country offers high-quality matched employer-employee data that I can leverage for my analysis.

2.3.1 Labor market institutions in Brazil

Several features in the Brazilian labor market make it well-suited for my analysis. In particular, minimum wage adjustments are frequent but modest, and legal regulations can interfere with discriminating job amenities across employees. This subsection reviews relevant institutions and discusses additional empirical evidence in Brazil supporting the approach.

Minimum wage. According to the *Consolidated Labor Laws* (CLT, for its Portuguese acronym), full-time workers (44 hours per week) with a formal contract cannot earn less than the statutory minimum wage. Along the same lines, the CLT stipulates that, for part-time workers, firms must adjust the sum proportionally to contracted hours. The Brazilian government supports enforcement through mandatory annual filings and routine labor inspections. As a result, compliance is relatively high among formal firms. And, more importantly, the minimum wage has a lighthouse effect, with a near full pass-through to informal employees working at formal firms and relatively small formal-to-informal reallocations (Saltiel and Urzúa, 2022; Derenoncourt et al., 2025).

Brazil adopted a unified federal minimum wage in 1984, although 5 out of 27 states (*unidades federativas*) have set independent regional wage floors above the federal rate.⁶

⁶ These five states comprise Rio de Janeiro and Rio Grande do Sul (2001), Paraná (2006), São Paulo (2007), and Santa Catarina (2010). Among them, Paraná, Rio Grande do Sul, and Santa Catarina implemented multi-layer regional wage floors that vary with occupation, industry, or qualifications. Over time, state-level minimum wages have ranged from parity with the federal rate to roughly 40 percent above it.

Between 2012 and 2019, minimum wage increases occurred on the first of January and followed a rule-based formula. More precisely, laws 12.382/2011 and 13.152/2015 tied annual updates of the minimum wage to past inflation and real GDP growth.

While the federal minimum wage remains the primary wage-setting institution in Brazil, unions routinely negotiate wage floors above the federal rate through collective bargaining agreements (CBA). Similar to the statutory minimum, coverage is universal. Therefore, negotiated terms apply to all employees within the same firm or industry, regardless of union membership. That said, only about 10 percent of CBA clauses concern wage-related bargaining, such as wage adjustments and floors (Lagos, 2025). Instead, a significant part of negotiations over collective agreements focuses on job amenities.

CBAs and the CLT. CBAs are binding contracts between unions and employers that apply collectively to all workers under the union’s representation. Meaning that individual contracts in Brazil cannot suppress or limit rights specified in the agreement, irrespective of union membership. Moreover, CBAs can be established at the firm level or across entire economic sectors. In sectoral CBAs, unions and employer associations negotiate general floors and conditions that build up the requirements stipulated in the CLT. A firm-level CBA, on the other hand, entails direct negotiations with the employer and addresses workplace-specific issues.⁷

The CLT is the most comprehensive labor framework in Brazil, so it often takes precedence over CBAs. It establishes minimum standards for formal employment relationships in the country and, while it does not explicitly prohibit differentiation across workers, it emphasizes uniform treatment of employees within establishments. The *Supreme Labor Court* (TST, for its Portuguese acronym) has adopted the same interpretation, ruling against firms unable to demonstrate objective, verifiable criteria for special gratifications or privileges.⁸

In comparison with labor markets in other countries, these institutional features appear to limit firms’ ability to differentiate job amenities across workers, supporting the single-amenity assumption.

⁷. For more information on Brazilian CBAs, see Lagos (2025). There, the author describes the nature of unions and collective agreements in Brazil, showing that, on average, up to 76% of clauses focus on job amenities unrelated to wages or separations. Furthermore, 94% of unions operate at the municipal level, negotiating sectoral and firm-level CBAs that cover around 48% and 16% of private-sector workers, respectively.

⁸. For example, Summary 451/2014 ruled against firms excluding workers from profit-sharing bonuses according to the separation date. Similarly, Case Law Guidance 383/2011 established equal conditions for outsourced and in-house workers who performed similar tasks.

2.3.2 Data

The following subsection describes the data used in the empirical analysis, consisting of the *Annual Social Information Report* (RAIS, for its Portuguese acronym), and the sample selection procedures used to identify firm-level job amenities from worker flows.

RAIS. Brazil offers high-quality data on employment relationships, establishments, and worker characteristics. Every year, the Brazilian government collects mandatory filings from registered firms, which cover the universe of formal employees. These matched employer-employee administrative records, known as RAIS, report many features central to the recent empirical literature on Brazilian labor markets (Alvarez et al., 2018; Gerard et al., 2021; Engbom and Moser, 2022; Van Doornik et al., 2023).

I focus on the period ranging from 2012 to 2019. Firms must report the exact hiring and separation dates for every worker employed in a given calendar year, allowing me to reconstruct workers' full employment histories during this period. Every employment spell details the modality, contractual weekly work hours, six-digit occupational codes, and average monthly wages, including variable pay components such as performance bonuses, profit-sharing, and overtime. More importantly, RAIS includes information about the reason notified for every hire and separation, allowing me to distinguish between layoffs, quits, and transfers. For more information about the properties of this variable, see Subsection 4.2.2 or Dobbin et al. (2025).

RAIS also contains unique individual and employer identifiers, as well as personal data (e.g., gender, age, and educational attainment) and firm characteristics (e.g., five-digit industry codes, location, and legal status). My analysis is at the establishment level, so that my definition of firm coincides with that of CBAs and the CLT. Throughout the chapter, I use the terms firm, employer, and establishment interchangeably.

Sample. The analysis centers on workers aged 18 to 54 with urban open-ended contracts in the private sector. I use RAIS to construct quarterly employment histories, assigning workers with multiple contracts to their highest-earning job.

Consistent with previous empirical work, I further restrict the sample to the largest connected set of workers and firms. I estimate firm-specific wage premiums in the weakly connected set of movers (henceforth, the AKM sample) and firm-level job amenities in the strongly connected set (SCC) of actively recruited workers. A connected set defines a cluster of workers and firms linked through employment transitions over a time window. A firm

belongs to the AKM sample if any of its workers were previously employed at or later moved to another firm in the weakly connected set (see Abowd et al., 1999; Card et al., 2013, 2018; Gerard et al., 2021). Similarly, a firm belongs to the SCC if it has both poached an employee from and lost an employee to another firm in the SCC through active recruiting (see Sorkin, 2018; Lehmann, 2025).

As I mentioned, estimating firm-specific wage premiums and job amenities requires defining a time window, which poses a dilemma. Longer windows increase worker mobility and improve connectedness, so the analysis covers more workers and firms, but they reduce the number of samples available to track dynamics over time. I prioritize representativeness and, therefore, split the sample into two equal periods of 16 quarters: 2012Q1-2015Q4 and 2016Q1-2019Q4.

Table 2.1 shows that focusing on the SCC (Columns 3 and 6) restricts the analysis to establishments that, on average, employ three to four times as many workers as the representative establishment in Brazil (Columns 1 and 4).⁹ Furthermore, workers in the SCC earn higher hourly wages and show greater wage dispersion, while establishments display less dispersion in firm homogeneity.¹⁰ Beyond these differences, employment characteristics remain broadly similar across samples.

Comparing samples over time reveals that most employment characteristics exhibit only minor shifts: average monthly hours decline by roughly one hour, female participation rises by about 2 percentage points, and the average employee is one year older. By contrast, the average real hourly wage increases by 4 log points, accompanied by a 0.04 decline in its variance. These changes are stronger in the SCC, where the average rises by 7 log points and the variance declines by 0.03. Subsection 2.4.1 explores these distributional wage patterns in more detail.

Table 2.1 also shows a sharp decline of about 40 million quarterly employment relationships across the two samples. That being said, job destruction does not account for the contraction: layoff rates decline uniformly across cohorts during this period (Figure 2.1a). Instead, Figure 2.1b shows that workers born after 1991 (aged 28 or younger in 2019) exhibit

⁹. Compared to existing work, four years of worker mobility in Brazil capture a larger proportion of establishments than a six-year window in the US (Sorkin, 2018) and an eight-year window in Austria (Lehmann, 2025). In Brazil, the SCC covers roughly 7 pp (percentage points) more establishments, 9 pp fewer employment relationships, and 18 pp fewer workers than the US. Compared to Austria, 12 pp more establishments, 21 pp more employment relationships, and 15 pp more workers.

¹⁰. Firm homogeneity refers to workplace homogeneity in Handwerker (2023) and describes the overall distribution of occupations by establishment. It takes a value of 1 when employers hire only one occupation, and 0 when no two workers within the firm have the same occupation.

Table 2.1 – *Summary statistics and sample selection*

	2012-2015			2016-2019		
	RAIS	AKM	SCC	RAIS	AKM	SCC
Establishments	4,291,363	3,323,529	765,419	4,354,262	3,126,974	498,859
	100.00%	77.45%	17.84%	100.00%	71.81%	11.46%
Workers	50,416,673	49,079,671	39,338,017	46,092,427	44,137,460	30,704,508
	100.00%	97.35%	78.03%	100.00%	95.76%	66.62%
Employment spells	510,008,563	497,308,284	372,771,472	470,445,459	451,144,482	291,773,507
	100.00%	97.51%	73.09%	100.00%	95.90%	62.02%
Firm size ($\mathbb{E}$)	11.3	13.4	35.1	10.3	12.9	42.0
Firm homogeneity ($\mathbb{E}$)	0.31	0.31	0.30	0.31	0.31	0.30
Firm homogeneity ($\mathbb{V}$)	0.060	0.058	0.048	0.061	0.058	0.048
Log hourly wage ($\mathbb{E}$)	2.10	2.11	2.19	2.14	2.16	2.26
Log hourly wage ($\mathbb{V}$)	0.394	0.396	0.432	0.349	0.353	0.401
Hours ($\mathbb{E}$)	185.9	185.9	185.4	184.8	184.8	183.9
Monthly paid (%)	0.91	0.91	0.89	0.93	0.92	0.90
Female employment (%)	0.41	0.41	0.40	0.43	0.42	0.42
Age ($\mathbb{E}$)	33.7	33.6	33.6	34.5	34.4	34.4
Years of schooling ($\mathbb{E}$)	12.3	12.3	12.4	12.5	12.5	12.7

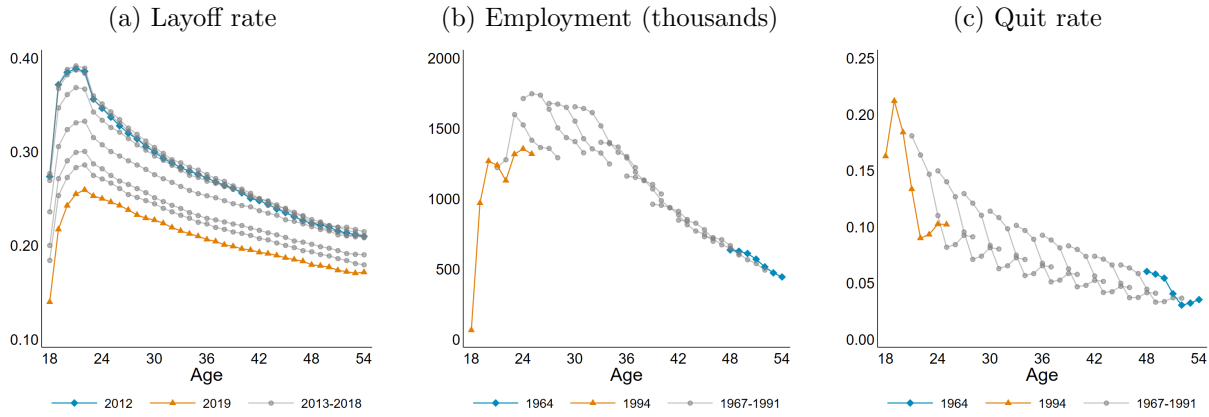
Notes: The table reports characteristics over time for employment relationships, workers, and establishments in Brazil between 2012 and 2019. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. The AKM sample centers on employers in the weekly connected set of movers and is used to estimate firm-specific wage premiums. The SCC focuses on employers in the strongly connected set of actively recruited employees and is used to estimate firm-level job amenities. Firm homogeneity refers to workplace homogeneity, as defined by Handwerker (2023), and displays the normalized occupational Herfindahl-Hirschman index (HHI). It measures the average occupation size within a firm, taking a value of 1 when employers hire only one occupation, and 0 when no two workers within the firm have the same occupation.

lower entry rates into the formal labor market compared to older workers.

At the same time, I observe a decline in worker mobility over time: quits fall uniformly across cohorts until 2016 and do not recover afterwards (Figure 2.1c). As a result, the SCC sample covers a smaller fraction of employment relationships, workers, and establishments in 2016-2019 compared to 2012-2015.

2.4 Empirical analysis: Taking predictions to the data

In this section, I turn to the empirical analysis. Results mirror the three testable implications from Section 2.2. Namely, that higher wage floors compress the wage distribution, increase dispersion in job amenities, and raise wages above the binding minimum wage.

Figure 2.1 – *Employment dynamics in Brazil (2012-2019)*

Notes: The figure reports average employment dynamics in Brazil between 2012 and 2019. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. Panel (a) displays the annualized layoff rate by age across different years. Panel (b) shows headcounts (in thousands) of workers by age across different cohorts. Panel (c) shows the annualized quit rate by age across different cohorts. In Panel (a), each line corresponds to a year, with each dot representing the average outcome for workers of a given age that year. In Panels (b) and (c), each line conveys a cohort, with each dot representing the average outcome for workers of a given age born in that cohort. I report cohorts every three years, ranging from 1964 (48 years old in 2012) to 1994 (18 years old in 2012).

2.4.1 Reduction in wage inequality

The first prediction states that higher wage floors reduce wage inequality. Correspondingly, Brazil has experienced a long-lasting decline in wage disparities since the 1980s. As a result, numerous researchers have investigated potential explanations for this trend, although the literature has concluded that the minimum wage played a central role. Consistent with similar findings, I document a decline in wage inequality between 2012 and 2019 across a wide range of measures.

For these empirical exercises, I use the entire RAIS sample to construct five annual measures of wage dispersion. The first two variables follow standards in the literature. I compute the lower-tail hourly wage inequality as the ratio of the median to the 10th percentile (P50:P10), and the upper-tail hourly wage inequality as the ratio of the 90th percentile to the median (P90:P50). For the next two measures, I deviate from the standard definition by replacing the median with the average hourly wage. Lastly, I use the variance of log monthly wages.

Figure 2.2 uncovers that wage disparities decreased over time across all five measures. For example, Figures 2.2a and 2.2b show that the upper-tail and lower-tail hourly wage inequality fell by about 9 and 7 log points, respectively. In relation to previous studies using longer time series (Alvarez et al., 2018; Engbom and Moser, 2022), the trends during this

decade indicate a more pronounced wage compression at the top of the distribution rather than at the bottom.

Figure 2.2d further contributes to this idea, showing that minimum hourly wage increases have had a significant passthrough to the average hourly wage. Both hourly rates moved closely together over the period, with the average hourly wage broadly tracking changes in the minimum hourly wage. By 2019, both series had accumulated roughly 8 log points in real terms. Conversely, Figure 2.2c displays a sharp decline in the variance of log monthly wages as the real statutory minimum wage rose.

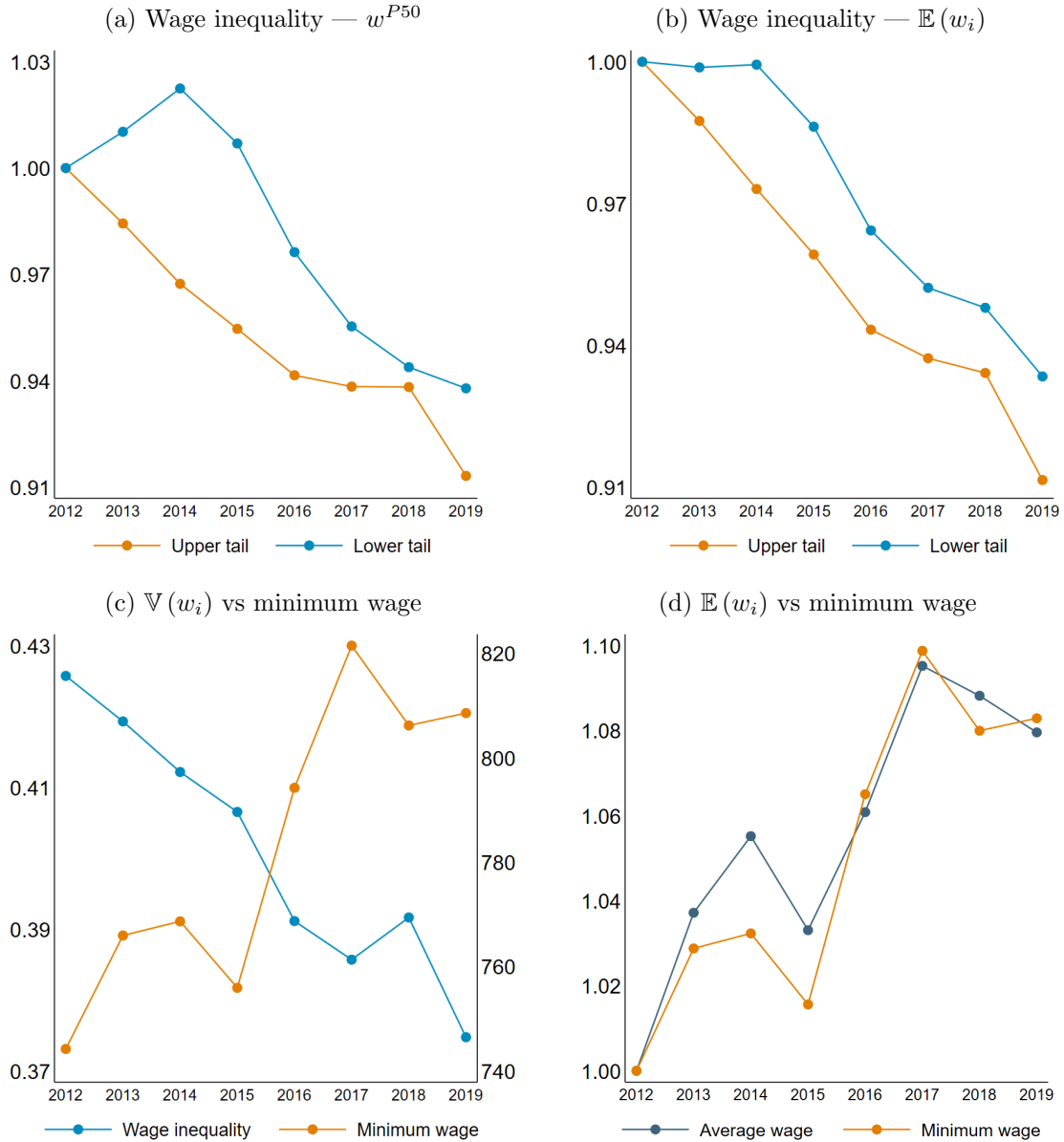
Firpo and Portella (2019, 2025) enumerate a series of potential explanations from the literature to account for these particular trends in Brazil. Researchers have covered many alternatives, including economic growth, productivity shocks in the tradable sector, higher educational attainment, formalization of employment, discrimination, and unions' bargaining power, among others. Nonetheless, the consensus indicates that the minimum wage accounts for at least half of the observed decline in wage and earnings inequality.

2.4.2 Increase in job amenities inequality

The second prediction poses that higher wage floors increase the variance of job amenities. In this subsection, I first discuss the interpretation of non-wage attributes of labor contracts. Next, I present my implementation of the empirical strategy in Lehmann (2025), which I use to recover firm-specific job amenities separately for periods 2012-2015 and 2016-2019. My findings indicate a significant increase in the dispersion of job amenities in Brazil over time.

Job amenities. When workers assess the value of jobs, they weigh contracts as a combination of wage and non-wage compensation attributes. Beyond traditional employee benefits (e.g., health and life insurance, childcare, paid time off, training), there is plenty of evidence that workers value other items in their employment relationships: remote work and flexible schedules (Mas and Pallais, 2017), job stability (Wiswall and Zafar, 2018), working conditions (Maestas et al., 2023; Collis and Van Effenterre, 2025), job safety (Ferreira et al., 2025; Anelli and Koenig, 2025), or even dignity (Dube et al., 2022). For further details on the existing literature, see Mas (2025).

Along the same lines, workers may loathe certain non-wage attributes of jobs. For instance, Humlum et al. (2025) show that higher-paying firms supply worse net job amenities. Even if these employers offer better perks and flexibility to employees, these jobs also entail

Figure 2.2 – *Minimum wage and inequality in Brazil*

Notes: The figure displays summary statistics from 2012 to 2019 comparing the evolution of the minimum wage and several measures of wage inequality in Brazil. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. Panel (a) compares lower- and upper-tail hourly wage inequality over time, using the standard definition in the literature: lower-tail inequality as the ratio of the median to the 10th percentile of the outcome (P50:P10), and upper-tail inequality as the ratio of the 90th percentile to the median (P90:P50). Panel (b) repeats the previous empirical exercise by replacing the median with the average wage. Panel (c) compares the real minimum wage with the variance of log monthly wages over time. In the data, the correlation between the two time series is -0.906. Panel (d) compares average and minimum hourly wages over time in logs. I normalize all variables in Panels (a), (b), and (d) to one in 2012.

higher turnover, more stress, and a faster work pace. Similarly, frameworks that incorporate the wage-effort tradeoff into their models (Board and Meyer-Ter-Vehn, 2015; Wang, 2025b) reach conclusions consistent with the literature on job amenities: workers are willing to pay for less demanding jobs.

The existence of *disamenities* connects to Weil (2014), who describes a shift away from direct hiring toward alternative work arrangements. An increasing share of workers act as independent contractors, keep on-call contracts, or are employed by temporary agencies and subcontractors. Such arrangements not only preclude access to regular job amenities but can also deny benefits tied to standard employment relationships. In settings like Brazil, informality plays a similar role, putting workers in precarious conditions compared to those in formal employment.

Informal workers and independent contractors. Derenoncourt et al. (2025) demonstrate that informal workers at formal firms in Brazil almost fully benefit from minimum wage increases, despite the CLT not covering them. I argue that the wage-amenities tradeoff also applies to these individuals, but in reverse order. When firms reduce firm-level amenities to adjust for higher wage floors, they must not only compensate higher earners but also informal workers employed at the firm. Otherwise, these individuals would have incentives to leave the firm for a formal job or another better-paying informal one.¹¹

I could wield a similar argument for independent contractors and workers at informal firms, even if these individuals were to keep the same working conditions in response to minimum wage increases. Nonetheless, RAIS does not cover informal workers and independent contractors in Brazil, and there is no empirical evidence at hand to support my argument. Thus, I cannot formally extend my analysis to these workers, although I do not expect this to affect the overall interpretation of the results.

Estimation of firm-level amenities. The backbone of the empirical strategy is Sorkin (2018). As in the framework in Section 2.2, it assumes a partial equilibrium model with posting. These assumptions imply that firm heterogeneity is taken as given and that there is no bargaining. In addition, the model introduces further assumptions about how workers search for jobs. First, employed and non-employed workers receive the same number of offers

¹¹. Several studies in Brazil show that formal workers eligible for unemployment insurance often agree to temporarily transition to informality (Carvalho et al., 2018; O’Leary et al., 2020; Van Doornik et al., 2023). This is a different example of the interplay between wages and the disamenities associated with informality. Here, the worker who transitions to an informal job receives a compensating differential paid by the government and not the firm: unemployment insurance.

from an identical distribution—meaning, a non-directed, on-the-job search with exogenous effort. Second, workers value firms the same, up to i.i.d. idiosyncratic preferences, which translates into a unique firm ranking—or a uniform job ladder across all workers.

Employers offer contracts $\langle w_{it}, a_j \rangle$ and differ according to size g_j , exogenous separation rates δ_j and ρ_j , as well as exogenous hiring rates f_j .

A worker i values a job at firm j from the flow payoff $v_j(w_{it}, a_j)$ and the corresponding continuation value, which they discount at rate β . With probability δ_j , the firm lays off the worker. Then, with probability ρ_j , the worker finds a job before the end of the next quarter. Otherwise, the layoff leads to nonemployment. If there is no exogenous job destruction, the worker receives an offer from firm k with probability λ_E and decides whether to quit and accept it or to stay at their current firm. With probability $1 - \lambda_E$, the worker does not receive any new offer but can still quit and move to nonemployment if preferred.

$$\begin{aligned}
 \underbrace{V_j^e(w_{it}, a_{it})}_{\text{value of being at } j} &= \underbrace{v_j(w_{it}, a_{it})}_{\text{flow payoff}} + \beta \cdot \mathbb{E} \left[\underbrace{\delta_j(1 - \rho_j) \int_{\iota_1} \{V^n + \iota_1\} dI}_{\text{exogenous job destruction}} + \underbrace{\delta_j \rho_j \sum_k \int_{\iota_2} \{V_k^e + \iota_2\} dI \tilde{f}_k}_{\text{exogenous employer-to-employer}} \right. \\
 &+ \underbrace{(1 - \delta_j)}_{\text{no exogenous job destruction}} \times \left(\underbrace{\overbrace{\lambda_E}^{\text{offer}} \sum_k \int_{\iota_3} \int_{\iota_4} \max \left\{ \overbrace{V_k^e + \iota_3}^{\text{accept}}, \overbrace{V_j^e + \iota_4}^{\text{reject}} \right\} dI dI f_k}_{\text{endogenous employer to employer}}} \right. \\
 &\quad \left. \left. + \underbrace{\overbrace{(1 - \lambda_E)}^{\text{no offer}} \int_{\iota_5} \int_{\iota_6} \max \left\{ \overbrace{V^n + \iota_5}^{\text{leave}}, \overbrace{V_j^e + \iota_6}^{\text{stay}} \right\} dI dI f_k}_{\text{endogenous job destruction}} \right) \right]
 \end{aligned}$$

The decision to quit a firm to take a job at a new one is what reveals workers' preferences for employers. As Sorkin (2018) shows, the relative flows across any pair of firms determine the relative value between the two employers. Moreover, one does not need to observe flows across every pair of firms; it is sufficient for the firm to be part of the SCC.

$$\underbrace{\frac{M_{jk}}{M_{kj}}}_{\text{relative flows}} = \underbrace{\frac{f_j}{f_k}}_{\text{relative offers}} \underbrace{\frac{g_k(1 - \delta_k)}{g_j(1 - \delta_j)}}_{\text{effective relative firm size}} \underbrace{\frac{\mathbb{P}(j \succ k)}{\mathbb{P}(k \succ j)}}_{\text{relative firm values}}$$

From this point forward, I follow Lehmann (2025) and assume that workers employed at firm j derive utility from wages and job amenities as

$$V_{it}^j = V_j^e(w_{it}, a_{it}) = w_{it}^\gamma \cdot a_{it} \quad (2.1)$$

where γ represents the elasticity of substitution between wages and job amenities. Furthermore, I assume that a_{it} depends on firm-specific job amenities $\chi_{J(it)}$ and idiosyncratic preferences ι_{it} ,

$$\log a_{it} = \chi_{J(it)} + \iota_{it}, \quad \iota \sim N(0, \sigma_\iota^2) \quad (2.2)$$

while wages are jointly determined by worker and firm characteristics, following an AKM model specification (Abowd et al., 1999; Card et al., 2013, 2018)

$$\log w_{it} = \alpha_i + \psi_{J(it)} + X_{it} \cdot b + \nu_{it}, \quad \nu \sim N(0, \sigma_\nu^2) \quad (2.3)$$

I estimate this model of wage determination separately for 2012-2015 and 2016-2019 using the AKM sample. In Section A.2 in the Appendix, I provide more details on the procedure and compare estimations across samples and over time.

By replacing Equations (2.2) and (2.3) into (2.1), I can express the worker's utility function in terms of fixed worker and firm characteristics, as well as idiosyncratic i.i.d. shocks to wages and preferences for job amenities, $\varepsilon_{it} = \gamma\nu_{it} + \iota_{it}$.

$$\log V_{it}^j = \gamma(\alpha_i + \psi_j + X_{it}\beta) + \chi_j + \underbrace{\gamma\nu_{it} + \iota_{it}}_{\varepsilon}, \quad \varepsilon_{it} \sim N(0, \underbrace{\gamma^2\sigma_\nu^2 + \sigma_\iota^2}_{\sigma_\varepsilon^2})$$

A worker i who receives an offer from firm k must make a maximizing decision by comparing V_{it}^j and V_{it}^k , which is why economists consider these transitions as *endogenous*.¹² When workers make voluntary moves from firm j to k , it reveals that they rank total compensation in firm k higher in the job ladder compared to firm j .

$$\mathbb{P}(k \succ j) = \mathbb{P}(\log V_{it}^k > \log V_{it}^j)$$

^{12.} A benefit of RAIS is that I can directly tell quits apart from layoffs, solving the problem of distinguishing endogenous from exogenous job-to-job transitions. For instance, Lehmann (2025) relies on unemployment benefit records, and Sorkin (2018) targets excess layoffs during downturns to calibrate the probability that any job-to-transition was a quit.

$$\begin{aligned}
&= \mathbb{P}\left(\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j) > (\varepsilon_{1it} - \varepsilon_{2it})\right) \\
&= \Phi\left(\frac{\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j)}{2\sigma_\varepsilon}\right)
\end{aligned}$$

Identification relies on the wage-amenities tradeoff discussed in Section 2.2. In essence, if a worker values compensation at two firms equally, $V_{it}^j = V_{it}^k$, any difference in firm-specific pay premiums, $\psi_j - \psi_k$, can be attributed to compensating differentials due to uneven job amenities, $\chi_j - \chi_k$, or idiosyncratic taste differences, $\varepsilon_{1it} - \varepsilon_{2it}$.

In equilibrium, better-skilled workers, on average, earn higher wages and benefit from more generous job amenities at the firm. However, the contracts of individuals with similar skills may vary greatly across employers. Intuitively, all these contracts lie over the same indifference curve, as workers ultimately derive the same utility from any of them. Note that this is based on a partial equilibrium model; as such, I abstract from why contracts differ across employers and treat these choices as given.

To take this relationship to the data, I specify the following likelihood function

$$\mathcal{L} = \prod_{n=1}^N \Phi\left(\frac{\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j)}{2\sigma_\varepsilon}\right)^{\frac{1}{g_j} \frac{1}{f_k}}$$

where g_j and f_k account for an important heterogeneity across firms. Consider the case where firms j and k are identical except that k makes twice as many job offers as j . Then, one should expect twice as many workers moving towards firm k as to firm j . Likewise, if k employed twice as many workers as j , one should expect twice as many workers leaving firm k compared to firm j .

The estimation does not yield estimates for γ and χ_j , $\forall j \in \mathcal{J}$. Notice that without assumptions on the scaling parameter σ_ε , the identification of these parameters becomes unfeasible. Instead, I obtain the following estimands

$$\hat{\gamma} = \frac{\gamma}{2\sigma_\varepsilon} \quad \text{and} \quad \hat{\chi}_j = \frac{\chi_j}{2\sigma_\varepsilon}$$

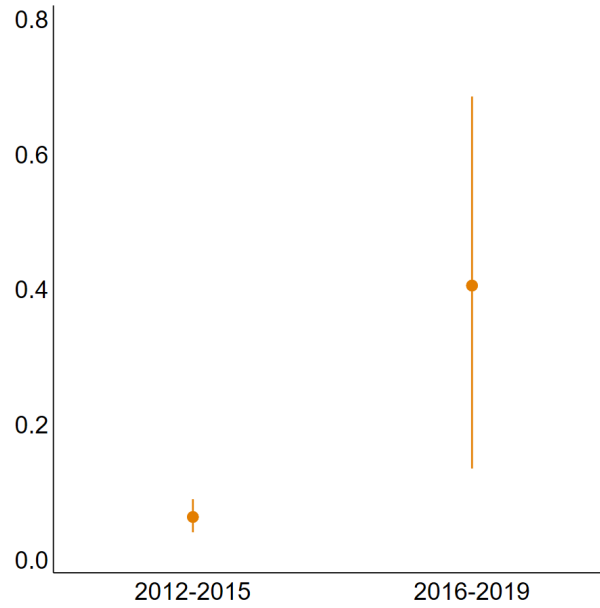
However, the coefficient given by the ratio between the two

$$\hat{\chi}^{BRL} = \hat{\chi}/\hat{\gamma} = \chi/\gamma$$

does not depend on σ_ϵ and, as such, is exempt from this limitation. In addition, the expression offers a nice interpretation, as $\hat{\chi}^{BRL}$ represents the value of firm-specific job amenities in the same currency used to express wages: log-BRL (*reais brasileiros*). For more details on the empirical estimation of firm-specific job amenities and how to account for noisy estimates in the plug-in estimator of the variance, see Section A.3 in the Appendix.

Figure 2.3 plots the variance of job amenities in Brazil, adjusted for firm size, across 2012-2015 and 2016-2019, showing a significant increase in the dispersion of firm-specific job amenities over time. Point estimates rose from 0.0628 in 2012-2015 to 0.4052 in 2016-2019. I bootstrap the likelihood function 50 times to compute confidence intervals for the variance estimations. The difference remains statistically significant at the 95% level, even when accounting for the later period being less precisely estimated. Although this result does not imply causality, the empirical exercise supports the framework's intuition that firms reduce job amenities in response to wage floor increases, thereby increasing job amenities inequality nationwide.

Figure 2.3 – *Variance of job amenities in Brazil*



Notes: The figure compares the variance of firm-level job amenities in Brazil over time. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. I estimate the variance of firm-level job amenities for establishments in the SCC, separately for the periods 2012-2015 and 2016-2019. Since estimates of job amenities contain noise, I obtain the variance of the estimation error by bootstrapping the likelihood function 50 times and use it to adjust the variance of firm-level job amenities using a formula (Sorkin, 2018; Lehmann, 2025). To obtain confidence intervals, I correct variance estimates across all 50 bootstraps and report the 5th and 95th percentiles of the distribution. See Section A.3 in the Appendix for details.

2.4.3 Spillovers across the entire wage distribution

The third prediction indicates that medium-paying firms generate wage spillovers throughout the wage distribution. The empirical strategy in this subsection leverages cross-state variation in the bindingness of the federal minimum wage, namely the *effective minimum wage*. I follow Engbom and Moser (2022) to estimate the marginal effects of minimum wage increases across different percentiles of the wage distribution. In particular, I perform the analysis separately for the periods 2012-2015 and 2016-2019, finding that higher wage floors propagate up to the 85th and 95th percentile of the wage distribution, respectively.

Consider a firm that pays the minimum wage to a fraction of its workforce, while the remaining employees earn above that level. When the firm reduces job amenities to offset the increase in labor costs, workers who earned above the new statutory minimum now perceive a loss in their total compensation. In consequence, the firm has to offer compensating differentials, and wage increases propagate across the wage distribution.

Lee (1999) develops an empirical strategy to measure wage spillovers separately from changes in the latent wage distribution. That is, the underlying wage distribution that one would observe in the absence of wage floors. This is important; otherwise, one could attribute wage changes to the minimum wage when they are actually due to other concurrent events.

The strategy defines measures for the effective minimum wage and wage dispersion relative to an arbitrary point in the wage distribution, p' , and exploits cross-state variation in the former. As long as p' remains completely unaffected by the minimum wage, co-movements between the two variables imply direct effects or wage spillovers related to its changes.

The first step is to establish the effective minimum wage in Brazil. I define the variable $kaitz_{st}(p') \equiv \log w_t^{min} - \log w_{st}(p')$, which stands for the log difference between the federal minimum hourly wage w_t^{min} and the p^{th} percentile of the hourly wage distribution in state s and quarter t . Thus, as the bindingness becomes tighter, $kaitz_{st}(p')$ becomes less negative.

The next step is to generate different measures of wage dispersion across wage percentiles $p \in \{5, 10, \dots, 95\}$. Thus, I define $y_{st}(p; p') \equiv \log w_{st}(p) - \log w_{st}(p')$, which represents the log difference between wage percentiles p and p' in state s and quarter t . Intuitively, as the wage distribution compresses, $y_{st}(p; p')$ moves towards zero.

I repeat this process separately for the periods 2012-2015 and 2016-2019, which allows me to construct two balanced panels at the state-quarter level, each comprising 27 states and 16 quarters.

Then, I use the following specification

$$y_{st}(p; p') = \sum_{n=1}^2 \beta_n(p) \times kaitz_{st}(p')^n + \delta_s(p) + \sum_{n=1}^2 b_s^n(p) \times t^n$$

to estimate non-linear effects of minimum wage's bindingness on wage dispersion, separately for each wage percentile p . I include linear and quadratic measures of effective minimum wage, as well as state fixed effects and state-specific linear and quadratic time trends. Since the specification is quadratic, I compute the marginal effects $\tau(p)$ for each wage percentile $p \in \{5, 10, \dots, 95\}$ using

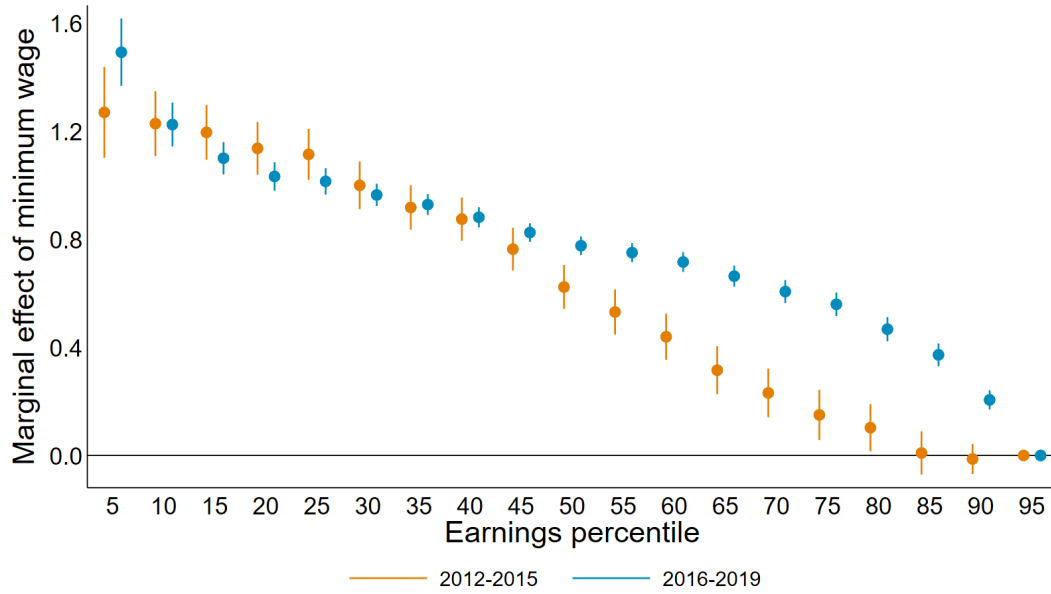
$$\tau(p) = \beta_1(p) + 2 \cdot \beta_2(p) \mathbb{E}[kaitz_{st}(p')]$$

where $\mathbb{E}[kaitz_{st}(p')]$ is the average $kaitz_{st}(p')$ across states and quarters.

Before moving forward, there are two things worth mentioning. First, previous studies using the same empirical approach in the US set $p' = 50$, arguing that the median wage is unlikely to be directly or indirectly affected by the minimum wage (Lee, 1999; Autor et al., 2016). On the contrary, Engbom and Moser (2022) show that spillovers in Brazil are significant and opt for $p' = 90$. In this chapter, I lean toward $p' = 95$.

Second, Autor et al. (2016) claim that the specification is weak to measurement error and potential transitory shocks correlated to $kaitz_{st}(p')$. Therefore, they propose an IV strategy that uses three instruments: the log real minimum wage, its square, and its interaction with the state-specific average over time of the percentile p' of the log real wage distribution, $\mathbb{E}_s[w_{st}(p')]$. According to the authors, the minimum wage level should affect wage dispersion only through the concurrent minimum wage's bindingness at the state level, which satisfies the exclusion restriction. I implement their IV strategy by replacing the log real minimum wage with its hourly counterpart.

Figure 2.4 reports the estimated marginal effects $\tau(p)$ for $p \in \{5, 10, \dots, 95\}$, showing significant spillovers up to the top of the wage distribution. For example, during 2012-2015, my findings indicate positive spillovers up to the 80th percentile of the hourly wage distribution. Between 2016 and 2019, spillovers appear to be more significant than in the preceding years. Notice that the blue line (2016-2019) lies above the orange line (2012-2015) for all percentiles from the median to the 90th percentile. A plausible explanation is that compensating differentials became more pronounced at the top of the distribution due to larger reductions in job amenities in the last four years.

Figure 2.4 – *Estimated minimum wage effects on the wage distribution*

Notes: The figure plots the estimated marginal effects of minimum wage increases across different wage distribution percentiles in Brazil. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. I follow Lee (1999) and Engbom and Moser (2022) by exploiting cross-state variation in wage floor bindingness. The baseline specification includes state fixed effects and state-specific linear and quadratic quarterly time trends. I replicate the instrumental variables (IV) strategy in Autor et al. (2016) by predicting wage floor bindingness using the log minimum real hourly wage, its square, and its interaction with the state-specific average over time of the percentile p' of the log real hourly wage distribution. I estimate marginal effects using the entire RAIS sample, separately for the periods 2012-2015 and 2016-2019. Each dot represents a marginal effect and corresponds to a separate regression of the dependent variable, ranging from the 5th to the 95th percentile of the wage distribution. Bars represent 95 percent confidence intervals based on regular standard errors.

Chapter 3

Overlapping ownership in vertical relations: From fostering to tunneling

In the past, some have mistakenly assumed that our predominantly passive management style suggests a passive attitude with respect to corporate governance. Nothing could be further from the truth

William McNabb III

Vanguard's CEO, 2008-2018

3.1 Introduction

Several theoretical and empirical studies analyze the connection between the ownership structure of strategically interacting firms and market outcomes. However, much of the literature still assumes that managers maximize the value of the company that appointed them. While reasonable at first glance, this view overlooks that shareholders—who appointed managers in the first place—prioritize the total value of their portfolios rather than the individual value of each firm they own. Therefore, either through direct investor influence or because managers internalize other benefits from pleasing pivotal shareholders, it is undeniable that a company's ownership structure can alter its goals, market strategies, and business rela-

tionships. In this chapter, I investigate the issue by exploring the role of shareholders in forming and sustaining trading partnerships over time in the US supply chain.

The recent proliferation of large asset management institutions has accentuated the role of diversified minority shareholders in corporate strategy (Schmalz, 2018). The rapid expansion of institutional financial intermediaries and a renewed preference for passive indexing strategies have led a few institutional investors, such as Vanguard, BlackRock, and State Street, to rank among the top shareholders of most publicly listed companies in the US.¹

A significant strand of the literature has explored the impact of *universal ownership* and *common ownership*. The former refers to the presence of one or more institutional shareholders that simultaneously hold positions in two companies (henceforth, *overlapping shareholders*), while the latter focuses on companies within the same market. As such, common ownership often refers to close competitors. While a growing body of evidence suggests that overlapping shareholders can hinder or, in particular cases, promote competition (Azar et al., 2018; Torshizi and Clapp, 2021; Koch et al., 2021; Antón et al., 2025a; Newham et al., 2025), their role in vertical relations remains underexplored. But, if overlapping shareholders drive executives to assess the externalities they impose on competitors, what would stop them from incorporating effects across upstream and downstream markets?

My findings, on the one hand, support previous findings in the literature. Higher degrees of overlapping ownership increase the likelihood of active trading partnerships.² A potential explanation is that common investors can help firms circumvent contractual and informational frictions that often affect potential customers and suppliers. On the other hand, my empirical work reveals an alternative explanation. Overlapping shareholders may be channeling resources from one firm to another, extracting private benefits from supply chain relationships.

The research centers on publicly listed companies in the United States and Canada between 1999 and 2013 by combining several publicly available data sources. I rely on previous research papers that compiled several public filings that companies must disclose to the United States Securities and Exchange Commission (SEC) periodically. I obtain

¹. Although the US stands out in the process, the trend is growing rapidly among large firms and particular geographic areas, such as Ireland, the Netherlands, and Sweden. Interestingly, recent evidence suggests that institutional ownership has been steadily high in regions like China, South America, and Mexico (Antón et al., 2025b).

². Throughout the chapter, I use the terms vertical relations, trading partnerships, and supply chain relationships interchangeably. Given that the identities of the customer and the supplier matter, each pair of firms spans two different customer-supplier dyads.

ownership ties—covering large institutional investors and blockholders—from 13D, 13F, and 13G filings, which I use to construct measures of overlapping ownership according to the framework of Rotemberg (1984). Feasible supply chain relationships stem from previous work of Frésard et al. (2020), while active trading partnerships originate in 10-K filings, in which firms disclose the identities of all customers that comprise at least 10% of their total annual sales.

The analysis employs two-way fixed-effects regressions to examine the relationship between ownership structures and trading partnerships among firms. Due to endogeneity and sample selection concerns, I combine two strategies to address biases. First, I weigh each feasible customer-supplier dyad by its *vertical upstream relatedness*. That is, the degree to which the supplier’s output serves as input for the customer. Second, I instrument measures of overlapping ownership using predicted rather than actual shareholdings. This strategy focuses on companies and investors with more stable ownership structures and portfolios, based on panel data on shareholdings over a decade, as well as on additions and deletions from the S&P 500 index.

Overall, increasing the similarity between ownership structures among feasible customer-supplier dyads by one standard deviation raises the unconditional probability of an active trading partnership by 18% to 23%. Since vertical overlapping shareholders share several features with *vertical integration* and *vertical control*, previous studies have linked these findings to the typical tropes of partial vertical integration, such as contractual and informational frictions. Instead, I focus on a different mechanism.

My findings spotlight that trading partnerships become more likely when they primarily benefit one side of the relationship. Therefore, increasing the ownership structure similarity by one standard deviation can have disparate effects, depending on the relative shareholder concentration across the two firms. For instance, increasing the relative concentration from the bottom to the top decile of the distribution raises the baseline effect from 8.25% to 50.82%. In other words, increasing the similarity of ownership structures across two firms increases the likelihood of an active trading partnership by a factor of 6 when one firm has a more diluted ownership structure than the other.

What is the origin of such disproportionate effects if they stem from similar increases in ownership structure similarity? Throughout the chapter, I argue that managers in companies with relatively diluted ownership downplay outcomes within their own firms. In particular, because managerial decisions affect specific shareholders to a lesser extent, as gains and losses

spread across more of them. If the same decision benefited another company with overlapping shareholders, managers may put more emphasis on its consequences—particularly when ownership in that firm is more concentrated.

The opposite also holds. So, managers appointed in firms with more concentrated ownership minimize consequences for other firms and prioritize profits at their own company. Thus, every manager ends up paying closer attention to companies with more concentrated ownership structures, as long as they have common investors. As a result, increases in ownership structure similarity across customers and suppliers can create incentives for managers to transfer resources from the company with relatively diluted ownership to the other.

My findings point exactly in this direction: trading partnerships are more likely to occur if overlapping shareholders can privately benefit from one-sided relationships. Therefore, managerial decisions may result in direct losses or foregone profits, to the detriment of non-overlapping shareholders at the company with more diluted ownership.

This chapter contributes to the literature on common ownership. The concept dates back to the theoretical framework of Rotemberg (1984), in which companies collude to favor shareholders with more diversified portfolios. Nonetheless, the topic captured everyone's attention when numerous studies began reporting that common owners soften competition, thereby leading to welfare losses. For example, previous studies find an increase in market concentration (He and Huang, 2017) and prices (Azar et al., 2018; Park and Seo, 2019; Torshizi and Clapp, 2021), as well as negative effects on employment, wages (Azar et al., 2021), and market entry (Xie and Gerakos, 2020; Newham et al., 2025).

At the same time, other papers document positive aspects of overlapping ownership, such as spillovers in research and development activities (López and Vives, 2019; Antón et al., 2025a), increased product variety (Aslan, 2024), or faster innovation diffusion (Kostovetsky and Manconi, 2020). In particular, the few studies on vertical relations stand out as their most substantial supporters. For example, firms secure larger loans at lower interest rates or are more likely to secure syndicated loans when shareholders overlap (Ojeda, 2018). Similarly, they increase the likelihood of companies joining strategic alliances (Lindsey, 2008), discourage suppliers from engaging in upward earnings management (Gao et al., 2022), boost upstream partner-specific investments (Deng et al., 2025), and extend the lifetime of existing trading partnerships (Freeman, 2025).

I add to this literature in two different ways. First, my empirical work uncovers an unexplored and unfavorable mechanism that affects shareholders along the supply chain.

My findings add to an extensive list of warnings from researchers about the adverse effects of overlapping ownership. Although competition agencies invoke common ownership as an element of competitive context in merger decisions (Burnside and Kidane, 2021), authorities often overlook overlapping shareholders in vertical mergers. Although previous studies support the view by emphasizing benefits akin to partial vertical integration, my work highlights a potential drawback that warrants consideration.

My second contribution concerns the empirical strategy, which circumvents critiques for other sources of exogenous variation. For example, Berger (2023) challenges fire sales from mutual fund scandals while Lewellen and Lowry (2021) discusses issues related to mergers of financial institutions and index additions. Still, two reasons justify the call for new instruments rather than employing previous identification strategies in the literature.

First, vertical relations entail two-sided decisions. Therefore, previous approaches based on market outcomes from unilateral company decisions, such as prices, entry, or investment, fail the exclusion restriction when applied in two-sided decisions, such as forming or maintaining a trading partnership. Second, these strategies rely on shifts in the degree of overlapping ownership but cannot capture changes in the relative concentration of ownership structures (e.g., Azar et al., 2018; Antón et al., 2023). Therefore, while adequate for exploring symmetric effects on levels (such as market and firm outcomes), they fall short for asymmetric incentives between the two firms. As such, my empirical approach is crucial for exploring the role of relative shareholder concentration across customers and suppliers.

The remainder of the chapter proceeds as follows. Section 3.2 describes the data, the sample, and the overlapping ownership measures. Section 3.3 discusses the empirical approach, the use of weights in my analysis, and the identification strategy. Section 3.4 shows that overlapping owners increase the likelihood of active trading partnerships and can extract private benefits from supply chain relationships, at the expense of non-overlapping shareholders at firms with more diluted ownership structures.

3.2 Data: Ownership ties and the supply chain

The following section describes the data sources, the sample, and all key measures I use throughout the analysis. I start by introducing several publicly available datasets on ownership ties and supply chain relationships. I then detail the construction of the sample and present three measures that capture ownership patterns across customer-supplier dyads: the

degree of overlapping ownership, the relative shareholder concentration, and the implied profit weights.

3.2.1 Data

The analysis centers on publicly listed companies in the United States and Canada from 1999 to 2013, excluding those in the financial (SIC codes 6000-6999) and utility (SIC codes 4900-4999) sectors.

Publicly listed companies. I use *Compustat North America – Fundamentals*, which provides quarterly information about firms, including outstanding shares, market value, closing price, net sales, total assets, and research and development (R&D) investment. In addition, I compute the company age from the first report of book assets since 1976 and include the Herfindahl-Hirschman Index (HHI) from the text-based network industry classification (TNIC) by Hoberg and Phillips (2016). I winsorize all continuous variables at the 1st and 99th percentiles of their distributions except for net sales, market value, and the HHI. Furthermore, I adjust nominal values for inflation to 2012 US dollars using the Bureau of Economic Analysis' GDP deflator.

Ownership ties. The SEC requires that investment managers holding over USD 100 million in North American securities file 13F forms quarterly, irrespective of the fraction of shares they own in the companies. Blockholders, on the other hand, must file 13D and 13G forms once they acquire at least 5% of a public company's outstanding shares and subsequently on an annual basis. Other SEC filings include Forms 3 to 5, which corporate insiders must file to inform ownership and transactions in company securities.

For this chapter, the US ownership network relies on two publicly available datasets from previous studies. I compile 13F filings submitted to the SEC between 1999 and 2017 from Backus et al. (2021), and combine them with 13D and 13G filings from 1998 to 2016 from Schwartz-Ziv and Volkova (2025). Insider filings are not available and therefore are excluded from the analysis.

Active trading partnerships. The supply chain network data stems from Barrot and Sauvagnat (2016). Publicly traded companies file 10-K forms with the SEC every year. Following the Statement of Financial Accounting Standard (SFAS) N°131, which supersedes SFAS N°14, companies must disclose the identity of any customer contributing at least 10% to their total annual sales. *Compustat North America – Customer Segment* collects

this information, although it does not provide customer identifiers. However, the authors combine a phonetic string-matching algorithm with manual identification to recover trading partnerships among publicly listed companies in the United States and Canada from 1976 to 2013.³ I use a clean version of the data published by the authors, which constitutes the supply chain network of this chapter.

Vertical upstream relatedness. Frésard et al. (2020) propose a directed measure of *vertical upstream relatedness*. That is, the degree to which any given company produces outputs that serve as input for any other company. For example, the variable would take a value of one if all outputs from the supplier would serve as inputs to all goods produced by the customer. In practice, values close to one are extremely rare, and most firm dyads have values close to zero, meaning their products are completely unrelated.

The authors proceed by mapping product descriptions from 10-K filings to a broad range of US commodities. Then, they combine this information with input-output intensities across commodities and construct directed measures for every pair of publicly listed companies in the US. Along with their paper, the authors have published annual updates from 1989 to 2021, which I use to identify the subset of feasible customer-supplier dyads among all possible combinations of two companies.

3.2.2 Sample

Compustat reports 112,382 annual financial records between 1999 and 2013, corresponding to 15,433 unique companies across industries other than the financial and utility sectors. However, on average, only half are in business in any given year. Panel A of Table 3.1 reports this information, along with cross-year averages for several other relevant characteristics.

The chapter centers on the relationship between customers and suppliers, so hereafter I distinguish between companies that typically appear as customers or suppliers in the data. Notice that many publicly listed firms traditionally buy upstream and sell downstream of the US supply chain and, as such, appear in both categories. A good example of companies that show only as customers includes retailers, since they mainly offer goods and services to final consumers. In contrast, importing firms appear only as suppliers, as they source their goods outside the US.

³. The methodology employed by the authors follows the approach of previous studies in the literature. See, for example, Fee et al. (2006), Wu and Birge (2014), or Cheung et al. (2020).

Table 3.1 – *Cross-year summary statistics for companies in Compustat*

	N	3-digit SIC industries	Age	Market value	Annual sales	Total assets	R&D intensity	HHI	MB ratio
<i>Panel A - Compustat</i>									
All companies	7,492	222	11.24	2,847.41	2,270.46	1,819.59	25.37%	3,212	-0.10
<i>Panel B - Feasible supply chain links</i>									
Customers	3,690	127	15.06	4,362.03	3,762.88	2,732.82	12.11%	3,212	2.77
Suppliers	3,715	128	14.63	3,388.52	2,609.09	2,076.42	12.90%	3,208	2.83
<i>Panel C - Active trading partnerships</i>									
Customers	670	29	19.80	23,544.81	20,626.25	12,576.12	5.52%	2,757	3.39
Suppliers	1,949	60	14.59	3,642.25	2,340.77	2,082.19	13.78%	3,082	2.73
<i>Panel D - Disclosing suppliers</i>									
Customers	528	21	19.61	23,804.53	20,598.98	12,691.23	5.98%	2,759	3.14
Suppliers	1,420	40	15.41	3,757.86	2,450.71	2,095.14	13.34%	2,992	2.46
<i>Panel E - Non-disclosing suppliers</i>									
Customers	220	12	19.61	22,511.97	18,116.37	11,231.93	5.48%	2,749	3.82
Suppliers	530	20	12.54	3,327.43	2,097.54	2,037.07	15.22%	3,353	3.32

Notes: The table reports cross-year summary statistics for publicly listed companies in *Compustat North America – Fundamentals* from 1999 to 2013. Panel A includes all companies in industries other than the financial and utility sectors. Panel B is restricted to companies identified as customers or suppliers in feasible trading partnerships, while Panel C centers on customers and suppliers in active trading partnerships. Panel D includes only suppliers, and their corresponding customers, disclosing at least one customer below the 10% mandated by the SEC. Panel E, instead, reports non-disclosing suppliers and their associated customers.

Feasible trading partnerships. In 2011, Nokia Oyj announced it would ditch its flagship operating system, Symbian, and license Windows 7 and Bing from Microsoft Corporation. The example highlights the importance of capturing that Microsoft was a potential supplier of Nokia before 2011. More importantly, my sample should capture all potential supply chain relationships, regardless of whether I observe an active trading partnership at some point in time.

On the contrary, consider the case of Activision Blizzard Inc., a video game holding company, and Alexion Pharmaceuticals Inc., a subsidiary of the pharmaceutical and biotechnology company AstraZeneca PLC. Despite being publicly traded since 2006, it comes as no surprise that the two companies have not established a trading partnership since then. It would be unreasonable to expect them to do so, regardless of any favorable conditions one could think of, given the industries they operate in and the commodities they produce.

Unfeasible supply chain links, like the one between Activision Blizzard and Alexion Pharmaceuticals, can bias my results. Irrespective of their ownership structures, their products

make them incompatible trading partners.⁴ To tackle the problem, I keep only dyads with a positive vertical upstream relatedness or with an observed trading partnership.⁵

Therefore, the final sample comprises 193,795,735 annual observations, representing 43,212,666 unique feasible dyads among 7,798 customers and 8,014 suppliers, spanning 1999-2013. Panel B reports cross-year summary statistics for these companies. Roughly half of customers and suppliers are in business in a given year, exhibiting longer lifespans and higher sales, assets, and market values than the average firm in the Compustat dataset. Moreover, they exhibit lower market-to-book ratios and R&D intensities.

Active trading partnerships. Following Barrot and Sauvagnat (2016), I define trading partnerships as active in all periods from the first to the last year in which a company reports another firm as a significant customer in a 10-K filing. The decision yields 29,413 annual active trading partnerships between 1999 and 2013, comprising 8,081 unique dyads between 1,499 customers and 3,119 suppliers.

Note that there are some limitations to the data. First, matching customers by disclosed names can introduce noise into the data, as firms with similar names may refer to unrelated organizations despite their shared historical roots.⁶ Second, the wording of SFAS N°131 may lead to sample selection by making it more likely for small suppliers and large customers to exceed the mandatory 10% threshold.

The SEC requires all companies to disclose their customers when they file their 10-K forms, so I do not expect underrepresentation of large suppliers in the data. Panel C corroborates the intuition. Active suppliers, on average, exhibit cross-year characteristics similar to those of potential suppliers in Panel B. If anything, they have a slightly higher market value and lower annual sales.⁷

The wording of the Statement, however, can censor relatively small companies, even when they buy only from a few suppliers. Since these companies make smaller purchases, they

⁴. Even determining the direction of such bias can be challenging, as the sign depends, among other things, on the degree of overlapping ownership across feasible and unfeasible firm dyads.

⁵. The distinction stems from the fact that Frésard et al. (2020) lose data when mapping CUSIP (American Bankers Association) to GVKEY (Compustat) identifiers. Thus, supposedly unfeasible supply chain links still appear in the Customer Segment of Compustat. Subsection B.4.3 in the Appendix discusses this methodological decision and shows that the conclusions remain the same under alternative criteria.

⁶. For example, A.T. Massey Coal Company spun off Massey Energy Co. in 2000 and changed its name to Fluor Corp. Therefore, the string "Massey Coal Company" should be matched to "Fluor Corp." rather than "Massey Energy Co.," which most phonetic string-matching algorithms would miss.

⁷. For further discussion on the issue, see Barrot and Sauvagnat (2016) and Wu and Birge (2014). Both papers compare the Compustat - Customer Segment with alternative data sources, such as Capital IQ and Bloomberg SPLC.

become less likely to surpass the mandated 10% threshold. Panel C confirms this concern. Compared to suppliers, only a fraction of all potential customers are in business in any given year. Moreover, they exhibit a lifespan of nearly 5 additional years, lower engagement in R&D activities, and significantly higher sales, market value, assets, and market-to-book ratios than customers in Panel B.

Non-mandatory disclosure. Despite the Statement not requiring the disclosure of customers below the 10% threshold, several companies include their names when filing 10-K forms. To assess whether voluntary disclosure could become an issue, I compare suppliers that have disclosed a customer below the mandated threshold at least once since 1976 (Panel D) and those who have not (Panel E). Similarly, I classify customers in the same group as their associated supplier. Roughly 70% of all suppliers and 80% of all customers fall under the disclosing category. Notably, there are no significant differences across Panels C, D, and E, neither for customers nor suppliers, eliminating concerns about this potential issue.

3.2.3 Overlapping ownership

I combine all ownership ties from 13D, 13F, and 13G filings. After removing duplicates, the ownership network contains 10,781,778 annual observations with 3,316,767 unique ownership ties between 33,861 investors and 15,459 companies. Each entry indicates the fraction of outstanding shares owned by shareholders $s \in \mathcal{S}$ in company $j \in \mathcal{J}$.

Next, I compute *profit-weight values* as measures of overlapping ownership. In Rotemberg (1984), profit weights, *kappa*, arise because managers maximize a weighted average of the portfolios held by all shareholders in the company. The model implies that investors do not need to actively engage with managerial decisions. Instead, they influence managers passively through perceived control and stock ownership in the company. For a more detailed description of the model and its derivation, see Section B.1 in the Appendix.

Consequently, a manager in company j weighs the effects of their decisions in the profits of company k in year t as follows

$$\kappa_{jkt} = \overbrace{\cos(\beta_{jt}, \beta_{kt})}^{\text{Overlapping ownership}} \cdot \underbrace{\sqrt{\frac{IHHI_{kt}}{IHHI_{jt}}}}_{\text{Relative concentration}} \quad (3.1)$$

where β_{jt} and $IHHI_{jt} = \|\beta_{jt}\|^2$ depict the ownership structure and shareholder concentration

of company j in year t , respectively. Note that β_{jt} is an S -dimensional vector with elements in $[0, 1]$, representing the fraction of total shares held by shareholder $s \in \mathcal{S}$. Due to insider and minority shareholders, the elements of β_{jt} do not necessarily add up to 1.

Equation (3.1) shows that κ can be decomposed into two interpretable expressions. The first factor determines how similar the ownership structures of firms j and k are, as measured by the cosine similarity between their shareholding vectors.⁸ Throughout the chapter, I refer to this expression as the *degree of overlapping ownership*, which ranges from 0 to 1. A higher value indicates greater overlap among shareholders and more balanced shareholdings. For instance, imagine two investors, each controlling stock in a single company. If they both sold their shares to a third investor who previously had no holdings, the increase reflects greater overlap in ownership. On the other hand, consider two overlapping shareholders, each with a stronger position in one of the companies. If they exchanged shares, the degree of overlapping ownership would increase due to the more balanced shareholdings.

The second factor denotes the relative concentration of shareholders across firms. The numerator matches the Herfindahl-Hirschman index of investor concentration for the weighted firm, with the denominator serving as a normalization relative to the investor concentration at the firm that appointed the manager. Therefore, values greater than one immediately signal tunneling incentives, while values close to zero denote complete indifference (Backus et al., 2021).

The measure proves compelling because it captures the tradeoff between incentives and influence within companies' boards. That is, the more concentrated the ownership structure in company k (or the more diffuse in company j), the more inclined the manager at company j becomes in internalizing the externalities they may impose on company k . Importantly, they create asymmetries in profit weights, since $\kappa_{jk} = \kappa_{kj}$ only when ownership concentration is equal across the two firms.

Using the ownership network for publicly listed companies in the US, I construct annual measures of relative shareholder concentration, overlapping ownership, and κ between 1999 and 2013. For a discussion about alternative measures found in the literature and robustness checks, see Subsection B.4.1 in the Appendix.

⁸. Cosine similarity is the complement of the angular distance between two vectors in an inner product space, and it characterizes whether they point in roughly the same direction. Furthermore, the formula $\cos(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$ comes from the polar notation of the cross product of the vectors.

3.3 Empirical strategy

The analysis consists of panel data regressions using a linear probability model. My baseline results employ a two-way fixed-effects (TWFE) model to determine whether the degree of overlapping ownership among customer-supplier dyads affects the likelihood of their trading partnership. Namely,

$$Link_{sct} = \delta_{st} + \delta_{ct} + \tau OvrOwn_{sct} + \varepsilon_{sct} \quad (3.2)$$

where $Link_{sct}$ identifies whether supplier s and customer c engage in a trading partnership during year t , and $OvrOwn_{sct}$ denotes the degree of overlapping ownership between the two companies. I include time-varying company fixed-effects δ_{ct} and δ_{st} to control for idiosyncratic trends across firms and account for usual controls in the literature, such as age, assets, annual sales, market share, market-to-book ratio, or industry concentration measures, many of which would act as *bad controls* if I add them directly to the empirical model (Angrist and Pischke, 2009).

Alternatively, I also present results using a conventional TWFE approach

$$Link_{sct} = \delta_s + \delta_c + \delta_t + \tau OvrOwn_{sct} + \varepsilon_{sct} \quad (3.3)$$

which introduces time-invariant supplier and customer fixed effects, δ_s and δ_c , separately from time fixed effects, δ_t , which control for time trends common to all feasible trading partnerships.

To facilitate the interpretation of estimates, I normalize variables so that they report percentage changes in the unconditional probability of an active trading partnership given a one-standard-deviation increase in the degree of overlapping ownership. In this manner, I avoid reporting small percentage-point coefficients arising from the large number of inactive supply chain links in the sample.

To explore the role of asymmetric incentives, I extend the empirical models described in Equations (3.2) and (3.3), so that

$$Link_{sct} = \delta_{st} + \delta_{ct} + \tau_0 OvrOwn_{sct} + \tau_1 RelCon_{sct} + \tau_2 Kappa_{sct} + \varepsilon_{sct} \quad (3.4)$$

where $RelCon_{sct}$ and $Kappa_{sct}$ represent the highest relative concentration and the highest profit weight among customer c and supplier s in year t , respectively. As shown in Equation

(3.1), the profit weights of any two firms comove with the degree of overlapping ownership. Instead, differences in their relative shareholder concentration can bring their respective κ closer together or push them apart. Similar to before, I also report results using time-invariant fixed effects for customers and suppliers.

The underlying idea is that the degree of overlapping ownership affects κ more disproportionately the more disparate their relative shareholder concentrations. Equation (3.4) captures differences with respect to the situation in which the relative concentration is precisely 1, which allows me to decompose changes in the likelihood of trading partnerships originated in a greater ownership overlap or more asymmetric incentives across managers.

Consistent with the literature, the following subsections detail my approach to address two empirical challenges in my work that arise from sample selection and endogeneity. First, I describe the use of vertical upstream-relatedness to weight observations in the sample, accounting for unfeasible trading partnerships that remain in the sample. Then, I discuss my main identification strategy, which involves estimating predicted shareholdings to construct instrumental variables for relative shareholder concentration, overlapping ownership, and κ of customer-supplier dyads in my sample.

3.3.1 Using weights to account for unfeasible trading partnerships

Unlike previous studies, I do not condition the sample on the prior existence of trading partnerships, allowing me to incorporate the extensive margin into the analysis.⁹ That is, whether companies with a greater overlapping ownership prefer trading among themselves rather than with other potential customers and suppliers.

Notice that, despite focusing solely on customer-supplier dyads with a positive vertical relatedness, only a tiny fraction of dyads appear as active trading partnerships. Indeed, the unconditional probability of one is just 0.02%, which constitutes the first cue that the sample still contains many unlikely trading supply chain relationships, which can lead to biases in the estimates.¹⁰

⁹. For example, Fee et al. (2006) and Freeman (2025) focus on supply chain relationships already in place and, as such, can only explore the intensive margin or the decision to extend the partnership for an additional year.

¹⁰. A way to think about the issue is to notice that no company within the sample produces a single commodity, as publicly listed companies in the US tend to be multiproduct firms. Therefore, it is enough for the supplier to produce at least one commodity that the customer uses to produce at least one of its outputs for the vertical upstream-relatedness to yield a positive, but close to zero, value. This is a clear example of a customer-supplier dyad that could introduce biases in the estimation. Even more so when the input signifies a tiny contribution to the production process of that sole product.

Let me illustrate the issue with an example. Consider the following model is correctly specified

$$Link_{sct} = \alpha + \tau OvrOwn_{sct} + X_{sct} \beta + \varepsilon_{sct}$$

Then, the OLS estimates (α, τ, β) originate from minimizing the following expression

$$\mathbb{E} \left[\left(Link_{sct} - \alpha - \tau OvrOwn_{sct} - \beta X_{sct} \right)^2 \pi(F_{sct} = 1) \right]$$

where $\pi(F_{sct} = 1)$ denotes the inherent likelihood of a feasible supply chain link, which I cannot directly observe.

Companies cannot engage in trading partnerships that are ultimately unfeasible (like the one between Activision Blizzard and Alexion Pharmaceuticals from the example in Subsection 3.2.2). As a result, $Link_{sct}$ always takes the value of zero when $F_{sct} = 0$. Therefore, I can rewrite the expression for the error term as follows

$$\begin{aligned} \mathbb{E} \left[\mathbb{E} \left[(m_{sct} - f_{sct})^2 \omega_{sct} \middle| F_{sct} = 1 \right] \pi(F_{sct} = 1) \right. \\ \left. + \mathbb{E} \left[(\alpha + \tau OvrOwn_{sct} + \beta X_{sct})^2 \omega_{sct} \middle| F_{sct} = 0 \right] (1 - \pi(F_{sct} = 1)) \right] \end{aligned}$$

where m_{sct} , f_{sct} , and ω_{sct} represent the conditional mean, the best predicting function, and an arbitrary weight for each observation, respectively.

Since the estimation minimizes only the first term, failing to account for the feasibility of the customer-supplier dyad can lead to a sizable, systematic bias stemming from the second term. As $\pi(F_{sct} = 1)$ approaches zero, the targeted moment becomes irrelevant for predicting the likelihood of the trading partnership.

Since the issue arises around $\pi(F_{sct} = 1)$ close to zero, setting a strictly positive threshold to identify customer-supplier dyads as feasible can alleviate the problem. Moreover, incrementally raising the threshold further reduces concerns, although it could eventually lead me to discard unlikely—though still feasible—dyads.

A better alternative would be to set weights ω_{sct} , provided they appropriately capture the fact that companies seldom source inputs from multiple suppliers, except when the specificity of the production process demands it. So far, I have assumed that the vertical upstream relatedness is informative enough about this. So, under the same assumption, I can plug the variable in directly and use it to weight customer-supplier dyads, downweighting the second term towards zero and significantly reducing the bias.

3.3.2 Identification strategy: Predicted shareholdings

The literature on common ownership has paid special attention to the endogeneity problems associated with using ownership-related measures to explain outcomes at the firm- or market-level (Schmalz, 2018, 2021). Several examples suggest similar concerns in vertical settings. For instance, active investors are known for gathering thousands of data points before making investment decisions. Thus, several institutional investors would likely pay close attention to supply chain rumors, which could affect my interpretation due to reverse causality. Similarly, unobservable factors could simultaneously affect the firm's ownership structure and trading partnerships, leading to omitted-variable biases. For example, consider a substantial technological innovation from a company. On the one hand, the firm might enter new trading partnerships to source the inputs required to produce the new technology. On the other hand, a newly granted patent could attract the attention of informed investors, increasing their holdings not only in the innovator but also in companies they believe might benefit from the novelty along the supply chain. As a result, both overlapping ownership and trading partnerships would comove, without one implying the other.

Despite initial debates over valid strategies for addressing endogeneity in ownership structures, a few papers document exogenous shifts that effectively work around the problem (See Boller and Morton, 2020; Antón et al., 2023).¹¹ However, vertical settings cannot accommodate these empirical strategies. Trading partnerships require both parties to negotiate and agree on the terms of the relationship, thereby invalidating the exclusion restriction.

I propose a different approach, in which I instrument the three measures required for my analysis using analogous versions that use predicted rather than actual shareholdings.

The idea builds on the interpretation that sufficiently lagged measures of ownership can serve as a good instrument, provided the persistence in the outcome declines faster than that of the covariate. Thus, I construct predicted shareholdings based on over a decade of past holdings of institutional shareholders across publicly listed companies in the US, as well as differences in the market value of annual additions and deletions from the S&P 500 index.

Contrary to identification strategies that rely on shifts in the average degree of overlapping ownership (or common ownership for specific markets), my approach also allows me to

^{11.} A good example of the initial debate is the back-and-forth between Kennedy et al. (2017), Azar et al. (2018), Dennis et al. (2022), and Azar et al. (2022). Other examples include the critiques in Berger (2023), who highlights the challenges associated with using fire sales resulting from mutual fund scandals, and in Lewellen and Lowry (2021), who discusses issues related to mergers of financial institutions and companies added to indexes.

instrument for any known function of shareholdings, including profit weights and relative investor concentration.

I proceed as follows. First, I focus on shareholders with fewer portfolio perturbations and on companies with more stable ownership structures. That way, the approach addresses the sudden shifts in shareholding positions driven by investors anticipating changes in the supply chain. I exclude institutional investors purchasing shares in a company for the first time and treat them as retail shareholders—individual investors who buy and sell company shares in small amounts—when computing ownership structures for that year. Furthermore, I exclude companies entering and exiting the S&P 500 index from portfolios in the same year, as they would probably attract the attention of various institutional investors.

Second, for the remaining shareholders and companies, I predict shareholdings using a simple Markovian process

$$\beta_{sjt} = \alpha + \rho \beta_{sjt-1} + \gamma W_{jt} + u_{sjt} \quad (3.5)$$

where β_{sjt} denotes the fraction of shares owned by investor $s \in \mathcal{S}$ on company $j \in \mathcal{J}$ in year t and W_{jt} depicts shifts in the index-weights that company j faces in year t . Intuitively, S&P 500 additions and deletions mechanically affect the weights of all remaining index constituents, given differences in the market values of entrances and exits. For instance, if the market value of entering companies were higher than that of exiting ones, index weights for the remaining companies would adjust proportionally downwards to make room for the higher index weights of the entrants. The opposite would hold as well.¹² Therefore, γ should

^{12.} Because the index comprises the 500 most influential companies in the US according to S&P Indices, a typical change of its constituents would associate one addition with its corresponding deletion. However, publicly listed companies in the US engage in mergers, acquisitions, and spin-offs on a regular basis, prompting the S&P Indices Committee to respond when such transactions involve index constituents. Therefore, they do not always substitute low-market-capitalization companies for high-market-capitalization ones, making it challenging to anticipate the net change in the constituent's market value. For example, on December 14, 2011, S&P Indices announced that *TripAdvisor Inc.* would replace *Tellabs Inc.* in the S&P 500 index by December 20, 2011. According to the press release, the announced day corresponds to the expected date on which the S&P 500 constituent Expedia was to complete the spin-off of TripAdvisor. Assuming a fixed number of outstanding shares and their prices, the market capitalization of Expedia before the spin-off should amount to the sum of TripAdvisor and Expedia during the first quarter of 2012. Therefore, this change in constituents only reflects the market value lost at Tellabs' exit. Conversely, on March 27, 2000, Standard & Poor's announced that *Linear Technology Corp.* and *Pharmacia Corp.* would replace *Monsanto Company* and *Pharmacia & Upjohn* in the S&P 500 index. The press release explains that Pharmacia is the merger of Monsanto and Pharmacia & Upjohn. Therefore, the change in constituents solely reflects the added market value from Linear Technology's entry. There are many other examples, with 51 spin-offs, 158 mergers and acquisitions, and several changes of names and tickers between 1999 and 2013.

capture how institutional shareholders reallocate their passive investments within portfolios in response to shifts in index weights. For more details, Section B.2 in the Appendix provides an extensive description of the use of additions and deletions from the S&P 500.

I argue that the residual variation, u_{sjt} , captures most of the endogeneity that threatened the original estimation and, therefore, using predicted rather than actual shareholdings frees the analogous versions of overlapping ownership, relative concentration, and profits weights from the usual concerns in ownership measures.

A crucial identification assumption remains, that fixed effects effectively absorb supply chain persistence at both steps of estimation. For that reason, my preferred specification includes the time-varying supplier and customer fixed effects, δ_{st} and δ_{ct} , from Equation (3.2), which are more likely to account for trends among specific customers and suppliers. A potential argument against it would be that persistence depends on idiosyncratic characteristics of the trading partnership itself, rather than on the customer and supplier who constitute it. Unfortunately, that level of flexibility falls outside the scope of the chapter.

Since the procedure can introduce unintended non-linear interactions, I employ a first-order Taylor expansion. Subsection B.4.2 in the Appendix, however, shows that the results remain unchanged when using either a second-order Taylor expansion or the full functional form.

3.4 The supply chain as a vehicle for private gains

While overlapping owners can foster trading partnerships, the gains from these relationships can become unequally distributed. The following section examines whether institutional shareholders may influence managerial decisions by promoting trading partnerships that become one-sided.

My findings align with previous literature, showing that similarities in the ownership structure of customers and suppliers increase the likelihood of active trading partnerships. While prior studies argue that overlapping owners can foster trading partnerships by reducing contractual and informational frictions—such as holdup (Deng et al., 2025; Freeman, 2025), double marginalization (Riva, 2022), or corporate earnings management (Gao et al., 2022)—I focus on a different mechanism. Nevertheless, Section B.3 in the Appendix presents additional empirical exercises, in which my identification strategy roughly captures patterns similar to those in the literature.

Building on this baseline result, I present suggestive evidence that overlapping owners exploit supply chain ties to extract private benefits at the expense of non-overlapping shareholders in firms with more diffuse shareholdings. In particular, I show that increases in the ownership structure similarity are markedly stronger when the relative concentration deviates from 1.

Taken together, these findings point to managers of firms with diffuse ownership disproportionately weighing the interests of long-horizon shareholders, translating some degree of managerial influence into private gains.

3.4.1 Overlapping owners fosters trading partnerships

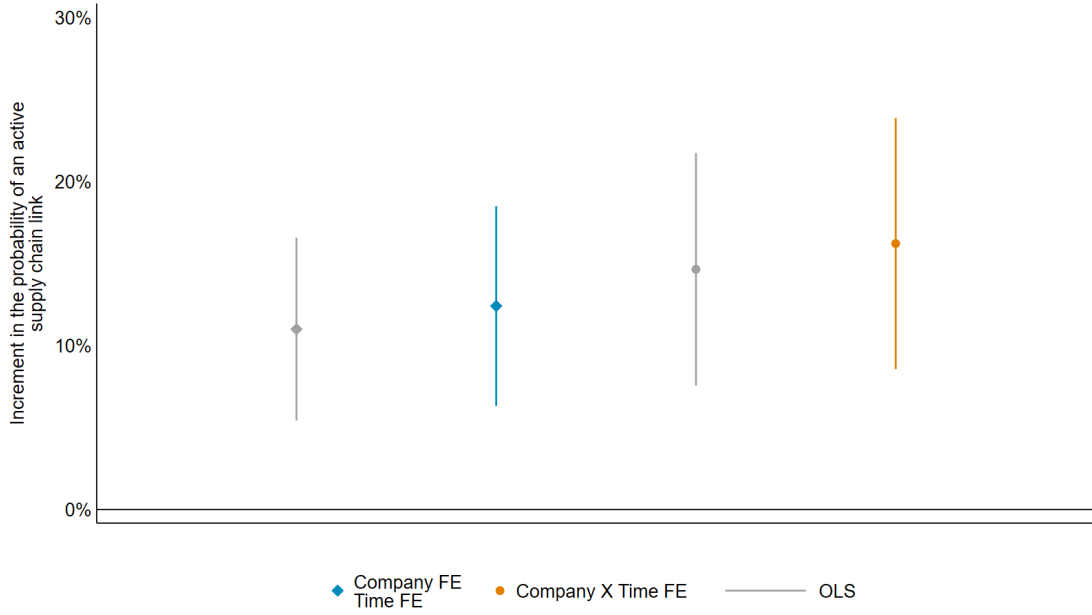
The following subsection shows that the likelihood of an active trading partnership increases with the similarity in the ownership structure of customers and suppliers.

I estimate Equations (3.2) and (3.3) using OLS and 2SLS, as described in Section 3.3. Figure 3.1 showcases the results, which indicate that a one-standard-deviation increase in the ownership structure similarity raises the unconditional probability of an active trading partnership by 14.93% to 16.21%, depending on the specification and the estimation method. Although the coefficients are slightly higher in the 2SLS model, they do not differ statistically from those in the OLS model. For more details, see Table B.1 in the Appendix, which reports OLS and 2SLS estimates, including the corresponding first stages.

Before moving forward, note that 2SLS estimates capture local effects related to the instrument. That is, average effects among compliers, weighted by the strength of the first-stage on each observation. On the one hand, the instrument excludes from the prediction changes in supply relationships for firms entering or exiting the S&P 500 and shareholders acquiring their first shares in a firm during the same year. On the other hand, the 2SLS approach downweights observations that substantially deviate from the predicted instrument. As a result, 2SLS coefficients emphasize effects driven by customers and suppliers with more stable ownership structures over time.

From the shareholders' standpoint, this definition fits the strategies followed by passive index investors—whose holdings tend to track and replicate known market indices—and by institutional investors with long-term stakes, such as strategic alliances, partial vertical integrations, cross-ownerships, and conglomerates.

The similarity between OLS and 2SLS estimates, however, does not suggest that endogeneity concerns were unsubstantiated. It appears unlikely that active investment funds

Figure 3.1 – *Overlapping ownership increases the likelihood of active trading partnerships*

Notes: The figure depicts the effect of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different fixed-effects specifications. I report the effect of increasing the average degree of overlapping ownership (0.0905) by one standard deviation (0.1361). I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). In blue, I report 2SLS estimates from Equation (3.3), and in orange, 2SLS estimates from Equation (3.2). Estimates for the OLS are reported in gray to the left of the corresponding 2SLS estimates. The 2SLS estimations use a first-order Taylor expansion of the ownership structure similarity based on predicted shareholdings, as described in Section 3.3. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification. For further details, refer to Table B.1 in the Appendix.

play little to no role in supply chain decisions, or that institutional shareholders put little weight on trading partners when choosing their portfolios.

An alternative explanation, and my preferred interpretation, is that long-standing shareholders at the firm are the ones with the most influence over managers. Not necessarily because they directly persuade executives, but because managers tend to please pivotal shareholders with longer investing horizons, and therefore factor in how managerial choices affect other companies in these long-horizon investors' portfolios.

3.4.2 Privately benefitting from trading partnerships

The framework in Rotemberg (1984) provides a broad interpretation of the previous findings. Overlapping shareholders influence managerial decisions through profit weights κ_{jkt} . In

particular, managers internalize the consequences of their actions upstream and downstream, incorporating changes in $\pi(x_{jt}, x_{-jt})$ into the maximization program, where $-j$ represents companies in $\mathcal{J}$ other than j . As a result, trading partnerships in the margin may change status due to common institutional shareholders. Either because overlapping shareholders prompt managers to create or prolong certain supply chain relationships, or because they induce managers to dissolve or deter other ones.

While Section B.3 in the Appendix explores mechanisms associated with the tropes of vertical integration, this subsection focuses on an alternative explanation. Overlapping shareholders use trading partnerships to extract private benefits at the expense of one of the companies, particularly its non-overlapping shareholders. The exact mechanism can take many forms. For example, managers could foreclose upstream or downstream markets for companies not belonging to the portfolio of shareholders, create tighter market-entry barriers, or bear incentives to tunnel value from one company to another, similar to the case where firms set different *transfer prices* to carry profits from one division into another.

Although the analysis cannot establish the exact channel, I present suggestive evidence that companies become more inclined to engage in trading partnerships when these entail tunneling opportunities. When a firm has more diffuse ownership than a potential trading partner, overlapping ownership increases the likelihood of a trading partnership by up to 4 times the baseline. I argue that these patterns coincide with managers weighing the gains in trading partners more heavily than the losses at the companies that appointed them.

Tunneling. That overlapping shareholders privately benefit from managerial decisions at the expense of other investors recalls the concept of *tunneling* (Johnson et al., 2000; Atanasov et al., 2007), in which a company transfers assets or cash flows from one firm to another. For example, consider a group of overlapping owners holding 60% of the equity in customer c and 20% in supplier s . If overlapping shareholders could prompt managers to set a lower input price than usual, then for every USD 100 million the customer saves on input costs, overlapping shareholders would pocket USD 40 million. Non-overlapping shareholders in the downstream firm would benefit as well, obtaining another USD 40 million, while non-overlapping shareholders in the upstream firm would lose the entire USD 80 million.

As Ehrhardt and Nowak (2015) note, tunneling can take many forms and does not necessarily entail money transfers. Thus, overlapping shareholders could pursue other goals, including greater control over firms, the supply chain, or the degree of competition across markets. In an analogous comparison with partial vertical integration, numerous studies have

found evidence of such mechanisms. For instance, Levy et al. (2018) presents a theoretical framework in which companies can acquire partial stakes in vertically related companies to foreclose rivals. According to the predictions, partial acquisitions can become profitable, depending on firms' ownership and control structures. Along the same lines, Boehm and Sonntag (2023) provide empirical evidence that vertical mergers increase the likelihood of foreclosure, and Newham et al. (2025) show that pharmaceuticals can deter entry of generics. In conclusion, overlapping shareholders could play a role similar to that of partially integrated firms, reaping private benefits as managers select certain trading partners over others.

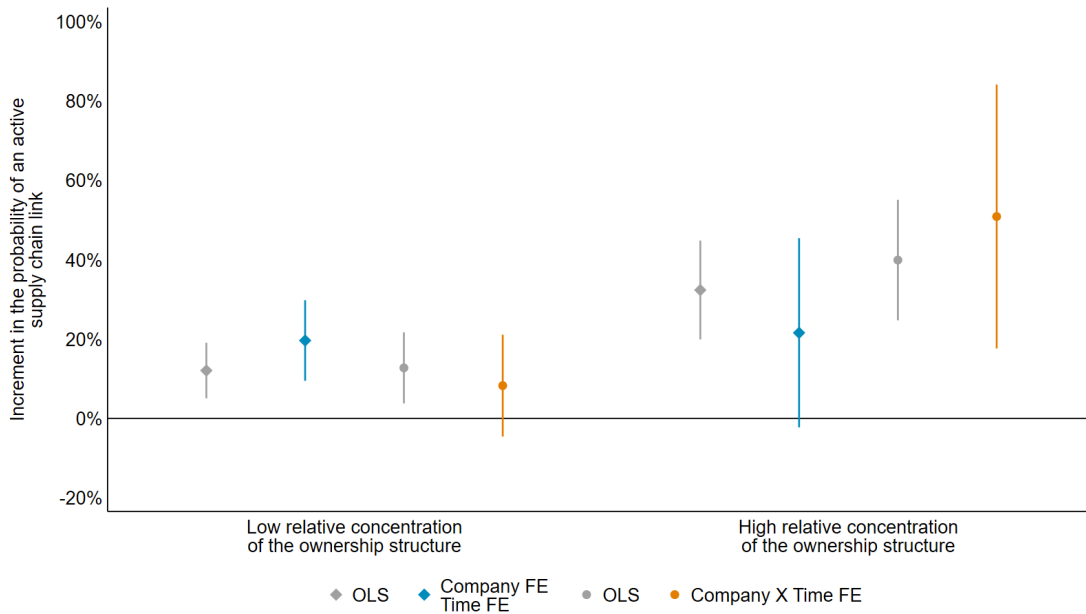
Relative shareholder concentration. Backus et al. (2021) connect tunneling to profit weights above 1, as managers would assign strictly greater weights to those companies than to the one firm that appointed them. In vertical settings, trading partnerships create aggregate value for customers and suppliers. Therefore, any positive profit weight can drive decisions to deviate from optimal choices, compared to situations where managers maximize the firm's own profits. In particular, changes in profit weights can shift how companies are willing to split the gains from doing business together.

I explore the role of disparate managers' incentives on the likelihood of active trading partnerships by estimating Equation (3.4) using OLS and 2SLS, as described in Section 3.3. Likewise, I estimate the analogous extension of Equation (3.2), though this specification is less effective at controlling for supply chain persistence.

Amel-Zadeh et al. (2022) highlight a substantial number of insider shareholders in the US economy.¹³ In particular, the authors find that insiders hold less diversified portfolios and constitute a considerable component of the largest firms' ownership structures. While the result does not raise concerns about the degree of overlapping ownership, it implies significant noise in the relative shareholder concentration. As a workaround, I drop customer-supplier dyads with a relative concentration above the 90th percentile, where holdings in the hands of insiders are mechanically more likely to appear.

Figure 3.2 displays the marginal effects of increasing the ownership structure similarity by one standard deviation. Coefficients on the left report marginal effects for customer-supplier dyads at the bottom decile of the relative shareholder concentration, while coefficients to the right replicate the analysis for customer-supplier dyads at the top decile.

¹³. The SEC considers officers, directors, and blockholders who hold more than 10% of any class of a company's securities to be insiders and requires them to file Forms 3, 4, and 5. Blockholders and large asset managers may appear in 13D, 13F, and 13G filings; however, these forms do not capture holdings for many officers and directors.

Figure 3.2 – *More diffuse shareholdings amplify the effects of increasing the degree of overlapping ownership*

Notes: The figure depicts the marginal effects of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership across different fixed-effects specifications and relative levels of shareholding concentration. I drop customer-supplier dyads with a relative concentration above the 90th percentile (5.8918). I report the effect of increasing the average degree of overlapping ownership (0.1427) by one standard deviation (0.1056). I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). The four coefficients to the left represent marginal effects for the bottom decile of the highest relative shareholder concentration (1.0838). The four coefficients to the right denote marginal effects for the top decile of the highest relative shareholder concentration (3.4820). In orange, I report 2SLS estimates from Equation (3.4), and in blue, 2SLS estimates from an analogous extension of Equation (3.2). Estimates for the OLS are reported in gray to the left of the corresponding 2SLS estimates. The 2SLS estimations use a first-order Taylor expansion of the ownership structure similarity, the relative shareholder concentration, and profit weights based on predicted shareholdings, as described in Section 3.3. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification. For further details, refer to Table B.2 in the Appendix.

Results for my preferred specification suggest that a greater similarity in shareholdings has no statistically significant effect on the likelihood of an active trading partnership when the relative shareholding concentration is close to one. That is, when managers assign profit weights close to or equal to the customer's and supplier's profits. The alternative specification and the OLS estimates report positive, significant effects. Since these coefficients are prone to capturing supply chain persistence, I refrain from interpreting them. If anything, the magnitudes—which are close to or exceed the baseline effects—may imply that asymmetric incentives among managers already drive effects discussed in Subsection 3.4.1.

Keeping the degree of overlapping ownership fixed, customer-supplier dyads with a relative shareholder concentration at the top decile of the distribution almost quadruple profits weights. Consistent with the interpretation that managers are more likely to engage in one-sided trading partnerships, increasing ownership structure similarity by one standard deviation roughly triples the baseline effect.

When comparing the coefficients for the specification with time-varying company fixed effects across the bottom and top deciles of relative shareholder concentration, the empirical findings indicate that the marginal effects increase by a factor of 6, from 8.25% to 50.82%. For a more comprehensive report of coefficients and standard errors, see Table B.2 in the Appendix.

Intuitively, I argue that supply chain relationships become more likely when managers weigh the externalities imposed on the potential trading partner more heavily. In particular, when their peers do not respond in kind. These decisions offer tunneling opportunities for managers and, more importantly, for overlapping shareholders, potentially harming non-overlapping shareholders at the firm that appointed the conflicted manager.

As the exact channel falls beyond the scope of this chapter, I cannot determine whether thwarted shareholders suffer from direct losses or unrealized gains. As mentioned above, supply chains often create aggregate value for customers and suppliers, so companies with diluted ownership may still reap some of the benefits of trading partnerships. However, all evidence points to overlapping owners capturing a surplus as private benefits, derived from an unequal distribution of the gains.

Chapter 4

Separations revisited: Do layoffs or quits drive lower separation rates in high-productivity firms?

I believed that an individual firm could save a significant number of jobs by reducing pay. This is rarely true, and the firms for which it is true are precisely the ones most likely to cut pay

Truman Bewley

4.1 Introduction

A well-established fact in labor economics is that more productive firms have fewer separations (Topel and Ward, 1992; Davis et al., 2013; Haltiwanger et al., 2018*a,b*)—a pattern we refer to as the negative productivity-separation correlation. Though theoretical models often link this relationship to efficient separations (Burdett and Mortensen, 1998; Postel-Vinay and Robin, 2002; Postel-Vinay and Turon, 2010), typically without distinguishing between quits and layoffs, recent empirical studies document widespread inefficiencies (Schmieder and von Wachter, 2010; Jäger et al., 2023; Davis and Krolikowski, 2025), underscoring the need to revisit the underlying drivers of the productivity-separation correlation.

The following chapter addresses the gap in the literature. First, we present empirical evidence that the negative correlation between firm productivity and overall separation rates is mainly driven by a negative relationship between *firm productivity and layoffs*, rather than quits. Second, we develop a parsimonious job search model in which wage rigidity and productivity uncertainty interact to generate inefficient separations. The model predicts that higher-productivity firms have larger markdowns and, as a result, lower layoff rates, laying down the theoretical foundations for our empirical findings. Third, we empirically validate the model’s key mechanisms. In particular, we show that firms facing stronger wage rigidity exhibit higher layoff rates and a steeper productivity-layoff correlation, and that markdowns are systematically higher in more productive firms and negatively correlated with layoffs.

Our empirical analysis draws on aggregate information from the Annual Manufacturing Survey (PIA), conducted by the Brazilian Institute of Geography and Statistics, as well as on administrative records from the Brazilian Ministry of Labor (RAIS). RAIS covers the universe of formal labor contracts in Brazil and provides uniquely detailed information on separations, including the exact date and explicit distinctions between quits and layoffs. It also reports contract wages separately from variable pay, which we exploit to construct a measure of wage rigidity. PIA provides estimates of value added per worker—our baseline measure of firm productivity—and markdowns.

We begin by describing key institutional features of the Brazilian labor market, which support the accuracy of the variable distinguishing quits from layoffs in the RAIS data. Most importantly, firms are required to provide severance pay to workers and a fine to the government in the case of layoffs, creating a financial incentive to classify separations as quits whenever possible. Conversely, workers benefit from reporting a layoff, as it grants access to severance payments and certain government benefits. These opposing incentives guarantee that both parties have strong motivations to report separations accurately.¹ Moreover, we show that post-separation outcomes for laid-off workers are consistently worse than for those who quit. This empirical pattern further reinforces the reliability of the distinction between quits and layoffs in the data.

Using this dataset, we document that layoff rates decline with firm productivity across multiple productivity measures—value added, firm size, and wage premium—yielding similar

¹. Workers and firms could collude to misreport quits as layoffs to access government benefits, but this would require large upfront firm payments (severance and fines) with uncertain compensation, making enforcement difficult. Subsection 4.2.1 offers empirical evidence that such arrangements are rare.

results. While quits also decrease with firm productivity, the slope is much flatter. Consequently, the negative correlation between firm productivity and overall separation rates is mainly driven by lower layoff rates at high-productivity firms. In fact, the slope of layoff rates with respect to firm productivity closely matches that of overall separation rates, with the ratio between these slopes ranging from 0.81 to 0.93, depending on the productivity measure used.

The negative correlation between separations and firm productivity may, in part, reflect worker heterogeneity, as high-skill workers tend to sort into high-productivity firms (Card et al., 2013; Gerard et al., 2021). Lower layoff rates at more productive firms could simply capture differences in workforce composition rather than firm characteristics. However, we show that this is not the case. Even after controlling for worker heterogeneity with a rich set of covariates, lower layoff rates at high-productivity firms still account for most of the productivity-separation correlation.²

To explain these empirical findings, we develop a partial-equilibrium wage-posting search model featuring two key ingredients: worker-level productivity uncertainty and wage rigidity. Our framework introduces wage rigidity by assuming that firms must post wages *before* observing worker-specific productivity shocks and cannot adjust them *ex post*. Therefore, firms face a tradeoff between worker retention, which increases with wages, and markdowns. That is, the gap between the average productivity and posted wages.

In the model, we denote firm productivity as the value added per worker. Nonetheless, we show that both wages and firm size increase with firm productivity in equilibrium, providing theoretical validation for our two alternative empirical productivity measures.

The main theoretical result is that higher-productivity firms have lower layoff rates. The mechanism operates as follows. Each firm's expected profit per worker equals the product of its retention rate and its expected markdown, making these factors supermodular: an increase in markdowns strengthens the firm's incentive to retain workers, and vice versa. As a consequence, higher-productivity firms optimally choose both larger expected markdowns and higher retention rates. Due to larger markdowns, more productive firms absorb negative productivity realizations more easily, resulting in fewer workers whose productivity falls short of the posted wage. Since layoffs occur when a worker's productivity is below the wage (i.e., when the markdown is negative), higher-productivity firms resort to fewer layoffs overall.

². Covariates for workers include age, tenure, education, gender, race, and occupation.

We then return to the data to empirically validate the two main mechanisms proposed in the model. First, we show that, consistent with the model’s predictions, higher-productivity firms exhibit larger markdowns and that these markdowns are associated with lower layoff rates.

Next, we investigate the model’s prediction that layoffs partly depend on wage rigidity. We construct a proxy measure for wage rigidity based on the share of total compensation that depends on contract wages rather than variable pay. Then, we validate the proxy by showing that a higher contract-wage share is associated with fewer wage reductions and more wage changes close to zero.

We next examine the role of wage rigidity in shaping the relationship between productivity and layoffs. We first document that firms facing greater wage rigidity exhibit higher layoff rates. Then, we define labor markets by industry and location and estimate the productivity-layoff correlation separately for each market. Our findings indicate that this correlation is stronger in markets with greater wage rigidity, suggesting that this is a key mechanism linking productivity to layoff decisions.

Taken together, our empirical findings support the mechanisms proposed by our theoretical framework: inefficient layoffs arise due to wage rigidity, and higher-productivity firms have lower layoff rates because they impose larger markdowns.

Our work contributes to the empirical literature on wage rigidity and its role in shaping firms’ layoff decisions (Schmieder and von Wachter, 2010; Jäger et al., 2023; Ehrlich and Montes, 2024; Davis and Krolikowski, 2025). We show that wage rigidity is associated not only with higher layoff rates but also with a stronger productivity-layoff correlation. Our measure of wage rigidity builds on extensive evidence that variable pay is more flexible than contract pay (Altonji and Devereux, 1999; Messina et al., 2010; Anger, 2011; Grigsby et al., 2021).³ Similar to our approach, Makridis and Gittleman (2022), Reizer (2022), and Sockin and Sockin (2025) examine the relationship between employment volatility and wage rigidity, using the prevalence of variable pay as a measure of wage flexibility.

Our theoretical framework builds on the literature on wage dispersion and turnover, particularly the foundational work of Burdett and Mortensen (1998), who accentuate the tradeoff between wage markdowns and retention rates. A substantial body of research has

³. Cardoso and Portugal (2005) and Card and Cardoso (2022) examine the “cushion” between wage floors and total wages. While the authors focus on wage floors determined by collective bargaining agreements, we consider firm-specific floors arising during the negotiation on individual contracts between employers and employees.

extended this framework, typically assuming separations are efficient and not discerning quits from layoffs (e.g., Postel-Vinay and Robin, 2002; Postel-Vinay and Turon, 2010). In contrast, we empirically show that lower layoff rates at high-productivity firms account for almost all of the decline in separation rates, and propose a model that incorporates wage rigidity and inefficient separations to explain the pattern.

The chapter also relates to previous work on the determinants of variation in layoff rates. Prior research has shown that layoffs decline with tenure (Jovanovic, 1979; Topel and Ward, 1992; Ureta, 1992) and has examined the dynamics of layoff rates (Hopenhayn and Rogerson, 1993; Mortensen and Pissarides, 1994; Mueller, 2017; Carlsson and Westermarck, 2022; Acabbi et al., 2024; Blanco et al., 2024). We offer a novel perspective by investigating firm-level heterogeneity in layoffs. Motivated by evidence that firm-level differences in job stability play an important role in shaping unemployment persistence (Pinheiro and Visschers, 2015; Jarosch, 2023), our model captures the fact that less productive firms exhibit lower job stability due to lower markdowns.

Finally, a large body of the literature estimates labor supply elasticities using the elasticity of separation rates with respect to wages, following the methodology proposed by Manning (2003).⁴ However, as many of these studies acknowledge, the relevant elasticity for this empirical exercise should be the elasticity of *quits* with respect to wages rather than the elasticity of total separations. We show that the slope of quit rates with respect to firm wage premiums is much more muted than that of overall separation rates, suggesting that labor supply elasticities may be substantially smaller than previously estimated.⁵

The remainder of the chapter proceeds as follows. Section 4.2 provides an overview of the institutional setting and presents evidence that supports the reliability of the quit-layoff distinction in our data. Section 4.3 documents our main empirical finding: layoffs account for most of the negative correlation between firm productivity and separation rates. Furthermore, Section 4.4 develops a theoretical framework to explain this pattern, while Section 4.5 empirically validates the mechanisms it proposes.

⁴ See, for example, Hirsch et al. (2010), Ransom and Oaxaca (2010), Ransom and Sims (2010), Webber (2015), Bachmann et al. (2022), Bassier et al. (2022), and Webber (2022), as well as Sokolova and Sorensen (2021) for a recent survey of this literature.

⁵ Relatedly, other studies document a positive correlation between tenure and wages and interpret this as evidence that workers prefer high-wage jobs, but do not estimate labor supply elasticities from this correlation (Krueger and Summers, 1988; Card et al., 2013; Drenik et al., 2023).

4.2 Distinguishing quits and layoffs in the data

4.2.1 Data and sample

Employer-employee data from Brazil. We utilize the *Relação Anual de Informações Sociais* (RAIS), an extensive administrative record from Brazil that captures formal employment relationships. Annually, companies submit RAIS filings, documenting all employees from the preceding year, including personal data such as gender, birth date, and education level, alongside contract specifics like earnings, contracted hours, and detailed occupation according to the *Classificação Brasileira de Ocupações* (CBO 2002), which encompasses 2,638 different occupation codes. Crucially, RAIS mandates reporting the dates and reasons for employee separations, distinguishing between quits and layoffs. Additionally, RAIS registers contract wages separately from a variable pay component, which includes bonuses, performance pay, and overtime.

Firm surveys. Our analysis uses industry-state aggregates from the *Annual Manufacturing Survey* (PIA, for its Portuguese acronym), conducted by the Brazilian Institute of Geography and Statistics, which provides detailed information on production, employment, and costs.⁶ Value added (VA) denotes the value of industrial transformation per worker, computed as the difference between the gross value of production and the costs of intermediate inputs, divided by the number of workers.⁷

Measuring firm productivity. Our baseline measure of firm productivity is value added, as described earlier. Additionally, we consider two alternative proxies for firm productivity commonly used in the literature: AKM pay premiums and firm size. We estimate AKM firm pay premiums following the model proposed by Abowd et al. (1999). Although coefficients obtained through this empirical method are unbiased, these are noisily estimated. Therefore, we cluster firms into 100 groups using a k-means clustering algorithm, as recommended by Bonhomme et al. (2019). Section C.2 in the Appendix details the estimation procedures and validates the AKM model assumptions in our sample. We define firm size as the total number of employees in each firm. Since our focus is on cross-sectional heterogeneity rather

⁶. PIA aggregates information using 3-digit industry codes (for a total of 285 industries), and Brazil has 27 states, yielding 7,695 unique industry-state combinations.

⁷. As defined, value added includes labor costs, such as salaries, other benefits, and mandatory social security contributions.

than time variation, we measure value-added and firm size in the first year a firm appears in the sample and hold them fixed throughout the analysis.⁸

Sample. Our sample spans the years 2010 to 2017, starting after the Great Recession and ending before Brazil’s 2018 labor market reforms. We focus on Brazilian men and women born between 1959 and 1987 with at least one year of potential experience.⁹ We restrict the sample to individuals employed on December 31 with at least one month of tenure, in open-ended contracts, and earning above the minimum wage in urban areas within Brazil’s private sector. For individuals holding multiple jobs, we select the position with the highest contracted hours or, in case of a tie, the one with the highest hourly wage. We classify non-separated workers who remain with the same firm across consecutive years as *stayers* and use the reported cause of separation to distinguish between *layoffs* and *quits*.

Given the significant presence of informal employment in Brazil, which is not captured in our data, we follow Gerard et al. (2021) and restrict our sample to the Southeast region, comprising the states of Espírito Santo, Minas Gerais, Rio de Janeiro, and São Paulo. This region accounts for nearly half of the country’s formal employment and has relatively lower informality rates. Additionally, its minimum-to-median wage ratio is comparable to those in other developing and developed economies.¹⁰ Furthermore, to estimate firm and worker effects, we include only the largest connected set of firms and workers, as proposed by Abowd et al. (1999).

Table 4.1 provides descriptive statistics comparing our final sample with the broader national and regional datasets. Workers in our sample have profiles similar to those of the broader Brazilian workforce in terms of education, age, and tenure. Hourly wages are slightly higher in the Southeast, reflecting its economic status. Our sample includes 74% of the firms in this region but captures 99% of worker-year observations. Notably, layoffs constitute 81% of all separations in our sample. This proportion is even higher across the broader national landscape, suggesting that the predominant role of layoffs in separation dynamics would be even more pronounced in the whole sample.

Finally, the PIA focuses on the manufacturing sector and covers about 8% of firms and 14% of workers in RAIS. Furthermore, within this sample, firms in the Southeast region ex-

⁸. AKM pay premiums are fixed by construction.

⁹. We require individuals to be at least one year older than the theoretical age of completion of the highest degree they have attained, e.g., 18 years for a high school degree or 22 years for a college degree.

¹⁰. Related to this, Davis and Krolkowski (2025) finds that the minimum wage is not a primary driver of wage rigidity in the US.

Table 4.1 – *Summary statistics*

	Brazil	Southeast region	Sample
<i>Relação Anual de Informações Sociais (RAIS)</i>			
Number of firms	4,307,522	2,105,805	1,523,100
Average firm size	7.9	8.5	9.8
Number of worker-year observations	146,878,704	78,877,496	74,910,200
Number of workers	35,403,116	19,023,076	14,868,221
Average age (years)	37.3	37.6	37.4
Average log-hourly wage	2.181	2.282	2.332
Average tenure (months)	46.5	48.0	46.4
Average schooling (years)	10.9	11.0	10.4
Average annual layoff rate (%)	20.17	20.18	17.03
Average annual quit rate (%)	3.34	3.34	4.03
<i>Pesquisa Industrial Anual (PIA)</i>			
Share of firms covered (%)	8.11	9.98	10.68
Share of workers covered (%)	13.85	17.24	17.43
Average log-value added	11.28	11.32	11.70

Notes: The table presents characteristics of Brazilian workers and firms from 2010 to 2017. The first three panels report summary statistics on the RAIS dataset. The first column reports annual employment spells for workers with urban open-ended contracts in the private manufacturing sector in Brazil. The second column further restricts the sample to firms located in the Southeast region of the country, while the third column limits the sample to firms within the largest connected set, as detailed in Subsection 4.2.1. The last panel provides summary statistics on the PIA dataset, which is restricted to the manufacturing industry. The *share of workers covered* and the *share of firms covered* indicate the fraction of workers and firms, respectively, in the RAIS administrative records that correspond to the manufacturing sector and therefore also appear in the PIA survey.

hibit higher value added than the national average. Similarly, our sample, further restricted to larger firms, portrays a higher value added than the regional average.

4.2.2 Context: Quits and layoffs in Brazil

The RAIS dataset distinguishes between quits and layoffs, a critical distinction given that the government uses this data for administrative purposes. In the case of a layoff, the firm must pay a fine to the government and provide severance pay to the worker. Additionally, the worker becomes eligible for unemployment benefits and can withdraw funds from their public pension account, which is otherwise typically reserved for retirement or buying a

home. Given the low quit rate in the data, a natural concern is whether these policies can create incentives for workers and firms to misclassify quits as layoffs. In this subsection, we briefly discuss why such incentives are unlikely to lead to systematic misreports. For more details on these policies, see Section C.3 in the Appendix.

A separation that is reported as a quit benefits the firm; when reported as a layoff, it benefits the worker. Thus, both parties have strong incentives to ensure that the separation is accurately reported.

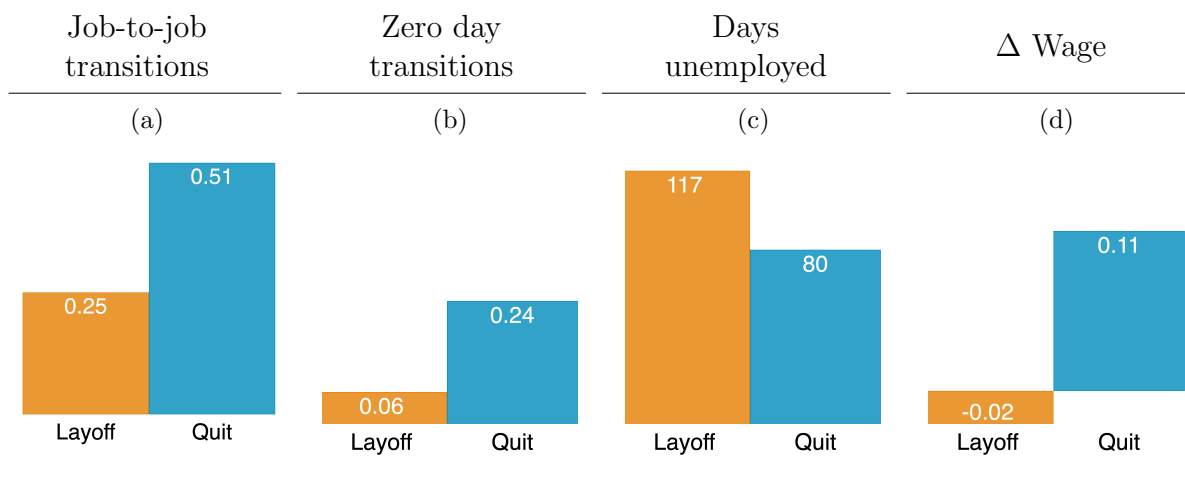
However, there is a potential issue. Upon a layoff, the firm incurs a cost by paying a government fine, while the worker benefits from unemployment insurance and early access to their pension account. If a worker highly values immediate liquidity—such as accessing their pension account early—the total insurance received from the government could outweigh the costs to the firm. This scenario might create incentives for collusion between the employer and the employee. That is, they agree to misclassify the separation as a layoff in exchange for side payments that benefit both sides.

Yet, such collusion is unlikely in practice. When a separation is classified as a layoff, the firm must make substantial payments to both the government and the worker. For collusion to succeed, the firm would need to trust that the worker will return a portion of these payments after accessing their pension account, an arrangement with little enforcement, given its illegal nature.

Furthermore, empirical evidence suggests that such collusion agreements are rare. Since 2018, firms and workers in Brazil have had the option to terminate contracts by mutual agreement. Under this arrangement, the worker receives severance pay and can access 80% of their pension account, while the firm avoids the government fine. Therefore, this form of separation leaves collusive parties strictly better off.

If a strong motive to misreport quits as layoffs come from workers willing to withdraw funds from the pension account, mutual agreement separations should have become widespread. However, they account for only 0.5% of all separations after becoming legal.

Another potential motive for misreporting relates to unemployment insurance. Using the same RAIS administrative records, Van Doornik et al. (2023) find that workers eligible for unemployment insurance are 11% more likely to be laid off. Despite this, their analysis shows that excess layoffs are not merely misclassified quits, further suggesting that misreporting seldom occurs.

Figure 4.1 – *Quitting workers make better moves than laid-off individuals*

Notes: The figure compares labor outcomes and mobility patterns between laid-off and quitting workers. Panel (a) reports the share of separated workers who find a job within a year (job-to-job transitions). Panel (b) reports the share of job-to-job transitions with no gap between the two jobs. Panel (c) reports the average number of days spent out of employment during a job-to-job transition. Panel (d) reports the difference in wage growth between workers who change jobs and those who stay at the same firm. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

Finally, we conduct further validation of the classification by comparing post-separation outcomes of workers who quit versus those who were laid off. Figure 4.1 presents compelling evidence: workers who quit are significantly more likely to secure employment within a year compared to those who were laid off—51% versus 25%, respectively. Moreover, among those who found jobs, quitting workers tended to secure new positions more quickly (46% found immediate employment, compared to 25% of those laid off) and experienced more favorable wage growth, with an average increase of 11% in wages compared to a 2% decrease among those laid off. These patterns align with the hypothesis that separations categorized as quits are indeed voluntary and initiated by the workers, while those labeled as layoffs are not, further substantiating the accuracy of the reporting in our data.

4.3 The productivity-separation correlation: Empirics

In this section, we empirically investigate the determinants of the productivity-separation correlation. First, we demonstrate that the negative relationship between firm productivity and separation rates is primarily driven by high-productivity firms having lower layoff rates, namely the layoff-productivity correlation. Second, we show that this result is unconfounded with the sorting of high-skill workers into high-productivity firms.

Figure 4.2 documents our main empirical finding: the layoff-separation correlation. Panel 4.2a uses value added, our preferred productivity metric; however, value added is available only for the manufacturing sector and aggregated at the industry-state level, as detailed in Subsection 4.2.1. To extend the analysis to the full sample and explore firm-level variation, we complement our analysis with two additional proxies of firm productivity: firm size (Panel 4.2b) and firm pay premiums (Panel 4.2c). Reassuringly, the same patterns emerge regardless of the measure used.

Our key finding is that layoff rates decline sharply with firm productivity. The magnitude of this relationship is substantial. The annual layoff rate for firms at the top 5% of value-added is 8%, compared to 23% for firms at the bottom 5%. Quit rates also decline with firm productivity. Firms in the top 5% of value-added have a quit rate of 1.5%, whereas those in the bottom 5% have a quit rate of 3.3%. However, since quits are relatively rare, their contribution to overall separations is limited.¹¹

To formally quantify the relative contributions of layoffs and quits to the overall correlation between productivity and separation rates, we estimate the following regression

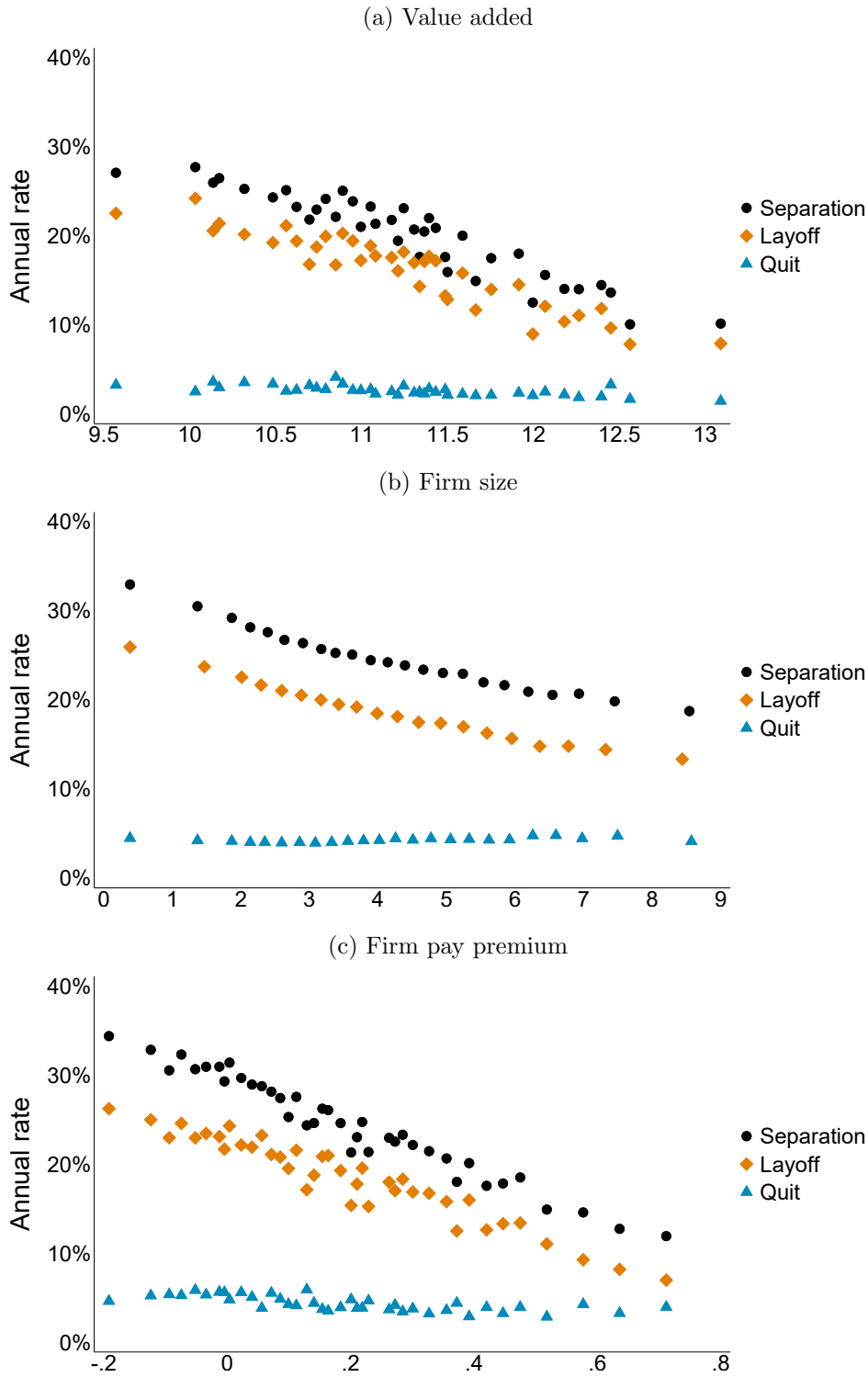
$$Y_{jt} = \beta_Y Q_{jt} + \epsilon_{Yjt} \quad (4.1)$$

where t represents the year, j denotes the firm, and Q_{jt} corresponds to each of the measures of firm productivity. We run three separate regressions, where the dependent variable Y_{jt} represents the firm's quit, layoff, or overall separation rate, and ϵ_{Yjt} captures residual variation. The parameters of interest, β_Y , measure the relationship between firm productivity and each rate.

Since the overall separation rate is the sum of the layoff and the quit rate, it follows that

$$\beta_{Separation} = \beta_{Layoff} + \beta_{Quit} \quad (4.2)$$

^{11.} The relationship between quits and firm productivity remains negative but weakens when using firm pay premiums rather than value-added, and it becomes not statistically significant when using firm size. However, these differences do not alter our main conclusion that layoffs primarily drive the productivity-separation correlation, as confirmed by Table 4.2.

Figure 4.2 – *Layoffs explain most of the productivity-separation correlation*

Notes: The figure illustrates the relationship between annual separation rates and firm productivity across three measures. Panel (a) uses value added, Panel (b) firm size, and Panel (c) firm pay premiums. For more details on firm productivity measures, see Subsection 4.2.1 and Section C.2 in the Appendix. Each figure depicts total separations (black circles), layoffs (orange diamonds), and quits (blue triangles), using firm-level binscatter regressions following Cattaneo et al. (2025), weighted by firm size. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1. Panel (a) uses value added from the PIA and, as such, it reports aggregates at the industry-state level and is further restricted to manufacturing firms.

Motivated by the decomposition in Equation (4.2), we assess the role of layoffs in the productivity-separation relationship using the following ratio

$$\text{Role of layoffs in productivity-separation relationship} \equiv \frac{\beta_{Layoff}}{\beta_{Separation}}$$

The first column of Table 4.2 reports estimates of the ratio, revealing that the negative relationship between firm productivity and overall separation rates is predominantly driven by layoffs. Notice the similarity in the slopes of annual layoff rates and separation rates with respect to firm productivity. The ratio between the two slopes ranges from 0.81 to 0.93, depending on the measure of firm productivity, supporting our conclusion.

Worker segregation and sorting. An alternative explanation for the patterns observed in Figure 4.2 is sorting. Substantial evidence shows that more skilled workers tend to sort into more productive firms (Card et al., 2013; Song et al., 2019; Gerard et al., 2021). As a result, the lower layoff rates among high-productivity firms could simply reflect the fact that skilled workers are less likely to be laid off in the first place.

Therefore, to assess the extent to which worker sorting influences our findings, we extend Equation (4.1) to account for worker heterogeneity.

$$Y_{it} = \beta_Y Q_{J(i,t)t} + \gamma X_{it} + \epsilon_{it} \quad (4.3)$$

where $J(i,t)$ depicts the firm employing an individual worker i in year t . In this case, we estimate two separate regressions. One in which the dependent variable, Y_{it} , indicates whether the worker i was laid off in t , and another that indicates any form of separation. The key independent variable, Q_{jt} , measures firm productivity, while X_{it} includes a set of individual characteristics that control for worker heterogeneity. The residual term, ϵ_{it} , captures unobserved factors. The coefficient of interest, β_Y , estimates the relationship between firm productivity and layoffs or separations, respectively, net of worker-specific differences.

We control for a comprehensive set of worker characteristics. First, tenure is a key determinant of layoff rates (Jovanovic, 1979; Topel and Ward, 1992; Ureta, 1992), so we include it as a quadratic term. Second, career trajectories may vary by gender, skill, and experience. As such, we follow Gerard et al. (2021) and control for education- and gender-specific quadratic age trends. Third, we account for potential discrimination in the labor market by further including race fixed effects. Fourth, we capture occupational sorting across firms—such as

Table 4.2 – *Worker sorting does not drive the productivity-layoff correlation*

	(1)	(2)	(3)	(4)
<i>Panel A - Value added</i>				
$\frac{\beta^{\text{Layoff}}}{\beta^{\text{Separation}}}$	0.871*** (0.0055)	0.852*** (0.0090)	0.776*** (0.0096)	0.804*** (0.0116)
Observations	9,308,341	9,307,701	9,308,341	9,307,701
<i>Panel B - Firm size</i>				
$\frac{\beta^{\text{Layoff}}}{\beta^{\text{Separation}}}$	0.928*** (0.0032)	0.950*** (0.0048)	0.946*** (0.0046)	0.957*** (0.0062)
Observations	49,835,818	49,830,114	49,835,818	49,830,114
<i>Panel C - Firm pay premium</i>				
$\frac{\beta^{\text{Layoff}}}{\beta^{\text{Separation}}}$	0.811*** (0.0018)	0.774*** (0.0025)	0.698*** (0.0034)	0.703*** (0.0033)
Observations	49,835,818	49,830,114	49,835,818	49,830,114
Worker covariates			✓	✓
Worker FE		✓		✓

Notes: The table reports the ratio between the slopes of layoff rates and separation rates with respect to firm productivity. We estimate each slope using OLS at the worker level, separately for three measures of firm productivity and across four specifications. Column (1) corresponds to Equation (4.1), while Columns (2) to (4) correspond to Equation (4.3). Panel A uses value added; Panel B, firm pay premiums; and Panel C, firm size. For more details on firm productivity measures, see Subsection 4.2.1 and Section C.2 in the Appendix. We obtain standard errors using the delta method. We control for worker heterogeneity using individual-level data from RAIS, including quadratic tenure, education- and gender-specific quadratic age trends, race and occupation fixed effects, and worker fixed effects from the AKM estimation. For separate estimates of β^{Layoff} and $\beta^{\text{Separation}}$, see Table C.2 in the Appendix. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1. Panel A uses value added from the PIA and, as such, it reports aggregates at the industry-state level and is further restricted to manufacturing firms.

variation in unionization rates—using occupation fixed-effects. Finally, we adjust for other potential sources of unobserved heterogeneity in worker ability by incorporating worker fixed effects, which are simultaneously estimated with firm pay premiums.¹²

As before, we estimate the ratio $\frac{\beta^{\text{Layoff}}}{\beta^{\text{Separation}}}$ and examine the role of layoffs in driving the negative productivity-separation correlation.¹³ Table 4.2 shows that even after controlling for

^{12.} Despite clustering firms into 100 groups, which reduces noise in the estimation, results using this covariate should be interpreted with caution. In particular, the literature has noted that noise in the estimation tends to overestimate the variance of firm pay premiums at the expense of worker sorting (Bonhomme et al., 2019; Kline et al., 2020). For more details on this, see Section C.2 in the Appendix.

^{13.} See Table C.2 in the Appendix for separate estimates of β^{Layoff} and $\beta^{\text{Separation}}$.

worker heterogeneity, layoffs explain most of the productivity-separation correlation. Panel A presents our baseline estimates, showing that $\frac{\beta_{\text{Layoff}}}{\beta_{\text{Separation}}}$ falls only slightly, from 0.87 to 0.80 in the most flexible specification. Moreover, Panels B and C yield similar results using alternative firm-level measures of productivity. This suggests that worker heterogeneity does not account entirely for the observed patterns.

4.4 The productivity-separation correlation: Theory

The following section introduces a job-search model to explain the empirical pattern we observe in the data. That is, that high-productivity firms exhibit a lower layoff rate. Although previous theoretical work has established that both quits and layoffs can occur within frameworks that assume fully efficient separations (McLaughlin, 1991), we draw on recent empirical research showing that inefficient layoffs are pervasive (Schmieder and von Wachter, 2010; Jäger et al., 2023; Davis and Krolkowski, 2025).

We build on these empirical findings and develop a model in which endogenous inefficient layoffs arise from the interaction of two key ingredients: wage rigidity and uncertainty about the worker’s productivity. Specifically, firms post wages before observing the worker’s productivity shock. Thus, there exists a threshold that defines the worst possible productivity shock before it becomes unprofitable for the firm to retain the worker at the posted wage. Even if firms and workers might prefer to cut wages instead, downward wage rigidity prevents them from doing so. As a result, firms proceed with the layoff to shield themselves from losses, which is the source of the inefficiency in the model. Aside from these key features, we keep the model as simple and tractable as possible.

The remainder of the section proceeds as follows. First, we present the economy in which our model operates and the timing of agents’ decisions. Second, we delve into the productivity-separation correlation and present our key theoretical result. Namely, that more productive firms have *both* fewer quits and fewer layoffs.

4.4.1 Setting

The framework consists of a partial equilibrium, posting, random search model with *ex-ante* homogeneous workers. Random implies that firms are matched with an exogenously determined type and number of workers. Posting means that there is no bargaining between workers and firms. Partial equilibrium and posting imply that we take firm productivity

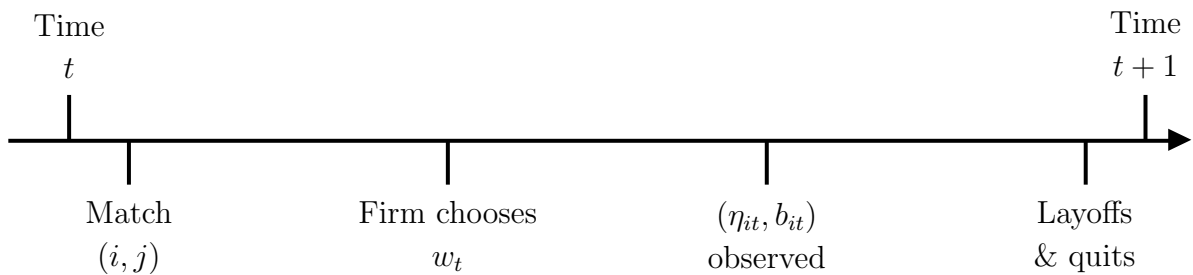
heterogeneity and the distribution of workers' outside options as given. In other words, the firm is atomistic, and its decisions cannot affect determinants of workers' outside options, including offers from other firms, government policies, or labor market conditions.

Section 4.3 motivates the assumption of a homogeneous workforce, given that worker heterogeneity cannot fully explain the negative relationship between layoff rates and productivity. In any case, one can interpret the framework as representative of a labor market for a specific worker type—e.g., for a certain age group, education level, or occupation.

The firm. Consider a single firm, characterized by productivity ψ . Every period t , the firm maximizes the present value of expected profits at a discount rate β . Henceforth, we define the equilibrium under $\beta = 0$. Nevertheless, Section C.1 in the Appendix demonstrates that all results hold for any $\beta \in [0, 1)$. Each period, the firm chooses a single wage rate w_t for all employees and whether to lay off each worker.

The workers. We assume a continuum of ex-ante homogeneous workers, indexed by i , with expected productivity α . Each period, workers receive a productivity shock η_{it} . When employed, worker i derives utility equal to the wage rate w_t . Otherwise, they receive a stochastic, exogenously determined outside option b_{it} . For simplicity, we assume workers discount the future infinitely and hence quit whenever $w_t < b_{it}$. Both shocks, η_{it} and b_{it} , are idiosyncratic and follow known distributions, F_η and F_b , respectively. We normalize α such that $\mathbb{E}_\eta[\eta_{it}] = 0$.

Timing. The model proceeds as follows



The firm starts each period with a size s_t of workers and immediately meets an additional unit mass of potential hires with an expected productivity α . Next, the firm chooses a wage w_t for all incumbent employees and new hires. Then, productivity η_{it} and outside option b_{it} shocks are realized and observed by both the firm and the workers. After that, the firm decides which workers to lay off, while each employee simultaneously decides whether to

quit. Hereafter, let $\delta_\psi(w_t)$ and $\rho(w_t)$ denote the layoff rate and retention rate functions, respectively, where the latter corresponds to one minus the quit rate function.

Last, payoffs take place, and the firm updates its size to $s_{t+1} = \rho(w_t) \cdot [1 - \delta_\psi(w_t)] (1 + s_t)$ workers. Each worker receives w_t when employed and b_{it} otherwise. Since the productivity shock is known at this point, the firm's total revenue can be characterized as $\psi + \alpha + \eta_{it}$, so that the firm's per-worker profits correspond to the markdown $\mu_{it}(w_t)$, which represents the difference between the worker-specific ex-post productivity and the wage. That is,

$$\underbrace{\mu_{it}(w_t)}_{\text{markdown}} \equiv \underbrace{\psi}_{\text{firm productivity}} + \underbrace{\alpha + \eta_{it}}_{\text{worker productivity}} - \underbrace{w_t}_{\text{wage}}$$

The timing creates wage rigidity by requiring firms to set wages before observing the productivity shock η , while allowing them to decide on layoffs later on. When a worker-specific markdown turns negative, $\mu_{it}(w_t) < 0$, the firm can lay off that particular worker. Since firms cannot adjust wages once posted, layoffs occur even in cases where an alternative wage w'_t exists such that both the worker and the firm would prefer to continue the match. In other words, even if $w'_t \geq b_{it}$ and $\mu_{it}(w'_t) > 0$. This mechanism becomes crucial in illustrating how wage rigidity often leads to inefficient layoffs.

One important note. Workers can quit the same period they meet with the firm. Therefore, the retention rate represents both the share of current workers who remain in the firm and the share of new matches who accept the offer. Since all shocks are independent across periods and workers are ex-ante homogeneous, the two shares coincide.

The equilibrium arises as the optimal solution of three decisions. First, the firm lays off a worker whenever its markdown is negative. Second, a worker quits whenever its outside option exceeds the posted wage. Third, the firm chooses the wage that maximizes expected profits. For a formal definition of the equilibrium, see Section C.1 in the Appendix.

4.4.2 Drivers of the productivity-separation correlation

Section 4.3 presents evidence that a lower layoff rate among high-productivity firms mainly drives the productivity-separation correlation. In the following subsection, we turn to the predictions in the theoretical model and delve into the mechanisms. To build intuition, we present a simplified version of the relevant theorem below; the full technical statement can be found in Section C.1 in the Appendix.

Assumption 4.1. *Let the distribution of workers' outside options shocks, F_b , be a log-concave function, and let the distribution of workers' productivity shocks, F_η , be a function such that*

$$\frac{\partial \mathbb{E}_\eta [\eta | \eta > x]}{\partial x} \leq 1$$

Ultimately, the firm's objective function depends on the functional forms characterizing the distributions of the workers' outside options shocks, F_b , and the workers' productivity shocks, F_η . Nonetheless, under standard assumptions about F_b and F_η , average markdowns and worker retention become supermodular.¹⁴ That is, an increase in the firm's average markdown strengthens the incentives to retain workers, and viceversa.

Definition (Equilibrium). *Let F_b and F_η satisfy the conditions in Assumption (4.1). Then,*

- (I) *Firm size is increasing in firm productivity;*
- (II) *Wages are increasing in firm productivity;*
- (III) *The separation rate is decreasing in firm productivity;*
- (IV) *The quit rate is decreasing in firm productivity;*
- (V) *The layoff rate is decreasing in firm productivity;*
- (VI) *The average markdown is increasing in firm productivity.*

Consistent with prior theoretical work, our model shows that wages and firm size increase with firm productivity (Postel-Vinay and Robin, 2002; Elsby and Gottfries, 2022), justifying the use of firm pay premiums and firm size as proxies for firm productivity in our empirical analysis in Section 4.3. Furthermore, the model's conclusions align with earlier research showing that separation rates decline with firm productivity.

We extend this literature by decomposing separations into quits and layoffs and by investigating how firm productivity relates to each margin. Our model predicts that *both* quits and layoffs decline as firm productivity rises, with a productivity-quit relationship that follows previous intuition in the literature (Burdett and Mortensen, 1998). Workers employed at more productive firms are less likely to quit since these firms pay higher wages.

Our main theoretical contribution centers on explaining why higher-productivity firms resort to layoffs less often. With that in mind, notice that layoffs occur when worker-

^{14.} Log-concavity in F_b implies a unimodal distribution, without heavy tails, and with an increasing hazard rate. Conveniently, several of these conditions hold for many standard distributions often used in search models and duration analysis, including normal, exponential, and uniform families. The condition for F_η is weaker; it requires a distribution with a non-increasing mean residual life (DMRL), i.e., that the average excess above the cutoff x declines with the cutoff. Beyond log-concave functional forms, other distributions without heavy tails satisfy this property.

specific markdowns are negative, which happens only for sufficiently low realizations of the productivity shock, such that they offset the firm's average markdown

$$\psi + \alpha + \mathbb{E}[\eta_{it}] - w_t = \psi + \alpha - w_t$$

As a result, layoff rates decline, everything else equal, as markdowns increase. Then, the key question is: do more productive firms systematically exhibit larger markdowns?

Markdowns. When choosing the optimal wage, the firm faces a tradeoff between worker retention and the average markdown. By posting a higher wage offer, the firm increases worker retention and achieves higher profits from the extensive margin.¹⁵ On the contrary, a higher posted wage shrinks the average markdown across all workers, reducing profits from the intensive margin and increasing the probability of a layoff.

Without further assumptions, the model is flexible enough to accommodate situations in which the firm has stronger preferences for the extensive or the intensive margin. Nonetheless, Assumption (4.1) implies that profits per worker increase with markdowns.

High-productivity firms post higher wages than their less productive counterparts, while simultaneously setting higher average markdowns and worker retention rates. As a result, more productive firms exhibit both lower layoff and quit rates in equilibrium.

Takeaway. Under minimal assumptions, the model shows that more productive firms have larger average markdowns and higher retention rates, thereby exhibiting fewer layoffs and quits. However, the actual levels of both rates and how sharp they decline with productivity depend, essentially, on the exact underlying distributions for b_{it} and η_{it} .

For instance, consider a labor market with a more elastic labor supply (e.g., Gumbel distribution with a small scale parameter) but large idiosyncratic productivity volatility (e.g., uniform distribution, or Weibull with shape $k > 1$ and a relatively large scale parameter). In such a case, both distributions would lead the firm to be more inclined towards worker retention rather than average markdowns, resulting in fewer quits and more layoffs. On the contrary, labor markets with a more inelastic labor supply (e.g., Laplace distribution with a large scale parameter) and more stable workers' productivity (e.g., normal distribution with a small variance) would yield opposite results for analogous reasons.

^{15.} When worker retention is the sole mechanism, more productive firms exhibit *lower* rather than higher markdowns. For that reason, models that assume homogeneous firms often feature a negative relationship between wages and markdowns in equilibrium. For a canonical example of such a framework, see Burdett and Mortensen (1998).

Our framework also emphasizes the interplay between wage rigidity and productivity uncertainty, and how this can lead to inefficient layoffs. In summary, firms post wages before observing workers' productivity shocks. Whenever realized productivity falls below a certain threshold, workers become unprofitable, and the firm lays them off. As we mentioned above, there are cases in which the firm and the worker would prefer to cut wages but are unable to do so due to wage rigidity. In Section 4.5, we further investigate the mechanisms and their implications.

4.5 Empirical validation of the proposed mechanisms

The following section empirically validates the mechanisms proposed by the theoretical framework described in Section 4.4. In Subsection 4.5.1, we first document that higher-productivity firms exhibit larger markdowns. Then, we also show that firms with larger markdowns lay off fewer workers. In Subsection 4.5.2, we show that the productivity-layoff correlation is stronger among firms with tighter constraints on wage adjustments, which directly connects to wage rigidity. Together, these patterns closely align with the mechanisms proposed in the model.

4.5.1 Higher-productivity firms have larger markdowns

The following subsection describes the empirical relationship between markdowns, firm productivity, and layoffs in Brazil, showing that the predictions in our framework align with the patterns observed in the data.

Our empirical definition for markdowns uses the PIA.¹⁶ We define markdowns as the proportion of value added (VA) remaining after labor costs are paid.¹⁷ Namely,

$$\text{Markdown} = \frac{\text{VA} - \text{Labor Costs}}{\text{VA}}$$

Figure 4.3 illustrates the relationship between markdowns and firm productivity across three different measures: value added, firm pay premiums, and firm size. We find that, on average, more productive firms set higher markdowns, as predicted by our framework. For

^{16.} As described in Subsection 4.2.1, the PIA dataset covers only manufacturing firms and provides aggregates at the state-industry level. Hence, all results in Subsection 4.5.1 are restricted to this sample.

^{17.} The literature often defines markdowns as the ratio of marginal product to wages (Estefan et al., 2024). However, marginal product estimates are not available in our setting.

our preferred productivity measure, valued added (Panel 4.3a), firms in the bottom 5% of productivity have an average markdown of 32%, compared to those in the top 5%, which have an average markdown of 68%—more than double. Systematically, we find similar patterns using firm pay premiums (Panel 4.3b) and firm size (Panel 4.3c).

Moving forward, Figure 4.4 depicts a negative correlation between markdowns and annual layoff rates. Higher-markdown firms layoff fewer workers. Each 10 percentage-point increase in markdowns is associated with roughly a 1 percentage-point reduction in the annual layoff rate. Moreover, the relationship is economically meaningful: firms in the bottom 5% of markdowns have an average annual layoff rate of around 18%, compared to just 11% for firms in the top 5%. The findings corroborate two of the main predictions in our framework. First, that higher-productivity firms set larger markdowns (Figure 4.3), and that the layoff rate declines with markdowns (Figure 4.4).

In the following subsection, we turn our attention to another important mechanism in the model: wage rigidity.

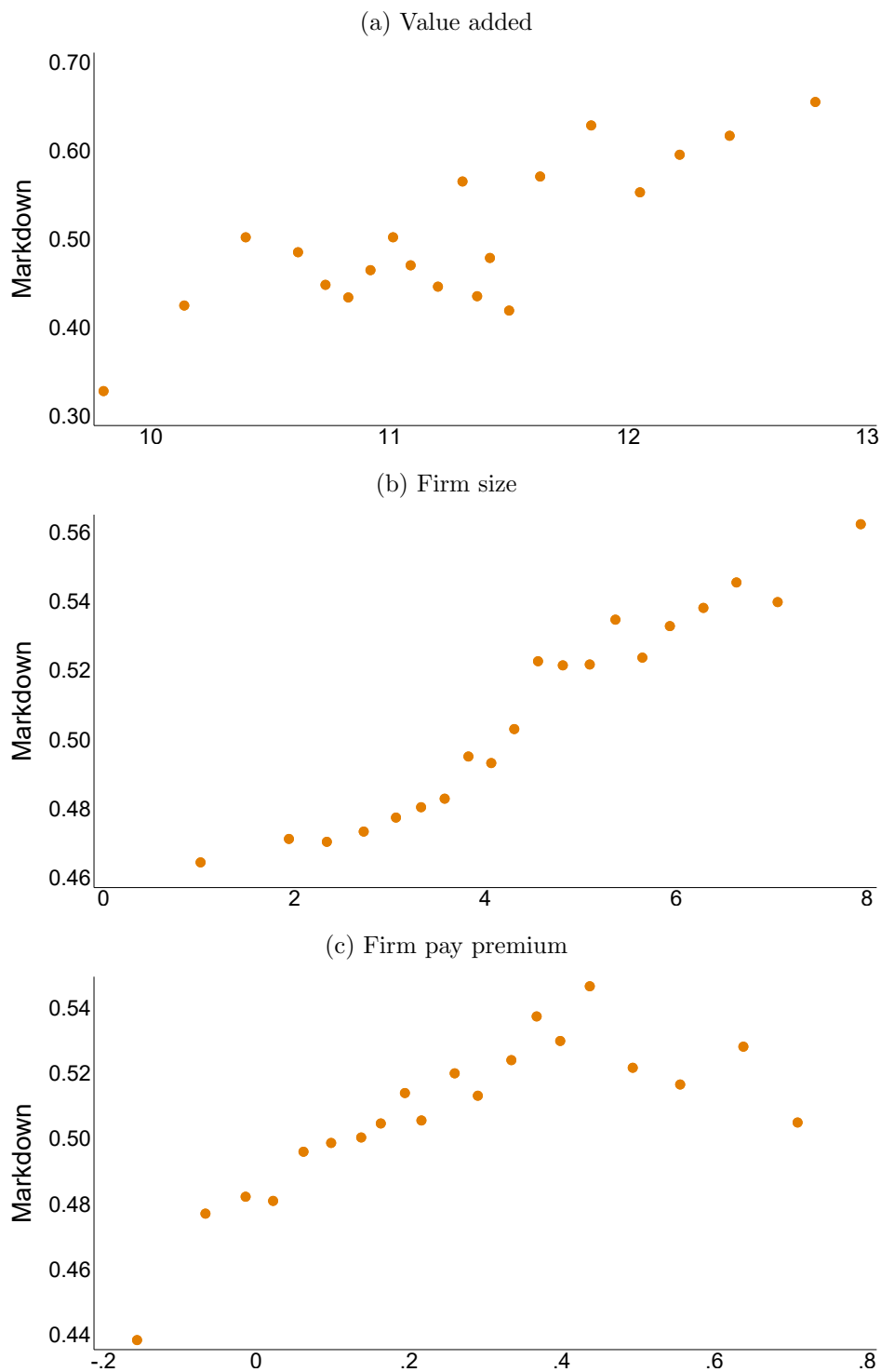
4.5.2 Wage rigidity amplifies the productivity-layoff correlation

In Section 4.4, wage rigidity plays a central role and connects directly to inefficient layoffs. Imagine a worker who faces a negative productivity shock, resulting in a markdown below zero. If the firm cannot adjust wages downward due to wage rigidity, then it lays off the worker. As a result, layoffs can occur even when both the firm and the worker would agree on a different, lower wage to keep the employment relationship going. As such, wage rigidity renders most layoffs inefficient.

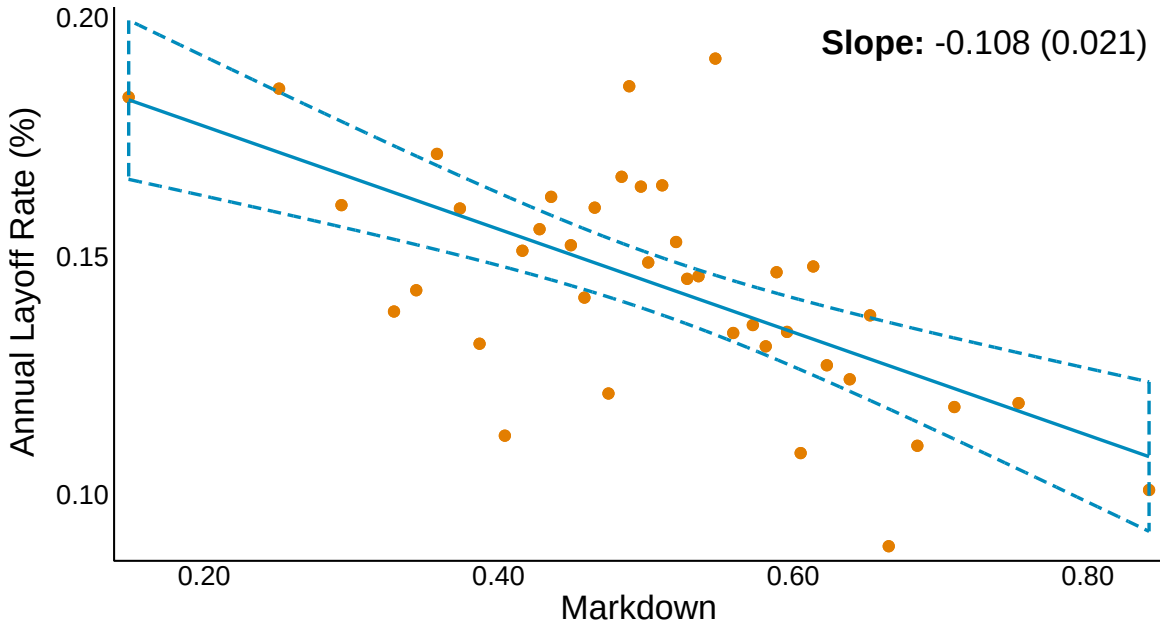
The following subsection further investigates the relationship between wage rigidity and layoffs in Brazil. First, we document a substantial wage rigidity in Brazilian labor markets. Second, we construct a proxy for wage rigidity at the firm level and use it extensively to examine whether wage rigidity is associated with higher layoff rates and a stronger productivity-layoff correlation.

Documenting wage rigidity. To study wage rigidity, we leverage the fact that RAIS reports contract wages separately from a variable wage component.¹⁸ The variable component encompasses bonuses, performance pay, and overtime. Firms face substantial rigidity

¹⁸. Throughout this chapter, we have used *wage* interchangeably with *total wage*, which is the sum of the contract wage and the variable pay component.

Figure 4.3 – *More productive firms have larger markdowns*

Notes: The figure illustrates the relationship between markdowns and firm productivity across three measures. Panel (a) uses value added, Panel (b) uses firm size, and Panel (c) uses firm pay premiums. For more details on firm productivity measures, see Subsection 4.2.1 and Section C.2 in the Appendix. We define markdowns as the proportion of value added (VA) remaining after labor costs are paid. We obtain markdowns and value added from the PIA and, as such, results report aggregates at the industry-state level and are further restricted to manufacturing firms. Orange dots correspond to a firm-level binscatter regression following the method in Cattaneo et al. (2025), weighted by firm size. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private manufacturing sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

Figure 4.4 – *Firms with larger average markdowns have lower layoff rates*

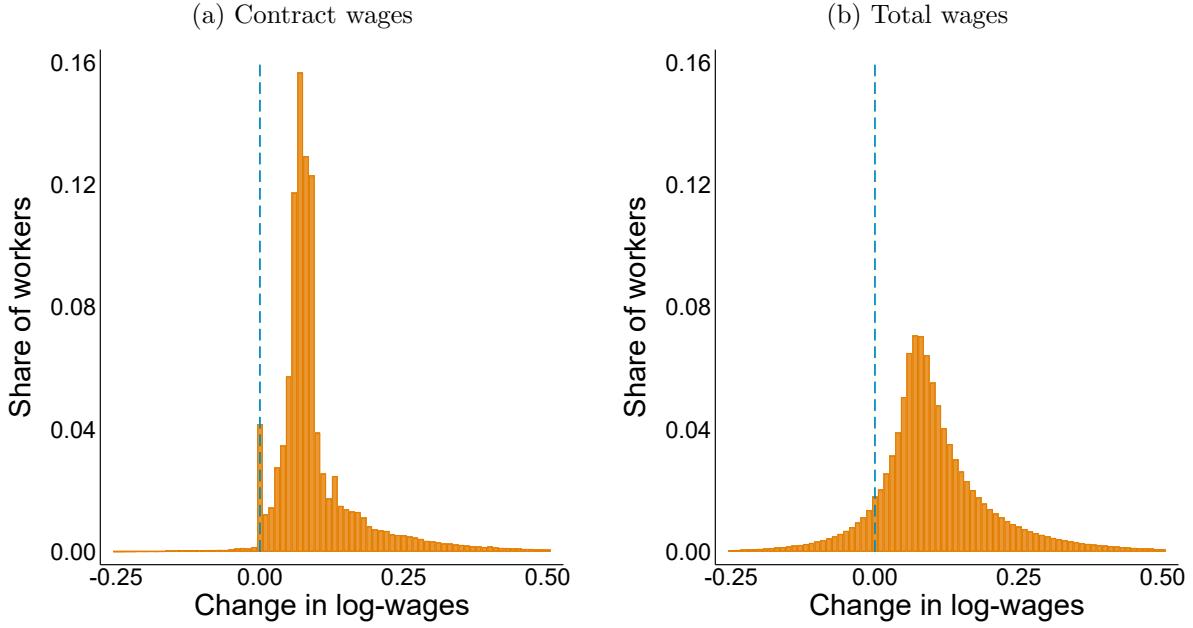
Notes: The figure illustrates the relationship between markdowns and layoff rates. We define markdowns as the proportion of value added (VA) remaining after labor costs are paid. We obtain markdowns from the PIA and, as such, results report aggregates at the industry-state level and are further restricted to manufacturing firms. Estimates are weighted by firm size. The blue solid line plots the best linear fit, with the corresponding coefficient and robust standard errors (95% confidence) reported in the upper-right corner. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private manufacturing sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

in adjusting contract wages, as the Brazilian Constitution prohibits wage reductions unless authorized by a collective bargaining agreement.¹⁹

In contrast, regulations do not constrain the variable wage component. Figure 4.5 presents the distribution of interannual wage changes for *stayers* (workers who remain in the same firm across two consecutive years). Consistent with our argument, Panel 4.5a shows that only 1.34% of workers face a reduction in their contract wage, whereas Panel 4.5b shows that 9.28% experience a reduction in their total wage. In addition, contract wage changes bunch around zero, whereas total wages exhibit no such pattern. These results highlight the substantial rigidity of contract wages relative to the flexibility of variable pay, a phenomenon well-documented in other contexts (Altonji and Devereux, 1999; Messina et al., 2010; Anger, 2011; Grigsby et al., 2021).

Motivated by this, we construct a firm-level measure of wage rigidity based on its dependence on contract wages versus variable pay, an approach similar to Makridis and Gittleman

¹⁹. Title II, Chapter I, Article 7, Paragraph VI of the 1988 Constitution.

Figure 4.5 – *Distribution of interannual wage changes for stayers*

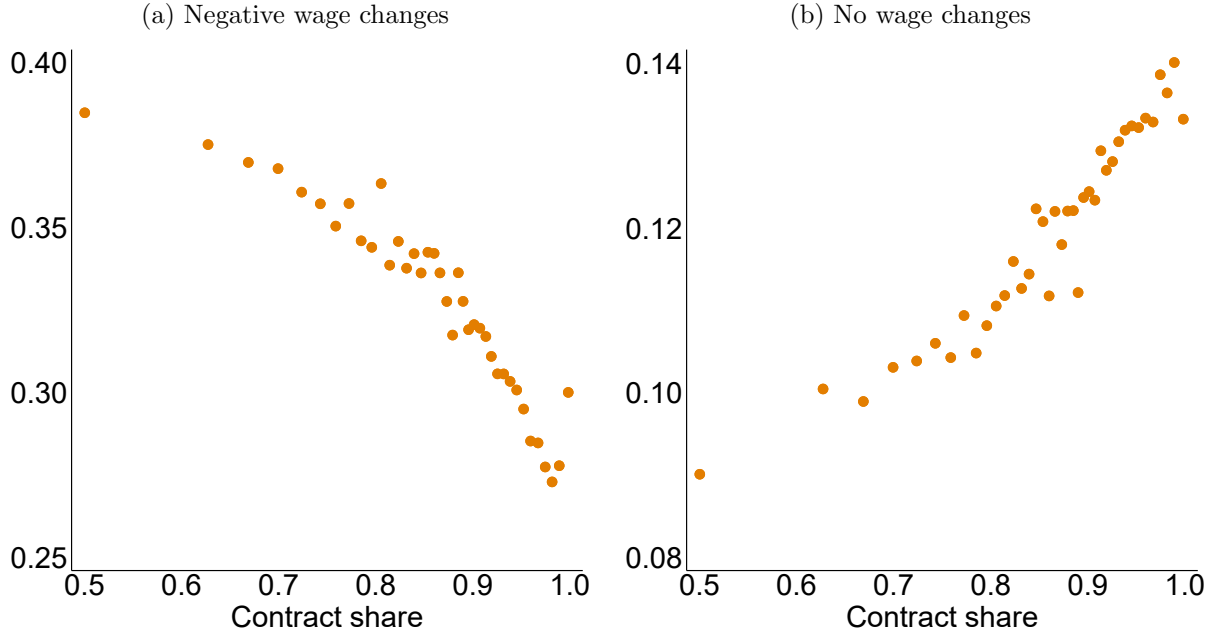
Notes: The figure compares the distribution of interannual nominal wage changes for workers who remain in the same firm across two consecutive years. Panel A depicts the distribution of contract wage changes, while Panel B shows the distribution of total wage changes (the sum of contract wage and variable pay components). The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the manufacturing sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

(2022), Reizer (2022), and Sockin and Sockin (2025). Specifically, we proxy for wage rigidity using the contract share (CS). That is, the share of the total wage bill attributable to contract wages

$$CS_j = \frac{1}{N_j} \sum_{i|j(i,t_{j0})=j} \frac{\text{ContractWage}_i}{\text{VariableWage}_i + \text{ContractWage}_i} \quad (4.4)$$

where $J(i, t_{j0})$ denotes the firm employing worker i in year t_{j0} , N_j the size of firm j , and t_{j0} the first year the firm appears in the sample.

The underlying idea is that firms can freely adjust variable pay but not contract wages; thus, a higher contract share indicates greater wage rigidity. Due to concerns about endogeneity and simultaneity—such as the contract share responding to productivity shocks—we compute CS using the year each firm first appears in RAIS, and hold it fixed throughout the analysis. Furthermore, for each firm, we exclude from the subsequent analysis all observations corresponding to the year we use to define CS . Panel C.2a in the Appendix depicts the distribution of CS across firms, revealing a substantial variation. The median share is roughly 88%, and the 5th and 95th percentiles are 65% and 99%, respectively.

Figure 4.6 – *A higher contract share is associated with greater wage rigidity*

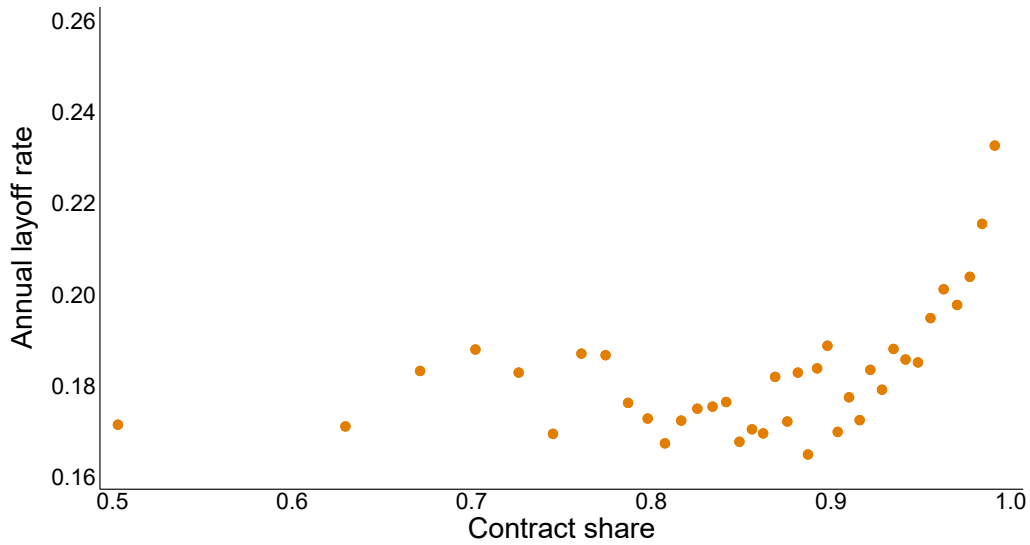
Notes: The figure illustrates the relationship between wage rigidity and the contract share (CS). The contract share denotes the fraction of the total wage bill attributable to contract wages, as defined in Equation (4.4). The panels depict the correlation between CS and the share of stayers in a firm that face interannual wage declines (Panel a) or no change (Panel b). Orange dots correspond to a firm-level binscatter regression following the method in Cattaneo et al. (2025), weighted by firm size. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

We validate the mechanism using the contract share as a proxy for wage rigidity by examining its correlation with interannual wage changes for stayers. Consistent with our interpretation that the contract share captures wage rigidity across firms, Figure 4.6 shows that CS negatively correlates with the share of wage declines among stayers within a firm and positively with the share of null wage changes. Furthermore, the relationships are significant. The fraction of stayers with interannual wage changes that bunch around zero is 40% larger among firms in the top 5% of CS than for firms in the bottom 5%. Furthermore, worker sorting cannot fully explain these patterns.

Figure C.3 in the Appendix shows the results remain significant after controlling for the same set of worker characteristics as in Section 4.3.²⁰

Greater wage rigidity is associated with higher layoff rates. After establishing that a higher contract share is associated with greater wage rigidity, we turn to its relationship with layoff rates. Figure 4.7 shows that firms with higher CS exhibit higher annual layoff

²⁰. Including quadratic tenure, race fixed effects, occupation fixed effects, education- and gender-specific quadratic age trends, and worker fixed effects from the same AKM estimation as firm pay premiums.

Figure 4.7 – *Layoffs are more prominent among firms with a higher contract share*

Notes: The figure depicts the relationship between annual layoff rates and the contract shares. The contract share denotes the share of the total wage bill attributable to contract wages, as defined in Equation (4.4). Orange dots correspond to a firm-level binscatter regression following the method in Cattaneo et al. (2025), weighted by firm size. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

rates. In particular, firms with a CS of 100% have an average annual layoff rate of around 28%, whereas those with a CS of around 90% depict an annual layoff rate of roughly 17%. Notably, the relationship flattens for CS below 90%, with annual layoff rates stabilizing at approximately 17% even for firms with CS below 60%.

These patterns establish two central implications. First, as CS approaches 100%, the elasticity between the layoff rate and the contract share increases significantly, underscoring the role that wage rigidity plays in explaining layoffs. Second, the persistence of layoffs at lower levels of CS indicates that wage rigidity cannot account for a significant number of layoffs. Through the lens of our model, these layoffs correspond to a productivity shock so severe that the employment relationship cannot accommodate it for any feasible wage adjustment. Alternatively, such layoffs could arise from exogenous job destruction shocks, which are typically standard in the literature (Burdett and Mortensen, 1998; Sorkin, 2018; Jarosch, 2023).

Nonetheless, worker heterogeneity once again raises concerns about the results. Any given firm is less likely to lay off more skilled workers. Now, imagine that high-skilled workers also tend to receive larger bonuses or benefit more from profit-sharing. Then, we would expect a positive relationship between the contract share and layoffs not driven by

wage rigidity. We address this issue in Figure A in the Appendix, which replicates the empirical exercise while controlling for the same set of worker characteristics as in Section 4.3.²¹ Results remain largely unchanged, which increases our confidence that the contract share captures wage rigidity beyond cross-firm differences in variable pay.

The productivity-layoff correlation strengthens with wage rigidity. To conclude this section, we investigate whether wage rigidity contributes to the productivity-layoff correlation. However, this empirical exercise poses a challenge. The correlation between the two objects of interest is an equilibrium object, observed at the market level rather than the firm level. Ideally, we would need a relatively large sample of disconnected labor markets, each with varying degrees of wage rigidity across firms, so we could directly compare the productivity-layoff correlation across them.

The closest approximation we can afford is to define labor markets by industry and location. Moreover, value added, our preferred productivity measure, is an aggregate at the state and 3-digit industry level. Thus, we define markets as combinations of state and 2-digit industry codes, ensuring sufficient variation in value added within markets.

As a result, we obtain 341 distinct labor markets. In each, we define wage rigidity as the average CS across firms, denoted as $\overline{CS}$. Panel A in the Appendix displays the distribution of $\overline{CS}$, revealing substantial variation in wage rigidity that we can exploit. The 5th percentile of $\overline{CS}$ is roughly 72%, while the 95th percentile is about 94%.

We test whether $\overline{CS}$ is a relevant proxy for wage rigidity by replicating the analysis in Figures 4.6 and 4.7 at the market level. Specifically, we estimate the following regression

$$Y_m = \chi_Y \cdot \overline{CS}_m + \epsilon_m^Y, \quad (4.5)$$

where $\overline{CS}_m$ represents the average contract share across firms operating in market m , χ_Y depicts our parameter of interest, and ϵ_m^Y denotes residuals. The outcome Y_m corresponds to either the share of negative wage changes among stayers in each market, the share of wage changes equal to zero among stayers, or the average annual layoff rate among all workers.

Since Figure 4.7 suggests a highly nonlinear relationship between the contract share and layoffs, we also estimate Equation (4.5) using as an outcome the average log of the annual layoff rate for each market.

²¹. Including quadratic tenure, race fixed effects, occupation fixed effects, education- and gender-specific quadratic age trends, and worker fixed effects from the same AKM estimation as firm pay premiums

Columns (1) to (4) of Table 4.3 report OLS estimates for Equation (4.5), confirming the patterns reported at the firm level. In particular, workers employed in markets with higher $\overline{CS}$ are less likely to experience interannual wage declines, more likely to see no wage changes, and face higher annual layoff rates. Consistent with our intuition, these results support that $\overline{CS}$ reliably captures wage rigidity across firms operating in different markets.

Next, we quantify the strength of the productivity-layoff correlation within each labor market by separately estimating the following regression on each market

$$\text{Layoff Rate}_{jt} = \beta_{M(j)} \cdot Q_{jt} + \epsilon_{jt}^M \quad (4.6)$$

where t denotes a year, $M(j)$ identifies the market in which firm j operates, and ϵ_{jt}^M captures residual variation. Moreover, Layoff Rate_{jt} denotes the firm's annual layoff rate. We estimate the same set of regressions separately across our three measures of firm productivity, Q_{jt} : value added, firm size, and firm pay premiums.

To account for potential nonlinearities, we follow our approach in Equation (4.5) and estimate the model using both annual layoff rates and their log as outcomes. The parameter of interest, β_m , captures the market-specific productivity-layoff correlation. Figure C.5 in the Appendix displays the distribution for estimates of $\hat{\beta}_m$ across markets.²²

Moving forward, we examine whether the productivity-layoff correlation strengthens in labor markets with greater wage rigidity by estimating the following regression

$$\hat{\beta}_m = \chi_\beta \cdot \overline{CS}_m + \epsilon_m^\beta \quad (4.7)$$

where χ_β denotes the parameter of our interest and ϵ_m^β captures residual variation.

Columns (5) to (10) of Table 4.7 report OLS estimates for Equation (4.7), corroborating that the productivity-layoff correlation strengthens in a market with greater wage rigidity. Columns (5), (7), and (9) report results for estimates of $\hat{\beta}_m$ under a linear specification of annual layoff rates. Columns (6), (8), and (10) follow the same logic for estimates under a log specification. Findings are consistent across the two specifications and two of the three measures of firm productivity.

In particular, we do not find a significant relationship between $\hat{\beta}_m$ and $\overline{CS}$ when using firm pay premiums. Since pay premiums are estimated objects, and measurement error

²². Notice that $\hat{\beta}_m$ is an estimated parameter and thus subject to measurement error. Interpreting its distribution should be done with caution.

Table 4.3 – *The productivity-layoff correlation strengthens in markets with greater wage rigidity*

	$\hat{\beta}_m$									
	$\Delta W < 0$			$\Delta W = 0$			Layoff			Firm pay premium
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	
$\overline{CS}_m$	-0.437*** (0.0393)	0.106*** (0.0208)	0.178** (0.0774)	0.148** (0.0634)	-0.211** (0.0883)	-0.175** (0.0724)	-0.032** (0.0156)	-0.027** (0.0130)	0.057 (0.0987)	0.045 (0.0841)
Observations	341	341	341	341	71	71	336	336	336	336
Avg. Outcome	0.276	0.092	0.191	0.191	-0.036	-0.036	-0.013	-0.013	-0.194	-0.194
Specification	Linear			Log	Linear	Log	Linear	Log	Linear	Log

Notes: The table examines the relationship between $\overline{CS}$ and several labor market outcomes. We define labor markets as the combinations of the 27 Brazilian states and 2-digit industry codes. The contract share denotes the share of the total wage bill attributable to contract wages, as defined in Equation (4.4). Columns (1) to (4) present OLS estimates of Equation (4.5). Columns (1) and (2) use the share of negative and null interannual wage changes among stayers, respectively. Columns (3) and (4) use average annual layoff rates, in levels and logs, respectively. Columns (5) to (10) present OLS estimates of Equation (4.7). The outcome $\hat{\beta}_m$ captures market-specific productivity-layoff correlations, estimated via OLS using Equation (4.6). *Specification* notes whether $\hat{\beta}_m$ has been estimated in levels or in logs. Columns (5) and (6) use value added as a measure of productivity. Columns (7) and (8) use firm size, while Columns (9) and (10) use firm pay premiums. For more details on firm productivity measures, see Subsection 4.2.1 and Section C.2 in the Appendix. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the manufacturing sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1. Columns (5) and (6) are further restricted to manufacturing industries.

accumulates across the multiple stages of estimation, we are inclined to think that estimates of Equation (4.7) can be particularly biased towards zero, especially for this specification.

Nonetheless, the magnitudes for the remainder of the coefficients in Table 4.7 are substantial. As an example, estimates in Column (5)—with value added as the firm productivity measure and a linear specification for the annual layoff rate—indicate that $\hat{\beta}_m$ is 34% larger (in absolute terms) than the median among markets in the top 5% of $\overline{CS}$, while 81% smaller among markets in the bottom 5%. Similar patterns emerge for Columns (6) to (8).

In conclusion, this section links wage rigidity to differential layoff patterns across firms and markets. In particular, we show that firms with greater wage rigidity showcase both higher annual layoff rates and a more pronounced productivity-layoff correlation. Taken together, these findings suggest that layoffs partly stem from firms being constrained to adjust wages downward, which empirically validates the mechanisms proposed in our theoretical framework.

Conclusions

In the last three chapters, I have challenged certain economic concepts rooted in the literature. Throughout my empirical and theoretical work, I have deconstructed certain notions about markets, firms, managements, and contracts. It is not about watching the world burn; it is only about reassembling them in light of new empirical findings that have piled up over the last two decades.

This dissertation is both a contribution and an attempt. A contribution to revisit old debates and questions that pertain to what is considered *known*. It is an invitation to wonder *what if*. But my work is also an attempt to answer these same questions to the best of my capabilities, combining economic intuition, theory, and empirics to learn something more about the world we live in.

The first chapter supports the thesis that the minimum wage in Brazil has triggered adjustments beyond wages at the bottom of the distribution. First, the higher dispersion in job amenities between 2016 and 2019 is consistent with firms reducing non-wage compensation in response to higher wage floors. And, second, the significant spillovers at the very top of the hourly wage distribution suggest that firms paid compensating differentials to high earners so to offset the decline in job amenities.

These results contribute to the literature on the minimum wage. On the one hand, it spotlights the role of *job amenities* as an important margin that firms can resort to instead of employment terminations. This channel joins other explanations in the literature—monopsony power and passthrough to prices—capable of bridging the economic intuition that firms adjust to higher wage floors with the empirical findings that show no employment declines. On the other hand, the chapter offers a reasonable explanation of why minimum wage increases in Brazil propagate through the entire wage distribution, but similar patterns do not emerge in other economies. Since Brazilian employers cannot tai-

for job amenities to different employees, wages at the top of the distribution rise due to compensating differentials within the firm.

Nevertheless, it remains unclear which job amenities firms may have cut back on over the decade. Notwithstanding, available evidence points to certain job amenities staying unscathed. First, reductions probably did not extend to non-wage attributes covered in CBAs. As Lagos (2025) explains, unions accumulated substantial bargaining power in 2012, following a TST ruling that automatically extended benefits from overdue CBAs. Second, some perks appear to be costless or byproducts of broader organizational decisions, making them unlikely for firms to cut deliberately. For example, a 2014 survey in Brazil indicates that workers place great value on professional relationships at work, recognition, and flexibility (The Boston Consulting Group, 2016).

The same survey, however, enumerates other job characteristics that workers value and can be costly to provide, e.g., training, work-life balance, career progression, or paid timeoff. While all these job amenities provide firms with a natural margin of adjustment to offset higher wage floors, work-life balance covers broader aspects worth discussing. On the one hand, firms may be making jobs more demanding. Board and Meyer-Ter-Vehn (2015) anticipate in their model that the minimum wage acts as a coordination mechanism, creating a mass of jobs that require more effort in exchange for the higher wage floor. Although the mechanism is absent in my framework—by assuming fixed worker productivity—, my empirical approach captures disamenities related to overwork, such as stress, multitasking, and extra hours.

Along the same lines, jobs may stay equally challenging but shift towards routine and boredom. For instance, 40% of all independent contractors and 29% of all employees surveyed in Brazil report dissatisfaction with their current work situation, with higher figures among short-tenured workers and those reporting low motivation. Moreover, 67% of demotivated workers are actively seeking new positions, indicating their intention to pursue a more suitable job elsewhere (The Boston Consulting Group, 2016).

The second chapter turns to financial markets, contributing to the growing literature on common ownership. While early work has shown that overlapping shareholders can influence managerial decisions to soften competition among rivals (e.g., Azar et al., 2018; Torshizi and Clapp, 2021; Newham et al., 2025), I ask whether the same mechanism operates when firms sit on opposite sides of a supply chain. Recent studies on vertical overlapping owners emphasize the role of institutional shareholders in mitigating contractual and informational

frictions (Freeman, 2025; Deng et al., 2025); however, I find that trading partnerships are more likely when overlapping shareholders can privately benefit from them. That is, managers appointed by companies with diluted ownership become inclined to transfer resources out of the company toward other potential trading partners with more concentrated shareholdings.

My work proposes a novel identification strategy that does not rely on aggregate shifts in the degree of overlapping ownership at the market- or firm-level (Boller and Morton, 2020; Antón et al., 2023), which allows me to instrument for measures at the customer-supplier dyad using predicted rather than actual shareholdings. The strategy focuses on companies and investors with more stable ownership structures and portfolios, based on panel data on shareholdings over a decade, adjusted for additions and deletions from the S&P 500 index.

In the chapter, I present suggestive evidence that managers internalize the effects of their decisions on other companies along the supply chain, which can ultimately hinder the interest of non-overlapping shareholders at the firm that appointed them. This finding reintroduces the classical tension between ownership and control, suggesting that overlapping shareholders do not always align incentives—they can also exploit them.

My findings indicate promising directions for future research. For instance, it remains unclear whether managers engage in tunneling through cash flows, or whether overlapping shareholders pursue long-run goals, including the foreclosure of companies that rival those in their portfolios, deter entry by new entrants, or even gain further control at the firm. Either way, the question of who ultimately bears the costs remains wide open.

Furthermore, it remains to be seen whether the mechanisms identified in the literature apply in cases where trading partners have asymmetric incentives. In particular, channels related to the disclosure of private information (Boone and White, 2015; Pawliczek et al., 2022), say-on-pay (Fichtner et al., 2017; Bubb and Catan, 2022; Antón et al., 2023), and interlocking boards (Azar, 2022) seem more appropriate in softening competition than administering trading partners. Other promising prospects include managers on the payroll of institutional funds (Freeman, 2025), selective omission (Wells, 2016; Hemphill and Marcel, 2019), and, especially, tying the compensation of executives to ESG-based outcomes (Bebchuk and Tallarita, 2022; Cohen et al., 2023), which can restrict customers or suppliers to companies that meet very specific characteristics.

The third chapter investigates the determinants of one of the most well-documented patterns in labor economics: the negative correlation between productivity and separation

(Topel and Ward, 1992; Davis et al., 2013; Haltiwanger et al., 2018*a,b*). By leveraging high-quality administrative records in Brazil, we decompose separations by who terminated the employment relationship. In this way, we uncover that layoffs, rather than quits, explain most of the negative relationship.

Contrary to other canonical models that attribute this pattern solely to efficient quits, we develop a job search model in which wage rigidity and productivity uncertainty suffice to replicate the empirical patterns in Brazilian labor markets. The model reveals two key aspects of the mechanism. First, more productive firms set higher wages but also larger markdowns, which act as a buffer against negative productivity shocks. As a result, less productive firms, with narrower markdowns, tend to resort to layoffs more often. Second, wage rigidity prevents firms from renegotiating wages downward, even when workers prefer that to being laid off, implying that a significant fraction of the layoffs are inefficient.

We conclude the chapter by returning to the data and validating the mechanisms proposed in the model: higher-productivity firms exhibit larger markdowns, layoffs decline with markdowns, and wage rigidity strengthens the productivity-layoff correlation. Furthermore, we show that worker heterogeneity cannot fully account for any of the empirical findings.

Our findings open the door to further research, as cross-country evidence indicates the importance of distinguishing between layoffs and quits. European labor markets exhibit lower layoff rates than Brazil but comparable quit rates. Conversely, the US displays comparably higher quit rates but similar levels of layoffs. The immediate economic intuition suggests that stricter employment protection legislation in Europe and more dynamic labor markets in the US can accommodate the differences observed.

Nevertheless, our framework emphasizes the role of labor market conditions and productivity volatility. Labor markets with a more elastic labor supply and greater productivity uncertainty would lead the economy towards relatively higher wages and layoff rates. Contrarily, labor markets with significant monopsony power—more inelastic labor supply—but more stable productivity would yield opposite patterns. Taken together, these insights suggest that Brazilian, European, and American labor markets may differ around these dimensions, calling for further research on how economic conditions shape productivity, layoffs, and quits across countries.

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Appendix to Chapter 2

A.1 The framework

The framework consists of a static, random, posting, partial equilibrium model. Random means that matches happen by chance. Posting means that there is no bargaining. Partial equilibrium and posting mean that I take firm and worker heterogeneity as given. Static and random mean that firms cannot substitute labor but can exit to avoid losses. Firms design contracts including wages and job amenities, subject to binding minimum wages. Workers can accept or reject the offer. The following section describes the model's setting and assumptions, characterizes separate solutions depending on whether firms can tailor job amenities to each worker, and develops empirical predictions in the case that firms set firm-level job amenities.

A.1.1 The model

Setting. Consider an economy with J firms and $2J$ workers. Each firm employs a team of two workers, with individuals being heterogeneous in both productivity α and outside option u . Firms differ in productivity solely because of their internal organization, implying that better-run firms are matched with more productive teams.

For convenience, I index agents so that firm $j \in \mathcal{J}$ matches with workers j and $2j$, and relabel them so that $j = L(j)$ and $2j = H(j)$, with $\{L(j), H(j)\} \subset \mathcal{I}$. I impose that $\mathcal{I}$ is a partially ordered set, with $\alpha_{H(j)} > \alpha_{L(j)} > 0$ and $u_{H(j)} > u_{L(j)} > 0$, but without an established relationship across firms. Therefore, $L(j)$ and $H(j)$ designate low- and high-productivity workers, respectively, within the firm only.

At the start of the period, each firm j matches with its corresponding team, with workers indexed by $i \in \{L(j), H(j)\}$. Immediately thereafter, the firm posts worker-specific contracts $\langle w_i, a_i \rangle$, consisting of wages w_i and amenities a_i . Workers share preferences over these attributes, attaining a utility equal to

$$U_i = w_i^\gamma a_i$$

where $\gamma > 0$ symbolizes the intensity of preference for wages.

If both workers accept the offer, the firm generates revenues equal to $f(\alpha_{L(j)}, \alpha_{H(j)})$, pays wages, and provides amenities at a firm-specific unitary cost $p_a(j)$. So, the firm earns profits equal to

$$\Pi_j = f(\alpha_{L(j)}, \alpha_{H(j)}) - (w_{H(j)} + w_{L(j)}) - p_a(j)(a_{H(j)} + a_{L(j)})$$

Since revenues depend on workers' joint productivity, $f(0, \alpha_{H(j)}) = f(\alpha_{L(j)}, 0) = 0$. Therefore, whenever a worker rejects the offer, the firm exits and both workers receive their respective outside options, u_i .

The environment, as it is, admits a feasible, closed-form equilibrium. I introduce four additional assumptions that further discipline the framework's empirical predictions around the wage-amenities tradeoff.

Assumption A.1. *Worker productivity α_i is fixed.*

The assumption shuts down the productivity channel. That is, firms accommodate higher wage floors by exhorting greater effort from workers who earn the statutory minimum (Board and Meyer-Ter-Vehn, 2015; Wang, 2025b). On paper, it keeps amenities one-dimensional; allowing for endogenous productivity would introduce another dimension related to effort costs. At the same time, it seals similar margins for adjustment, such as pass-throughs to prices (Allegretto and Reich, 2018; Harasztosi and Lindner, 2019; Renkin et al., 2022).

Assumption A.2. *Firms earn strictly positive profits, even when paying the minimum wage to all workers.*

Wage floors raise total labor costs, incentivizing firms in the framework to exit the market in response. The assumption rules out exit by ensuring that firms can always accommodate higher wages; a reasonable restriction given prior evidence of muted employment

declines. Recent empirical evidence from the United States and Germany points to monopsony power, with firms adjusting markdowns rather than employment (Azar et al., 2024; Berger et al., 2025; Faia et al., 2026). In the context of Brazil, Dobbin et al. (2025) report significant markdowns among large, more productive firms, which provides some support for the assumption.

Before moving on to the next two assumptions, I classify firms into three categories based on their initial exposure to the binding minimum wage, regardless of their average wage level.

Definition (Firm categories). *I define a high-paying employer $\mathcal{H}$ as the firm that pays wages strictly above the minimum wage to all its employees. A medium-paying employer, instead, is the firm in which only the low-productivity worker earns the statutory minimum. In addition, I split medium-paying firms into two distinct categories. On the one hand, specialized employers $\mathcal{S}$ exhibit lower workplace heterogeneity, defined by the gap between u_H and u_L . On the other hand, vertical employers $\mathcal{V}$ employ a more heterogeneous workforce. Finally, I define a low-paying employer $\mathcal{L}$ as the firm that pays the minimum wage to all its employees. In addition, I define $\pi_{\mathcal{X}}$ as the share of firms in a given category $\mathcal{X}$, where $\mathcal{X} \in \{\mathcal{H}, \mathcal{S}, \mathcal{V}, \mathcal{L}\}$. By construction, $\mathcal{L} \cup \mathcal{S} \cup \mathcal{V} \cup \mathcal{H} = \mathcal{J}$, and $\pi_{\mathcal{H}} + \pi_{\mathcal{S}} + \pi_{\mathcal{V}} + \pi_{\mathcal{L}} = 1$.*

Assumption A.3. *The joint distribution of the unitary price of amenities p_a and workers' outside options u_L and u_H is atomless.*

In other words, no two firms simultaneously share the same firm-specific unitary price of job amenities and the same workers' outside options. Under the partial equilibrium assumption, these variables are taken as given and fully capture heterogeneity across firms.

Although optimal contracts ultimately depend on the firm's category, this classification is based solely on these three exogenous characteristics. As a result, there are no incentives for bunching. That is, no endogenous choice in the model can generate mass points at any particular combination of these parameters. In any case, the assumption works as a convenience device, ensuring that firms located at the margin between categories cannot disproportionately drive the results.

Assumption A.4. *Workplace heterogeneity at specialized firms $\mathcal{S}$ —defined as the outside options ratio between high-paid and low-paid workers—is bounded by 2^γ .*

$$\frac{u_{H(s)}}{u_{L(s)}} < 2^\gamma \quad \forall s \in \mathcal{S}$$

The assumption places an upper bound on workplace heterogeneity, ultimately disciplining within-firm wage differentials at specialized firms. In the empirical analysis, it serves as a sufficient but not necessary condition and therefore can be relaxed. Nonetheless, I return to it in Subsection A.1.4, where I derive a prediction that matches the empirical observation that wage inequality decreases following increases in the minimum wage.

Assumption A.5. *The outside options of high-paid workers employed at low-paying firms and of low-paid workers employed at specialized firms exhibit a sufficiently low dispersion. More specifically, I assume that*

$$\pi_{\mathcal{L}} \mathbb{V}_{\mathcal{L}}[u_H] + \pi_{\mathcal{S}} \mathbb{V}_{\mathcal{S}}[u_L]$$

is relatively small.

Intuitively, the assumption imposes that workers earning the minimum wage across firms in the same category do not exhibit large differences. As before, the condition acts as a sufficient but not necessary condition in the empirical analysis. Consequently, I discuss it further in Section A.1.4, when I derive a prediction about job amenities dispersion following increases in the minimum wage.

Worker reallocations. Engbom and Moser (2022) offer an alternative interpretation of the limited employment effects in Brazil: workers reallocate to more productive firms. Although a static framework cannot capture these dynamics, a small tweak to the model can align its interpretation with the author’s findings. Let the incumbent firm exit before contracts are posted, and let an entrant firm replace it and post its own offers. Moreover, let the entrant face a $p_a(j)$ sufficiently low to break even. Since optimal wages and job amenities for constrained firms do not depend on $p_a(j)$, both versions are isomorphic. More importantly, empirical predictions stay the same as long as entrant and exiting firms belong to the same firm category, which holds by construction.

Single firm's problem. For convenience, I drop the firm index j from wages w_i , job amenities a_i , outside options u_i , and the unitary cost p_a when discussing the single firm's problem. Given Assumption (A.1), optimal wages and job amenities arise from the firm's cost minimization problem, subject to each worker's participation and minimum-wage constraints.

$$\begin{aligned} \min_{\{(w_i, a_i)\}_{i=L}^H} \quad & (w_L + w_H) + p_a(a_L + a_H) \\ \text{s.t.} \quad & w_i^\gamma a_i \geq u_i, \quad i \in \{L, H\} \quad (PC) \\ & w_i \geq \underline{w}, \quad i \in \{L, H\} \quad (MW) \end{aligned}$$

I can solve the problem using the Lagrangian approach. Let λ_i denote the multiplier on (PC_i) and μ_i the multiplier on (MW_i) , so that

$$\begin{aligned} \mathcal{L} = & (w_L + w_H) + p_a(a_L + a_H) \\ & + \lambda_L(u_L - w_L^\gamma a_L) + \mu_L(\underline{w} - w_L) \\ & + \lambda_H(u_H - w_H^\gamma a_H) + \mu_H(\underline{w} - w_H) \end{aligned}$$

with the following first-order conditions

$$\begin{aligned} 1 - \lambda_H \gamma w_H^{\gamma-1} a_H - \mu_H &= 0 & (w_H) & \quad 1 - \lambda_L \gamma w_L^{\gamma-1} a_L - \mu_L = 0 & (w_L) \\ p_a - \lambda_H w_H^\gamma &= 0 & (a_H) & \quad p_a - \lambda_L w_L^\gamma = 0 & (a_L) \\ \lambda_H (u_H - w_H^\gamma a_H) &= 0 & (\lambda_H) & \quad \lambda_L (u_L - w_L^\gamma a_L) = 0 & (\lambda_L) \\ \lambda_H &\geq 0 & & \quad \lambda_L \geq 0 & \\ u_H - w_H^\gamma a_H &\leq 0 & & \quad u_L - w_L^\gamma a_L \leq 0 & \\ \mu_H (\underline{w} - w_H) &= 0 & (\mu_H) & \quad \mu_L (\underline{w} - w_L) = 0 & (\mu_L) \\ \mu_H &\geq 0 & & \quad \mu_L \geq 0 & \\ \underline{w} - w_H &\leq 0 & & \quad \underline{w} - w_L \leq 0 & \end{aligned}$$

In the following subsections, I derive the solutions for two separate situations. First, when firms can tailor job amenities specifically to each worker, $a_L \neq a_H$. Second, when firms must commit to common levels of job amenities, $a_L = a_H = a$.

A.1.2 Flexible amenities

With worker-specific amenities, each contract can be solved separately. Equation (a_i) determines that $\lambda_i^* = p_a w_i^{-\gamma} > 0$, which implies that participation constraints are always binding and

$$a_i^* = \frac{u_i}{w_i^{*\gamma}}$$

Replacing λ_i^* into equation (w_i) determines that $\mu_i^* = 1 - p_a \gamma a_i w_i^{-1} \geq 0$, so optimal contracts depend on whether the minimum wage condition binds for the worker.

Definition ($\underline{u}$). Let $\underline{u} \equiv \underline{w}^{1+\gamma} (\gamma p_a)^{-1}$ denote the threshold that determines whether the firm faces a binding (MW_i).

The threshold $\underline{u}$ maps a positive μ_i^* onto a condition based on the relationship between the worker's outside option and the minimum wage. Namely, employers pay the minimum wage to workers with $u_i \leq \underline{u}$.

Notice that $\underline{u}$ is a firm-specific threshold, emphasizing that firms with access to cheaper job amenities (lower p_a) pay lower wages and are, everything else equal, more exposed to changes in the minimum wage.¹

Lemma A.1. When $u_i \geq \underline{u}$, optimal contracts offer workers

$$w^* = u_i^{\frac{1}{1+\gamma}} (p_a \gamma)^{\frac{1}{1+\gamma}}$$

and job amenities

$$a^* = u_i^{\frac{1}{1+\gamma}} (p_a \gamma)^{\frac{-\gamma}{1+\gamma}}$$

Proof. Combining $\mu_i^* = 0$ and a_i^* suffice to determine optimal contracts. □

Comparative statics portray intuitive results. Both w_i and a_i are increasing and concave in u_i , consistent with empirical work showing a positive correlation between wages and job amenities (Sockin, 2022; Maestas et al., 2023). On the contrary, w_i is increasing and

¹. The claim may appear counterintuitive at first glance since more productive firms plausibly face lower costs of providing amenities and pay the minimum wage to a lesser fraction of workers. This intuitive reasoning, however, overlooks two key aspects. First, that more productive firms match more often with more productive workers, who have higher outside options. Second, the opportunity cost of relaxing working conditions declines with productivity (Board and Meyer-Ter-Vehn, 2015; Wang, 2025b). Consistent with the proposed mechanism, Humlum et al. (2025) find that high-paying firms offer jobs with better perks and employee benefits. However, these jobs also entail greater stress, turnover, and related workplace disadvantages.

concave in p_a , while a_i is decreasing and convex, reflecting how firms substitute the relatively expensive form of compensation for the cheapest one.

Lemma A.2. *When $u_i < \underline{u}$, optimal contracts offer the worker the minimum wage $\underline{w}$ and job amenities $a_i^* = u_i \underline{w}^{-\gamma}$.*

Proof. The expression for a_i^* and $w_i^* = \underline{w}$ immediately reveal the optimal solution. $\square$

Thus, a_i is decreasing and convex in $\underline{w}$, implying that higher wage floors lead to greater reductions in job amenities. Though a_i is still increasing in u_i , the relationship is no longer concave but linear, indicating that firms reduce job amenities among low-productivity workers more aggressively in the presence of wage floors. Most notably, a_i no longer depends on p_a , since job amenities for constrained workers become pinned down by (PC_i) .

Labor costs. Assumption A.2 eliminates the need to impose a firm participation constraint. Even so, the framework captures how amenities can attenuate layoffs following minimum wage increases. I take the difference in amenity expenditures across the two cases

$$\Delta LC = p_a u_i \left((p_a u_i \gamma)^{\frac{-\gamma}{1+\gamma}} - \underline{w}^{-\gamma} \right)$$

so that ΔLC denotes labor cost savings relative to a setting with fixed amenities.

The result shows that the mechanism strengthens as wage floors rise, since ΔLC increases with $\underline{w}$. Moreover, ΔLC is non-monotonic in p_a and u_i . Savings increase for $u_i < \underline{u}(1 + \gamma)^{-\frac{1+\gamma}{\gamma}}$ and decrease thereafter. Therefore, amenity-driven savings concentrate among low-productivity workers, especially when the unit cost of amenities is higher.

Thus far, the setting with flexible amenities has helped to clarify the mechanisms and the intuition behind the framework. The analysis in the next subsection considers a setting empirically closer to Brazil, in which firms cannot differentiate amenities across workers.

Before proceeding, note that wages increase in u_i and, by construction, in α_i . In consequence, I henceforth refer to low- and high-paid workers rather than low- and high-productivity workers, which better aligns with the intuition behind the empirical predictions.

A.1.3 Firm-level amenities

With firm-specific amenities, equations (a_L) and (a_H) collapse into a unique first order condition

$$2p_a - \lambda_L w_L^\gamma - \lambda_H w_H^\gamma = 0 \quad (a)$$

Unlike the previous case, contracts must be solved jointly, leading to four solutions of the program. Either way, the participation constraint for the high-paid worker remains always binding.

Lemma A.3. *(PC_H) is always binding.*

Proof. Let (PC_H) be slack and replace $\lambda_H = 0$ in equations (w_h) and (a) . The first one yields $\mu_H^* = 1$, which immediately implies that $a^* > u_H \underline{w}^{-\gamma}$. The second one determines that λ_L is positive, so $u_L = w_L^\gamma a^*$. Given that $w_L \geq \underline{w}$ and $u_H > u_L$ by construction, combining both statements leads to the following contradiction

$$u_L > \left(\frac{w_L}{\underline{w}} \right)^\gamma u_H$$

so that (PC_H) has to be binding. □

The firm selects the optimal contract from four characterizations based on its exposure to the wage floor, which, in turn, depends on workers' outside options and the unit cost of amenities. For notational convenience, let me define $P_a \equiv (2p_a \gamma)^{\frac{1}{1+\gamma}}$, which denotes a renormalization of the firm's amenity price.

I map each characterization to a distinct firm category to provide better intuition into the firm's choices. Each employer then faces three firm-specific thresholds relative to the minimum wage that identify the optimal contract. As the minimum wage increases and surpasses each threshold, the bindingness of the wage floor strengthens, and the firm adjusts its response to further increases.

The first threshold, ω_1 , classifies the firm as either a high-paying or a specialized employer. In the former case, the firm pays wages above the statutory minimum and is thus unaffected by further marginal increases. In the latter, instead, the firm employs workers with small differences in their outside options but pays the minimum wage to the low-paid employee. By having a less heterogeneous workforce, the specialized firm prefers to pass

along the minimum wage increase to the high-paid worker, even though she earns above the statutory minimum.

Definition (ω_1). *Let*

$$\omega_1 \equiv P_a u_L^{\frac{1}{\gamma}} \left(u_L^{\frac{1}{\gamma}} + u_H^{\frac{1}{\gamma}} \right)^{-\frac{1}{1+\gamma}}$$

denote the threshold that separates high-paying $\mathcal{H}$ from specialized firms $\mathcal{S}$.

Three things worth mentioning about this threshold. First, P_a relaxes the restriction by making amenities relatively more expensive, thereby pushing the firm to offer higher wages to everyone in the first place. Second, a more heterogeneous workforce tightens the restriction by pushing down the wage of the low-paid worker. Third, a higher outside option for the low-paid worker softens the constraint. Naturally, a higher u_L , by construction, implies that the firm faces a more homogeneous worker composition, though a simultaneous increase in both outside options would keep wages away from the statutory minimum.

The second threshold, ω_2 , distinguishes a medium-paying firm as having either a specialized or a vertical structure. The difference between the two is that a vertical employer has a more heterogeneous workforce, i.e., workers exhibit a higher gap in outside options. As a result, the firm prefers to absorb the minimum wage increase, and the low-paid worker pockets a differential.

Definition (ω_2). *Let*

$$\omega_2 \equiv P_a u_L^{\frac{1}{\gamma}} \left(u_H^{\frac{1}{\gamma}} \right)^{-\frac{1}{1+\gamma}}$$

denote the threshold that separates specialized $\mathcal{S}$ from vertical firms $\mathcal{V}$.

As before, P_a increases the threshold by raising the cost of amenities relative to wages. Workforce heterogeneity also tightens the restriction. In particular, because compensating differentials are costly in this scenario, so that it is more profitable for the firm to absorb the wage floor that affects the low-paid worker.

The last threshold, ω_3 , determines whether the firm falls under the definition of a vertical or low-paying employer, where the latter pays the minimum wage to both workers.

Definition (ω_3). *Let*

$$\omega_3 \equiv P_a u_H^{\frac{1}{1+\gamma}}$$

denote the threshold that separates vertical $\mathcal{V}$ from low-paying firms $\mathcal{L}$.

Two things worth mentioning here. First, ω_3 does not depend on u_L . By construction, the firm cannot pay a lower wage to the high-paid worker, so this condition suffices. Second, as workforce heterogeneity falls, $\omega_2 \rightarrow \omega_3$. Therefore, the bindingness of the wage floor depends solely on the unit cost of job amenities and the outside option of the matched team.

Proposition A.1 (High-paying firms). *When $\underline{w} < \omega_1$, the optimal contract offers workers firm-specific job amenities equal to*

$$a^* = P_a^{-\gamma} \left(u_L^\gamma + u_H^\gamma \right)^{\frac{\gamma}{1+\gamma}}$$

and worker-specific wages equal to

$$w_i^* = P_a u_i^{\frac{1}{\gamma}} \left(u_L^\gamma + u_H^\gamma \right)^{-\frac{1}{1+\gamma}}$$

Proof. By definition, high-paying firms have $\mu_L = \mu_H = 0$. Replacing these values in equations (w_H) and (w_L) shows that λ_L and λ_H are positive, further implying that (PC_L) also binds.

Therefore, I can isolate w_i on each (PC_i) to replace these expressions in λ_i , which results in the following expression

$$\lambda_i = u_i^{\frac{1-\gamma}{\gamma}} \left(\gamma a^{\frac{1}{\gamma}} \right)^{-1}$$

that I can replace in equation (a).

Isolating a yields the optimal level of job amenities; thus, I can obtain optimal worker-specific wages w_i^* by replacing a^* in each (PC_i) .

Finally, I need to check that the threshold's condition holds. Notice that the expression for ω_1 coincides with w_L , so by construction, $w_H > w_L = \omega_1 > \underline{w}$. $\square$

The comparative statics reveal interesting interplays in compensation. As with Lemma (A.1), a is decreasing and convex in p_a , whereas wages are increasing and concave. Job amenities increase with both u_L and u_H , suggesting that the firm sets a to target an average level of compensation and uses wages to satisfy participation constraints. In fact, wages w_i are increasing and concave in u_i , but decrease with u_{-i} . The condition $\gamma \geq 1$, which implies that workers do not have stronger preferences for amenities, is sufficient for job amenities to be concave in outside options and for wages to be convex in the coworker's outside option. Otherwise, the sign of the second derivatives are indeterminate.

Proposition A.2 (Medium-paying firms). *When $\omega_1 \leq \underline{w} < \omega_3$, the firm sets the low-paid worker's wage at the statutory minimum. If $\underline{w} < \omega_2$, the firm serves as a specialized employer and offers a compensating differential to the high earner. Then, the wage of the high-paid worker amounts to*

$$w_H^* = \underline{w} \left(\frac{u_H}{u_L} \right)^{1/\gamma}$$

and the optimal contract offers firm-specific job amenities equal to

$$a^* = u_L \underline{w}^{-\gamma}$$

On the contrary, when $\omega_2 \leq \underline{w}$, the firm falls under the definition of a vertical employer. Therefore, the optimal contract offers job amenities equal to

$$a^* = P_a^{-\gamma} u_H^{\frac{1}{1+\gamma}}$$

and the following wage to the high-paid worker

$$w_H^* = P_a u_H^{\frac{1}{1+\gamma}}$$

Proof. By definition, medium-paying firms have $\mu_H = 0$ and $\mu_L > 0$. By using the same expression for λ_H as in Proposition (A.1) and substituting it into equation (a), I can obtain the following restriction for λ_L

$$\lambda_L = \frac{P_a^{1+\gamma} - u_H^{\frac{1}{\gamma}} a^{-\frac{1+\gamma}{\gamma}}}{\gamma \underline{w}^\gamma} \geq 0$$

Therefore, there are two potential characterizations for optimal contracts. Let me assume that λ_L is positive. Then, it immediately follows that (PC_L) is binding, which immediately pins down $a^* = \underline{w}^{-\gamma} u_L$. Replacing the optimal level of job amenities in (PC_H) thereby yields the optimal wage for the high-paid worker, completing the characterization of the specialized firm's optimal contract.

Next, I assume that $\lambda_L = 0$, which immediately reveals the optimal level of job amenities for the vertical firm. In turn, I can replace a^* in (PC_H) and obtain w_H^* .

Lastly, I need to check that all thresholds' conditions hold. First, $\omega_1 \leq \underline{w}$ trivially follows from $\mu_L > 0$. Second, a specialized firm pays a w_H^* that lies strictly above the minimum wage and, in the case of a vertical employer, the high-paid worker's wage coincides with ω_3 .

Therefore, $\underline{w} < \omega_3$ holds by construction. Third, $\lambda_L = 0$ implies that

$$U_L > \underline{w}^\gamma P_a^{-\gamma} u_H^{\frac{1}{1+\gamma}} > u_L \implies \underline{w} > P_a u_L^{\frac{1}{\gamma}} \left(u_H^{\frac{1}{\gamma}} \right)^{-\frac{1}{1+\gamma}} = \omega_2$$

Therefore, the specialized firm's characterization of the optimal contract holds when $\underline{w} \leq \omega_2$, while the characterization for the vertical employer holds otherwise. $\square$

A specialized firm no longer uses job amenities to target an average level of compensation, but to satisfy the participation constraint of the low-paid worker. As a result, a increases linearly only in u_L and decreases convexly in $\underline{w}$. Regarding the wage of the high-paid worker, it increases linearly in $\underline{w}$ and, when $\gamma > 1$, it increases concavely in u_H , while decreasing convexly in u_L . Otherwise, the signs of the second derivatives become indeterminate. In conclusion, setting a lower a due to the wage floor pushes the firm to offer a compensating differential to the high-paid worker, which explains the upward spillovers in the wage distribution.

However, a vertical employer acts differently. The firm sets job amenities to satisfy the participation constraint of the high-paid worker, allowing the lower earner to keep a wage differential. As a result, the optimal contract offered to the high-paid worker no longer depends on u_L . Comparative statics reveal that a is decreasing and convex in p_a while increasing and concave in u_H . w_H , on the other hand, is increasing and concave in both p_a and u_H .

These results have two implications. Spillovers arise and grow with workplace heterogeneity, defined as the gap between u_H and u_L . However, too much heterogeneity dissipates the effect, since the firm weighs compensating differentials as too costly and thus prefers to directly absorb the minimum wage increase.

Proposition A.3 (Low-paying firms). *When $\omega_3 \leq \underline{w}$, the optimal contract offers both workers the minimum wage $\underline{w}$ and firm-specific job amenities equal to*

$$a = u_H \underline{w}^{-\gamma}$$

Proof. By definition, a low-paying firm has $\mu_H > 0$ and $\mu_L > 0$. Therefore, (PC_H) pins down a^* . Furthermore, the threshold's condition holds from Proposition (A.2). $\square$

Compared to a specialized firm, a increases linearly in u_H but still decreases convexly in $\underline{w}$. Compared to the vertical employer, the firm still uses job amenities to address the participation constraint of the high-paid worker. Thus, (PC_L) remains slack, and the low-paid worker pockets the full wage differential, $u_H - u_L$.

A.1.4 Empirical predictions

The analysis henceforth imposes firm-specific job amenities and uses results in Propositions (A.1) to (A.3). The subsection illustrates that empirical predictions for an increase in the minimum wage may not be trivial when firms cannot tailor job amenities to workers. In particular, wage inequality can increase, and disparities in job amenities can fall. The underlying reason connects to the third and most important prediction of the model: some firms may pass along the minimum wage increase to higher earners. Next, I present testable implications of the model and show sufficient conditions under which the predictions align directly with the empirical patterns observed in the Brazilian context.

Lemma A.4. *Let $\pi_{\mathcal{X}}(\underline{w})$ denote the share of firms in category $\mathcal{X}$, with $\mathcal{X} \in \{\mathcal{L}, \mathcal{V}, \mathcal{S}, \mathcal{H}\}$. Then, changes in the share of firms in each category following a minimum wage increase are negligible. That is,*

$$\frac{\partial \pi_{\mathcal{X}}(\underline{w})}{\partial \underline{w}} \approx 0$$

Proof. Without loss of generality, consider the share of specialized firms $\pi_{\mathcal{S}}(\underline{w})$. Proposition (A.2) shows that the share depends on the thresholds ω_1 and ω_2 , such that

$$\pi_{\mathcal{S}}(\underline{w}) = F_2(\underline{w}) - F_1(\underline{w})$$

where F_1 and F_2 represent the cumulative distribution functions of $\omega_1(p_a, u_L, u_H)$ and $\omega_2(p_a, u_L, u_H)$ across firms, respectively.

By taking derivatives with respect to the minimum wage $\underline{w}$, I obtain the following expression

$$\frac{\partial \pi_{\mathcal{S}}(\underline{w})}{\partial \underline{w}} = f_2(\underline{w}) - f_1(\underline{w}) \approx 0$$

where f_1 and f_2 correspond to the probability density functions, and are negligible at any point due to Assumption (A.3). $\square$

Lemma (A.4) serves as a convenience device to eliminate the need to introduce further assumptions about the distribution of firms across categories. Since the thresholds defined in Subsection A.1.3 do not depend on endogenous choices, the assumption that firms do not bunch around them, nor any other point in the distribution, appears reasonable.

In any case, none of the analytical results that follow depend on the exact magnitudes of any $\pi_{\mathcal{X}}$. Introducing small perturbations to the shares or, equivalently, small buffers to include thresholds, leaves all results unchanged, further reinforcing that neglecting changes in the compositions across categories is inconsequential.

Lemma A.5. *Let $\mathbb{E}[w] \geq 2\underline{w}$. Then,*

$$\mathbb{V}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right] + \mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right] \left(\mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right] - \frac{\mathbb{E}[w]}{\underline{w}} \right) \leq 0$$

Proof. Define

$$A \equiv \left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \quad \text{and} \quad B \equiv P_a u_L^{\frac{1}{1+\gamma}} \underline{w}^{-1}$$

Proposition (A.2) states that specialized firms $\mathcal{S}$ satisfy $\underline{w} \in [\omega_1, \omega_2]$. By rearranging terms in ω_1 and ω_2 , I can define the following interval for A

$$A \in [L(A), U(A)]$$

where $L(A) = B A (1 + A)^{-\frac{1}{1+\gamma}}$ and $U(A) = B A^{\frac{\gamma}{1+\gamma}}$. Next, let me define

$$T_{\mathcal{S}}(A) = \mathbb{V}_{\mathcal{S}}[A] + \mathbb{E}_{\mathcal{S}}[A] \left(\mathbb{E}_{\mathcal{S}}[A] - \frac{\mathbb{E}[w]}{\underline{w}} \right)$$

and

$$\hat{A} \equiv \underset{a \in [L(A), U(A)]}{\operatorname{argmax}} T_{\mathcal{S}}(a)$$

By construction, I can set the following inequality

$$\begin{aligned} T_{\mathcal{S}}(A) &\leq \mathbb{V}_{\mathcal{S}}[\hat{A}] + \mathbb{E}_{\mathcal{S}}[\hat{A}] \left(\mathbb{E}_{\mathcal{S}}[\hat{A}] - \frac{\mathbb{E}[w]}{\underline{w}} \right) \\ &\leq \mathbb{E}_{\mathcal{S}}[\hat{A}^2] - \mathbb{E}_{\mathcal{S}}[\hat{A}]^2 + \mathbb{E}_{\mathcal{S}}[\hat{A}]^2 - \mathbb{E}_{\mathcal{S}}[\hat{A}] \frac{\mathbb{E}[w]}{\underline{w}} \\ &\leq \mathbb{E}_{\mathcal{S}}[\hat{A}] \underbrace{\left(\mathbb{E}_{\mathcal{S}}[\hat{A}] - \frac{\mathbb{E}[w]}{\underline{w}} \right)}_{\text{negative}} \end{aligned}$$

which yields that T_S is negative under Assumption (A.4) and that the average wage is at least twice the minimum wage.

$$E_S \left[\hat{A} \right]^\gamma < 2^\gamma \quad \text{and} \quad \frac{\mathbb{E}[w]}{\underline{w}} \geq 2$$

□

Lemma (A.5) follows from the combination of $\mathbb{E}[w] \geq 2\underline{w}$ and Assumption (A.4). The former condition stems from the observed ratio of the average to the minimum hourly wage in Brazil. In particular, the ratio ranged from 2.05 (in 2016) to 2.11 (in 2014), which backs this condition as suitable.

Assumption (A.4), on the other hand, is a sufficient condition that I impose to prevent the minimum wage from skewing the wage distribution to the right. In the framework, workplace heterogeneity maps one-to-one with within-firm wage differentials among specialized firms, so both interpretations of the assumption are interchangeable. Intuitively, Proposition (A.2) shows that wages in specialized firms increase linearly in $\underline{w}$, though at different rates. The assumption places an upper bound on the growth rate of higher earners among these types of firms, so it cannot exceed the ratio of the average to the minimum hourly wage. Otherwise, wage compression could occur solely at the upper tail, leaving a wider lower-tail wage dispersion.

Proposition A.4 (Wage compression). *Wage dispersion $\mathbb{V}[w]$ is not monotonic in $\underline{w}$. Assumption (A.4) is a sufficient condition for wage dispersion to decrease with $\underline{w}$.*

Proof. Using Propositions (A.1) to (A.3), I can express the first and second moments of w as

$$\mathbb{E}[w] = \pi_{\mathcal{L}} \underline{w} + 1/2 \pi_{\mathcal{V}} (\underline{w} + \mathbb{E}_{\mathcal{V}}[w_H]) + 1/2 \pi_{\mathcal{S}} \underline{w} \left(1 + \mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right] \right) + \pi_{\mathcal{H}} \mathbb{E}_{\mathcal{H}}[w]$$

and

$$\begin{aligned} \mathbb{E}[w^2] = \pi_{\mathcal{L}} \underline{w}^2 + 1/2 \pi_{\mathcal{V}} (\underline{w}^2 + \mathbb{E}_{\mathcal{V}}[w_H^2]) + 1/2 \pi_{\mathcal{S}} \underline{w}^2 \left(1 + \mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{2}{\gamma}} \right] \right) \\ + \pi_{\mathcal{H}} \mathbb{E}_{\mathcal{H}}[w^2] \end{aligned}$$

where $\mathbb{E}_{\mathcal{X}}[\cdot]$ highlights heterogeneities in average wages and workplace composition within

firms in category $\mathcal{X}$. Next, I can take first derivatives of the previous expressions with respect to $\underline{w}$ and get

$$\frac{\partial \mathbb{E}[w]}{\partial \underline{w}} = 1/2 (2\pi_{\mathcal{L}} + \pi_{\mathcal{V}} + \pi_{\mathcal{S}}) + 1/2 \pi_{\mathcal{S}} \mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right]$$

and

$$\frac{\partial \mathbb{E}[w^2]}{\partial \underline{w}} = \underline{w} (2\pi_{\mathcal{L}} + \pi_{\mathcal{V}} + \pi_{\mathcal{S}}) + \underline{w} \pi_{\mathcal{S}} \mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{2}{\gamma}} \right]$$

where I employ Lemma (A.4) to ignore the derivatives of the share of firms across categories with respect to the minimum wage. Next, let me write the second centered moment of w as

$$\mathbb{V}[w] = \mathbb{E}[w^2] - \mathbb{E}[w]^2$$

and take derivatives with respect to $\underline{w}$. Thus,

$$\begin{aligned} \frac{\partial \mathbb{V}[w]}{\partial \underline{w}} &= \frac{\partial \mathbb{E}[w^2]}{\partial \underline{w}} - 2\mathbb{E}[w] \frac{\partial \mathbb{E}[w]}{\partial \underline{w}} \\ &= \left(\underline{w} - \mathbb{E}[w] \right) (2\pi_{\mathcal{L}} + \pi_{\mathcal{V}} + \pi_{\mathcal{S}}) \\ &\quad + \pi_{\mathcal{S}} \left(\underline{w} \mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{2}{\gamma}} \right] - \mathbb{E}[w] \mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right] \right) \end{aligned}$$

By adding and subtracting

$$\underline{w} \pi_{\mathcal{S}} \mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right]^2$$

and rearranging the second term, I obtain

$$\begin{aligned} \frac{\partial \mathbb{V}[w]}{\partial \underline{w}} &= \underbrace{\left(\underline{w} - \mathbb{E}[w] \right)}_{\text{negative}} (2\pi_{\mathcal{L}} + \pi_{\mathcal{V}} + \pi_{\mathcal{S}}) \\ &\quad + \pi_{\mathcal{S}} \underline{w} \underbrace{\left(\mathbb{V}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right] + \mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right] \left(\mathbb{E}_{\mathcal{S}} \left[\left(\frac{u_H}{u_L} \right)^{\frac{1}{\gamma}} \right] - \frac{\mathbb{E}[w]}{\underline{w}} \right) \right)}_{\text{Lemma (A.5)}} \end{aligned}$$

which is strictly negative by Lemma (A.5). □

The second term in Proposition (A.4) illustrates how spillovers across the wage distribution make the prediction about wage inequality less obvious. Once $\underline{w}$ approaches the average wage, increases in the federal minimum wage can increase the skewness of the wage distribution.

Notice that Lemma (A.5) undoes a step in the algebra of Proposition (2.2). However, the second term in the latter sheds light on the counter-effects of increasing the minimum wage when there are significant spillovers. On the one hand, the wages of high-paid workers grow faster than the minimum wage, so higher wage floors increase wage dispersion within specialized firms. On the other hand, when spillovers affect earners above the average wage, wage disparities across categories can increase significantly, leading to a wider lower-tail wage dispersion.

Assumption (A.4) serves as a sufficient condition for the model to match the decline in wage inequality over time in Brazil. So the faster wage growth among high-paid workers does not translate directly into a more unequal wage distribution. Even if the sufficient condition is not met, the empirical evidence suggests that the decline in wage variance across categories has absorbed the increase within specialized firms. However, the steady pass-through of minimum wage increases across the whole wage distribution should raise awareness regarding potential pitfalls in the policy's effectiveness. In particular, in local labor markets with relatively lower average wages.

Proposition A.5 (Amenities dispersion). *Job amenities dispersion $\mathbb{V}[a]$ is not monotonic in $\underline{w}$. Furthermore, Assumption (A.5) is a sufficient condition for amenities dispersion to increase with $\underline{w}$.*

Proof. Using Propositions (A.1) to (A.3), I can write the first and second moment of a as

$$\mathbb{E}[a] = \pi_{\mathcal{L}} \underline{w}^{-\gamma} \mathbb{E}_{\mathcal{L}}[u_H] + \pi_{\mathcal{V}} \mathbb{E}_{\mathcal{V}}[a] + \pi_{\mathcal{S}} \underline{w}^{-\gamma} \mathbb{E}_{\mathcal{L}}[u_L] + \pi_{\mathcal{H}} \mathbb{E}_{\mathcal{H}}[a]$$

and

$$\mathbb{E}[a^2] = \pi_{\mathcal{L}} \underline{w}^{-2\gamma} \mathbb{E}_{\mathcal{L}}[u_H^2] + \pi_{\mathcal{V}} \mathbb{E}_{\mathcal{V}}[a^2] + \pi_{\mathcal{S}} \underline{w}^{-2\gamma} \mathbb{E}_{\mathcal{L}}[u_L^2] + \pi_{\mathcal{H}} \mathbb{E}_{\mathcal{H}}[a^2]$$

Thus, I can take first derivatives of the previous expressions with respect to $\underline{w}$ and get

$$\frac{\partial \mathbb{E}[a]}{\partial \underline{w}} = -\gamma \underline{w}^{-(\gamma+1)} \left(\pi_{\mathcal{L}} \mathbb{E}_{\mathcal{L}}[u_H] + \pi_{\mathcal{S}} \mathbb{E}_{\mathcal{S}}[u_L] \right)$$

and

$$\frac{\partial \mathbb{E}[a^2]}{\partial \underline{w}} = -2\gamma \underline{w}^{-(2\gamma+1)} \left(\pi_{\mathcal{L}} \mathbb{E}_{\mathcal{L}}[u_H^2] + \pi_{\mathcal{S}} \mathbb{E}_{\mathcal{S}}[u_L^2] \right)$$

where I employ Lemma (A.4) to dismiss the derivatives of the share of firms across the categories with respect to the minimum wage. Then, the second centered moment of a can

be written as

$$\mathbb{V}[a] = \mathbb{E}[a^2] - \mathbb{E}[a]^2$$

with a first derivative with respect to $\underline{w}$, equal

$$\begin{aligned} \frac{\partial \mathbb{V}[a]}{\partial \underline{w}} &= \frac{\partial \mathbb{E}[a^2]}{\partial \underline{w}} - 2\mathbb{E}[a] \frac{\partial \mathbb{E}[a]}{\partial \underline{w}} \\ &= -2\gamma \underline{w}^{-(2\gamma+1)} \left(\pi_{\mathcal{L}} \mathbb{E}_{\mathcal{L}}[u_H^2] + \pi_{\mathcal{S}} \mathbb{E}_{\mathcal{S}}[u_L^2] \right) \\ &\quad + 2\gamma \underline{w}^{-(\gamma+1)} \mathbb{E}[a] \left(\pi_{\mathcal{L}} \mathbb{E}_{\mathcal{L}}[u_H] + \pi_{\mathcal{S}} \mathbb{E}_{\mathcal{S}}[u_L] \right) \end{aligned}$$

By adding and subtracting

$$2\gamma \underline{w}^{-(2\gamma+1)} \pi_{\mathcal{L}} \mathbb{E}_{\mathcal{L}}[u_H]^2 \quad \text{and} \quad 2\gamma \underline{w}^{-(2\gamma+1)} \pi_{\mathcal{S}} \mathbb{E}_{\mathcal{S}}[u_L]^2$$

and rearranging the first term, I obtain

$$\begin{aligned} \frac{\partial \mathbb{V}[a]}{\partial \underline{w}} &= -2\gamma \underline{w}^{-(2\gamma+1)} \underbrace{\left(\pi_{\mathcal{L}} \mathbb{V}_{\mathcal{L}}[u_H] + \pi_{\mathcal{S}} \mathbb{V}_{\mathcal{S}}[u_L] \right)}_{\text{Assumption (A.5)}} \\ &\quad + 2\gamma \underline{w}^{-(\gamma+1)} \pi_{\mathcal{L}} \mathbb{E}_{\mathcal{L}}[u_H] \underbrace{\left(\mathbb{E}[a] - \mathbb{E}_{\mathcal{L}}[a] \right)}_{\text{positive}} \\ &\quad + 2\gamma \underline{w}^{-(\gamma+1)} \pi_{\mathcal{S}} \mathbb{E}_{\mathcal{S}}[u_L] \underbrace{\left(\mathbb{E}[a] - \mathbb{E}_{\mathcal{S}}[a] \right)}_{\text{positive}} \end{aligned}$$

which is strictly positive by Assumption (A.5). □

The intuition behind Proposition (A.5) comes from understanding the effects of two opposing forces in play. On the one hand, low-paying employers and specialized firms reduce job amenities, widening the gap with respect to high-paying employers and vertical firms. This effect across categories, which is the most intuitive, increases overall variance. Second, the variance within specialized employers and low-paying firms tends to shrink, as job amenities fall proportionally to the minimum wage change. This effect at the bottom of the distribution lowers overall variance, strengthening as the initial dispersion in job amenities among these firms increases.

Assumption (A.5) curbs the second force in play by bounding the initial dispersion in job amenities at the bottom. A first argument is that workers constrained by the minimum wage are unlikely to differ much from one another. In particular, because differences in

productivity and outside options should have faded over time with worker flows. Second, after many years of wage-amenities tradeoffs in response to higher wage floors, the lower tail of the job amenities distribution should have compressed.

Nevertheless, note that the assumption is not a necessary condition and can be relaxed. In practice, the intuition suggests that the between-group effect is likely to dominate, leading to an overall increase in the variance of job amenities in response to minimum wage increases, even if within-group differences exert some non-negligible downward pressure.

Proposition A.6 (Wage spillovers). *Minimum wage increases propagate through the wage distribution by proportionally raising wages of high-paid workers specialized firms.*

Proof. The result follows immediately from Proposition (A.2), which shows the high-paid worker's wage increases linearly with $\underline{w}$. That is,

$$\frac{\partial \mathbb{E}_S[w_H]}{\partial \underline{w}} > 0$$

□

From Proposition (A.6), the framework anticipates positive spillovers up to the percentile occupied by the highest-paid worker in specialized firms. Namely, employers with access to more affordable job amenities but limited workplace heterogeneity, or vice versa. More importantly, it highlights two features worth mentioning. First, spillovers are unrelated to other moments often used in the literature to interpret wage-floor effects, such as the median or average wage. Second, monotonicity in spillovers across the wage distribution may not necessarily hold. In particular, if the workplace heterogeneity of firms paying higher wage premiums on average exceeds that of firms with lower wage premiums. A similar argument can be made for the unitary price of job amenities.

A.2 Firm-specific wage premiums

The procedure for retrieving firm-level job amenities in Subsection 2.4.2 requires estimating firm-specific wage premiums within the SCC sample. In this section, I describe the empirical approach, report the estimation results, and present evidence addressing potential concerns about systematic bias.

I employ a model for wage determination (henceforth, the AKM model) based on Abowd et al. (1999), Card et al. (2013, 2018), Song et al. (2019), and Gerard et al. (2021), where the log of the hourly wage w_{it} , paid to worker i in quarter t , depends on observed and unobserved permanent worker and firm components, and time-varying observable characteristics

$$\log w_{it} = \alpha_i + \psi_{J(it)} + X_{it}b + \nu_{it}, \quad \nu \sim N(0, \sigma_\nu^2) \quad (\text{A.1})$$

where the worker fixed effect α_i (WFE) captures returns related to the inherent characteristics of individual i , such as schooling, skills, or cohort. The establishment fixed effect ψ_j (EFE) represents the wage premium that firm j pays to all employees, including profit-sharing, rents, or compensating differentials. Similarly, b represents returns to time-varying worker characteristics, including an education-specific cubic polynomial in age and education-specific year fixed effects.² $J(it)$ is a mapping function indicating the employer of worker i in quarter t . The residual term ν_{it} captures the remaining transitory fluctuations.

I estimate Equation (A.1) in the AKM sample described in Subsection 2.3.2, separately for the periods 2012-2015 and 2016-2019. As in the rest of the chapter, the analysis centers on workers aged 18 to 54 with open-ended urban contracts in the private sector. I next detail the data construction and cleaning procedure, designed to render the sample appropriate for the estimation of firm-specific wage premiums.

AKM sample. I collect all employment spells to determine the modal legal status of firms, and the modal gender, nationality, and cohort of workers. I drop non-Brazilian individuals, observations with missing values for gender, cohort, schooling, hours, or wages, and employees with less than a year of potential experience. Thus, workers must be at least one year older than the theoretical graduation age for the highest degree they have attained.

I construct quarterly employment histories, determine transitions, and afterward assign workers with multiple employment relationships to the job with the highest quarterly earnings. That way, I can keep track of transitions in and out of the final sample separately from transitions in and out of non-employment. When the worker's identifier (*Cadastro de Pessoas Físicas*, CPF) appears in the *National Registry of Firms* (CNPJ, for its acronym in

². Worker fixed effects and time-varying characteristics cannot be separately identified without normalization. I follow Card et al. (2018) and Gerard et al. (2021) by centering the education-specific polynomials on the last quarter of the sample, around 40 years old for male workers and around 35 years old for female workers. Furthermore, I define five identical education dummies: incomplete elementary school, complete elementary school, middle school, high school, and college.

Portuguese), I can identify transitions to formal self-employment. Therefore, in my setting, non-employment consists only of unemployment and informal work. Finally, I remove the entire employment history of any individual whose log hourly wage changes by more than 100 log points from one quarter to the next.

The last two restrictions are tightly interconnected. First, firm-specific wage premiums are identified solely through worker flows, so I need to further restrict the sample to the largest weakly connected set of workers and firms. That is, the set of firms connected through at least one hire or separation. Second, I drop singletons (workers and firms I only observe once), which can introduce unnecessary noise into the analysis. Dropping singletons can create new partitions in the connected set, and removing observations from the connected component can, in turn, produce additional singletons. I therefore repeat this process iteratively until I remove no further employment spells.

A.2.1 Identifying conditions

The AKM framework relies on three identifying conditions. First, that worker and firm heterogeneity are roughly fixed over time. Second, that firms in the weakly connected set are sufficiently interconnected through worker mobility. Third, that worker mobility is uncorrelated with time-varying wage residuals in the estimation.

Fixed worker and firm heterogeneity. The AKM model assumes that worker and firm components of wage determination are time-invariant. Although it hardly holds in general, worker skills and firm pay policies tend to change gradually, making the restriction more plausible as the estimation window shortens. I estimate firm wage premiums over a four-year window, a relatively short panel compared to other studies.³ A shorter horizon has the drawback of reducing the number of moves, raising concerns about limited mobility.

Limited mobility. Among sparsely linked weakly connected sets, estimates of firm-specific wage premiums become noisy. Although individual coefficients in Equation (A.1) remain unbiased, noise leads to an overstatement of the variance of EFE at the detriment of the covariance between WFE and EFE. The literature has developed several methods to obtain unbiased variance estimates, such as shrinkage, leave-one-out estimators (Kline et al., 2020), and finite-mixture methods (Abowd et al., 2019; Bonhomme et al., 2019; Lentz et al.,

³. Five years in Engbom and Moser (2022), six in Card et al. (2013) and Sorkin (2018), seven in Song et al. (2019), eight years in Card et al. (2016), and thirteen in Gerard et al. (2021).

2023). Given that my interest centers on individual firm-specific fixed effects, which remain unbiased, I do not implement any of these workarounds.

Nonetheless, a second drawback of limited mobility concerns representativeness. A shorter horizon reduces the likelihood that smaller firms will hire from or employ workers moving to another firm in the weakly connected set. I take advantage of Brazil's high turnover among smaller firms, with most moves occurring within a single year. Thus, I construct quarterly rather than annual employment histories, which allows me to stick to the standard assumption in the literature of keeping a unique employment spell per individual per period (e.g., Card et al., 2013, 2016, 2018; Gerard et al., 2021).

Exogenous mobility. The AKM framework obtains biased estimates when moves systematically depend on other components beyond worker and firm fixed effects or the included time-varying controls. If that were the case, worker mobility would correlate with time-varying match-specific shocks, the worker's employment history, or other transitory wage components, so that the specification would suffer from omitted-variable bias.

Nonetheless, estimates can be unbiased even when individual workers respond to positive or negative shocks at their current jobs, or if wage changes around job transitions combine true firm-specific wage premiums with match-specific gains or losses included in ν_{it} . As long as these deviations average out across movers, estimated firm effects will capture pay premia differentials rather than sorting. Therefore, the exogenous mobility assumption requires that, conditional on individual and firm characteristics, mobility is as good as random.

Card et al. (2013) developed a series of checks that would flag clear violations of the exogenous mobility condition. Gerard et al. (2021) run this analysis using RAIS data from 2002 to 2014 and confirm that wage changes for movers in Brazil are broadly consistent with the assumption. Next, I replicate the analysis using quarterly employment spells between 2012 and 2019 and arrive at the same conclusion.

A first approach proposes fully saturating the AKM framework by including a match-specific component, m_{ij} . If match effects were essential in wage determination, this version of the model should display a significant improvement in fit. Table A.1 shows that the non-saturated version already reports an adjusted R^2 above 0.95 across samples; I therefore skip this step.

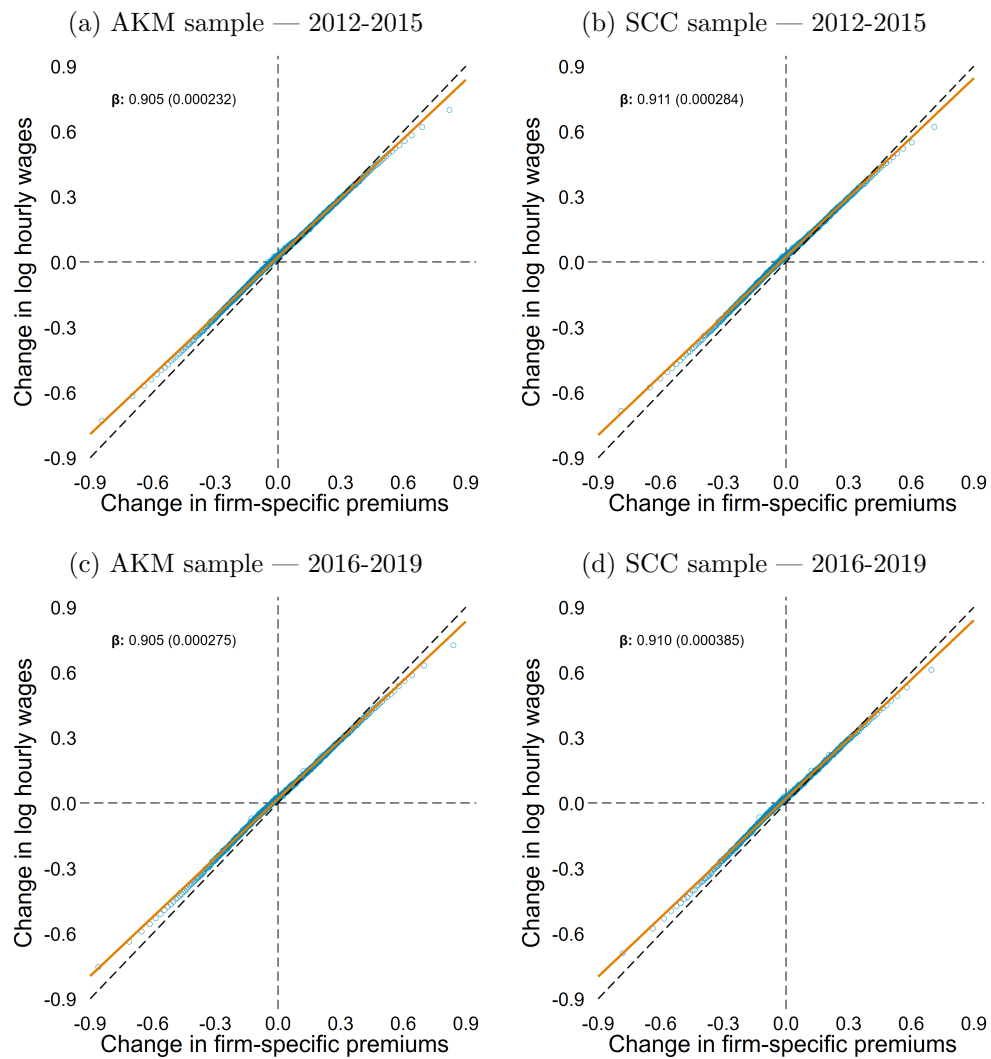
The second test follows from Equation (A.1). If the AKM model is correctly specified, then the following regression should satisfy an identity up to sampling error

$$\mathbb{E}[w_{it+1}^r - w_{it}^r] = \psi_{J(it+1)} - \psi_{J(it)} + \mathbb{E}[\nu_{it+1} - \nu_{it}]$$

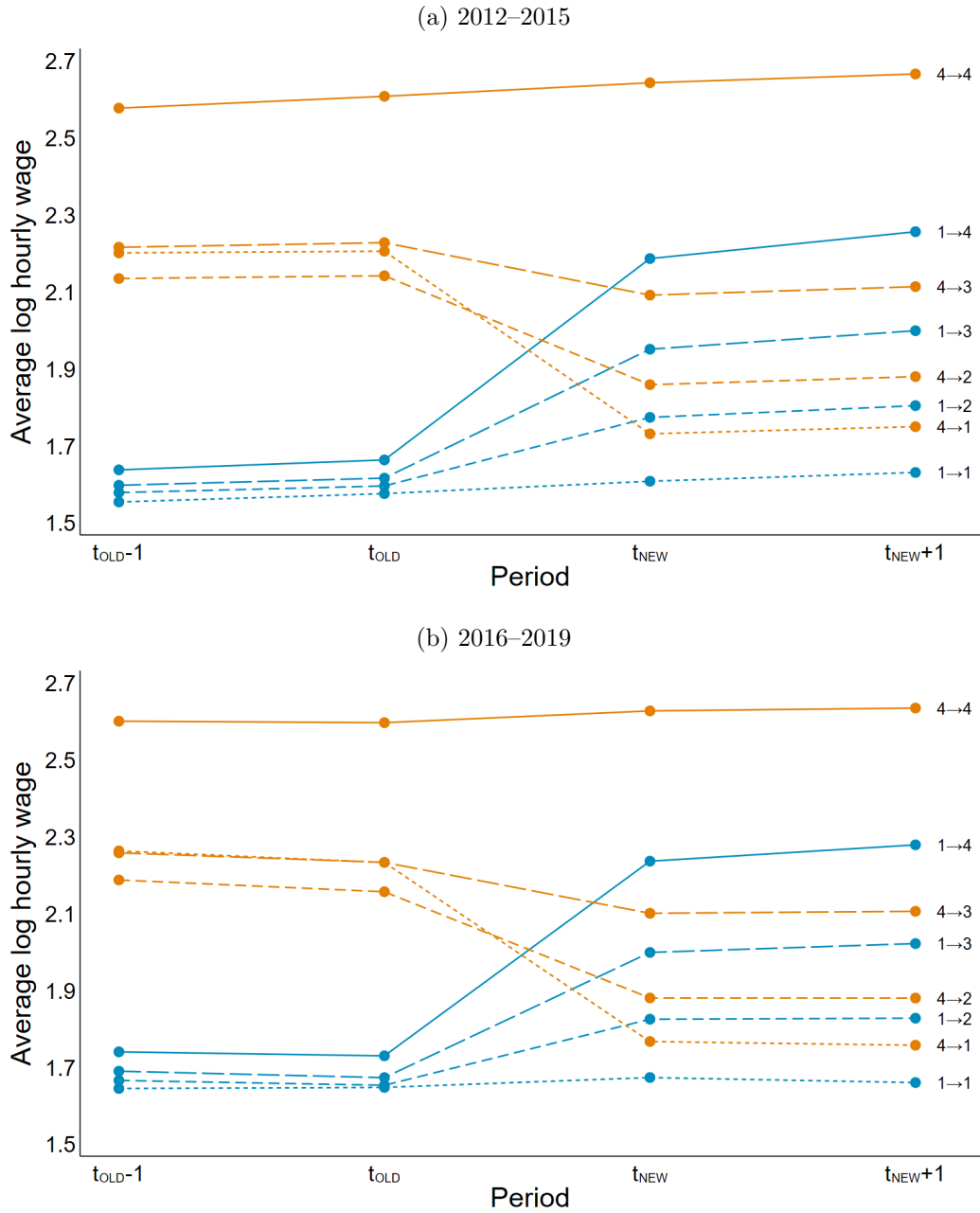
where $w_{it}^r = w_{it} - X_{it}\hat{b}$. Given that I focus on movers, t corresponds to the last quarter at the predecessor firm, and $t + 1$ corresponds to the first quarter at the successor firm. Figure A.1 plots the binscatter for this specification, separately for the AKM and the SCC samples, and the periods 2012-2015 and 2016-2019. Notice the relationship does not follow an exact identity, though the linear fit across individual transitions reports a slope above 0.9 in all cases. Therefore, one can conclude that firm-specific wage premiums likely contain some noise but do not immediately reveal large biases.

Finally, if systematic wage gains or losses drove worker mobility, one would expect at least one of two patterns. First, workers moving up and down the job ladder between the same two firms would exhibit wage changes of different magnitudes. Second, wage trajectories would display pre-trends at the predecessor firm or post-trends at the successor firm. Figure A.2 plots the canonical event study, which compares wage histories among workers moving across firms in different wage premium quartiles.

I do not restrict the analysis to job-to-job transitions, but focus on workers who stayed at the same firm for at least one year before and after moving. Thus, t_{OLD} denotes the last quarter at the predecessor firm, while $t_{OLD} - 1$ corresponds to four quarters before. Likewise, t_{NEW} denotes the first quarter at the successor firm, while $t_{NEW} + 1$ corresponds to four quarters after. The plot displays flat trends before and after the job transitions. In particular, across establishments within the same quartile, differences in log hourly wages are relatively small, suggesting no broad mobility premiums. Consistently, worker flows across establishments in different quartiles showcase roughly symmetric wage differentials.

Figure A.1 – *Wage and premiums changes for movers*

Notes: The figure plots changes in log hourly wages against changes in firm-specific wage premiums among movers. Log hourly wages are residualized with respect to time-varying characteristics using estimates from Equation (A.1) and correspond to hourly rates in the last quarter at the predecessor firm and in the first quarter at the successor firm. Panel (a) and (c) use all transitions among firms in the AKM sample. Panel (b) and (d) use all transitions among firms in the SCC sample. Blue dots correspond to a binscatter regression following the method in Cattaneo et al. (2025). The black-dashed diagonal corresponds to the 45-degree line. The solid orange line plots the best linear fit. The corresponding coefficient and standard error (95% confidence) are reported at the upper left corner. Estimations are based on quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector.

Figure A.2 – *Event studies of wage trajectories*

Notes: The figure plots wage trajectories for workers changing jobs from a predecessor firm in the bottom or top quartile to a successor firm in any of the other quartiles. I use quarterly RAIS employment spells in the AKM sample for workers aged 18–54 with open-ended contracts in the urban private sector. The analysis focuses on workers who stayed at the same firm for at least one year before and after changing jobs, although not necessarily to job-to-job transitions. Thus, t_{OLD} denotes the last quarter at the predecessor firm, while $t_{OLD} - 1$ corresponds to four quarters before. Likewise, t_{NEW} denotes the first quarter at the successor firm, while $t_{NEW} + 1$ corresponds to four quarters after. I define quartiles using estimates for firm fixed effects from Equation (A.1) and calculate average log hourly wages for each type of transition over time. Panel (a) performs the analysis for the period 2012–2015, while Panel (b) replicates the exercise for the period 2016–2019. Blue lines correspond to predecessors in the bottom quartile, while orange lines correspond to predecessors in the top quartile. The shorter the dash, the lower the quartile of the successor firm. For clarity, I report the quartiles next to each line for both the predecessor and successor firms.

A.2.2 Estimation results

The following subsection presents the estimation results for the AKM framework. Table (A.1) performs the canonical variance decomposition in Abowd et al. (1999). Namely,

$$\begin{aligned} \mathbb{V}[w_{it}] = & \underbrace{\mathbb{V}[\alpha_i] + \mathbb{V}[X_{it}b] + 2\text{Cov}[\alpha_i, X_{it}b]}_{\text{worker time-invariant and time-varying characteristics}} + \underbrace{\mathbb{V}[\psi_j]}_{\text{firm wage premia}} \\ & + \underbrace{2(\text{Cov}[\alpha_i, \psi_j] + \text{Cov}[\psi_j, X_{it}b])}_{\text{worker sorting}} + \mathbb{V}[\nu_{it}] \end{aligned}$$

These estimates quantify the relative importance of worker- and firm-specific components on wage determination. Notice, however, that as I mentioned before, the variance of firm pay premia is upward-biased at the detriment of $\text{Cov}[\alpha_i, \psi_j]$.

As mentioned above, using quarterly employment histories in Brazil increases the number of movers across samples, with a mover-to-worker ratio close to 1. Although the ratio decreases for the SCC samples, it remains relatively high compared to similar estimations, ranging from 0.48 to 0.70.⁴

EFE explains less of the variance in the SCC sample. Two immediate potential explanations arise. On the one hand, establishments in the SCC tend to be larger and may exhibit a more similar distribution of pay premiums. On the other hand, these employers are more interconnected through worker flows. Remember that, by construction, firms must have at least one hire and one separation that connects them with other employers in the SCC. As a result, firm fixed effects in the SCC may contain less noise and be more precisely estimated.

Table A.2 turns attention to an alternative variance decomposition following Song et al. (2019). Comparing results across SCC samples shows that within-firm wage inequality remained largely stable over time, with differences driven almost entirely by changes in the residual component. In contrast, most of the decline in wage inequality in Brazil over time is explained by determinants of the between-firm wage variance, including declines in worker segregation, worker sorting, and firm-specific pay premia.

⁴ For instance, Gerard et al. (2021), who explicitly report the number of movers in their samples, have a ratio of around 0.15.

Table A.1 – *Variance decomposition of log hourly wages*

	2012-2015				2016-2019			
	AKM		SCC		AKM		SCC	
$\mathbb{E}(w)$	2.108		2.188		2.155		2.260	
$\mathbb{V}(w)$	0.396		0.432		0.353		0.401	
$\mathbb{V}(\alpha)$	0.269	67.78%	0.302	69.77%	0.246	69.82%	0.289	71.99%
$\mathbb{V}(\psi)$	0.044	11.19%	0.037	8.49%	0.042	11.95%	0.032	8.04%
$\mathbb{V}(Xb)$	0.005	1.37%	0.006	1.29%	0.005	1.33%	0.005	1.21%
$\mathbb{V}(\nu)$	0.015	3.91%	0.017	3.94%	0.012	3.37%	0.013	3.31%
$2 \cdot \text{Cov}(\alpha, \psi)$	0.070	17.63%	0.078	18.15%	0.055	15.44%	0.069	17.13%
$2 \cdot \text{Cov}(\alpha, Xb)$	-0.007	-1.77%	-0.007	-1.67%	-0.007	-1.90%	-0.007	-1.81%
$2 \cdot \text{Cov}(\psi, Xb)$	-0.000	-0.11%	-0.000	-0.10%	0.000	0.00%	-0.000	-0.00%
$\text{Cov}(w, \psi)$	0.079	19.95%	0.076	17.51%	0.069	19.67%	0.067	16.60%
Establishments	3,323,529		765,419		3,126,974		498,859	
Workers	49,079,671		39,338,017		44,137,460		30,704,508	
Moves	49,198,880		27,714,180		40,228,874		14,821,121	
Employment spells	497,308,284		372,771,472		451,144,482		291,773,507	
R^2	0.9563				0.9624			

Notes: The table decomposes wage variation in Brazil, separately for the periods 2012-2015 and 2016-2019. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. The AKM sample centers on employers in the weekly connected set of movers. The SCC focuses on employers in the strongly connected set of actively recruited employees. The analysis follows the estimation of Equation (A.1) in the AKM sample and decomposes the variation in log hourly wages (w) into worker fixed effects (α), firm fixed effects (ψ), time-varying controls (Xb , including education-specific cubic polynomials of age and year fixed effects), residuals (ν), and the corresponding covariances. $\text{Cov}(w, \psi)$ reports the sum of all firm-related components, equal to $\mathbb{V}(\psi) + \text{Cov}(\psi, \alpha) + \text{Cov}(\psi, Xb)$. The last five rows report the numbers of unique establishments, unique workers, moves, employment spells (worker-year observations), and the R-squared of the estimation. The SCC columns use the same estimation but replicate the analysis only for worker-year observations in the SCC sample.

$$\begin{aligned}
& \overbrace{\mathbb{V}[w_{it}] = \underbrace{\mathbb{V}[\bar{\alpha}_j] + \mathbb{V}[\bar{X}_j b] + 2\text{Cov}[\psi_j, \bar{X}_j b]}_{\text{worker segregation}} + \underbrace{\mathbb{V}[\psi_j]}_{\text{firm wage premia}}}_{\text{Between-firm component}} \\
& \quad + \underbrace{2\left(\text{Cov}[\bar{\alpha}_j, \psi_j] + \text{Cov}[\bar{\alpha}_j, \bar{X}_j b]\right)}_{\text{worker sorting}} \\
& \quad \underbrace{+ \mathbb{V}[\alpha_i - \bar{\alpha}_j] + \mathbb{V}[X_{it}b - \bar{X}_j b] + 2\text{Cov}[\alpha_i - \bar{\alpha}_j, X_{it}b - \bar{X}_j b] + \mathbb{V}[\nu_{it}]}_{\text{Within-firm component}}
\end{aligned}$$

A.3 Firm-specific job amenities

I estimate firm-specific job amenities following Sorkin (2018) and Lehmann (2025). The underlying job search model shares the assumptions listed in Section 2.2, but introduces two

Table A.2 – *Variance decomposition between and within establishments*

	2012-2015				2016-2019			
	AKM		SCC		AKM		SCC	
$\mathbb{E}(w)$	2.108		2.188		2.155		2.260	
$\mathbb{V}(w)$	0.396		0.432		0.353		0.401	
Between-firm variance	0.229	57.71%	0.237	54.87%	0.199	56.30%	0.212	52.79%
$\mathbb{V}(\bar{\alpha})$	0.118	29.85%	0.126	29.19%	0.105	29.71%	0.114	28.49%
$\mathbb{V}(\psi)$	0.044	11.19%	0.037	8.49%	0.042	11.95%	0.032	8.04%
$\mathbb{V}(\bar{X}b)$	0.001	0.16%	0.001	0.12%	0.001	0.19%	0.001	0.14%
$2 \cdot \text{Cov}(\bar{\alpha}, \psi)$	0.070	17.63%	0.078	18.15%	0.055	15.44%	0.069	17.13%
$2 \cdot \text{Cov}(\bar{\alpha}, \bar{X}b)$	-0.004	-1.01%	-0.004	-0.98%	-0.004	-0.99%	-0.004	-1.00%
$2 \cdot \text{Cov}(\psi, \bar{X}b)$	-0.000	-0.11%	-0.000	-0.10%	0.000	0.00%	-0.000	-0.00%
Within-firm variance	0.168	42.29%	0.194	44.99%	0.154	43.70%	0.189	47.07%
$\mathbb{V}(\alpha - \bar{\alpha})$	0.150	37.93%	0.175	40.57%	0.142	40.11%	0.175	43.50%
$\mathbb{V}(Xb - \bar{X}b)$	0.005	1.21%	0.005	1.17%	0.004	1.14%	0.004	1.08%
$\mathbb{V}(\nu)$	0.015	3.91%	0.017	3.94%	0.012	3.37%	0.013	3.31%
$2 \cdot \text{Cov}(\alpha - \bar{\alpha}, Xb - \bar{X}b)$	-0.003	-0.76%	-0.003	-0.69%	-0.003	-0.91%	-0.003	-0.82%

Notes: The table decomposes wage variation within and across firms in Brazil, separately for the periods 2012-2015 and 2016-2019. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. The AKM sample centers on employers in the weekly connected set of movers. The SCC focuses on employers in the strongly connected set of actively recruited employees. The analysis follows the estimation of Equation (A.1) in the AKM sample and decomposes the variation in log hourly wages (w) into between- and within-firm components, as in Song et al. (2019). The between-firm component includes firm fixed effects (ψ), firm-level averages of worker fixed effects ($\bar{\alpha}$), and time-varying controls ($\bar{X}b$, including education-specific cubic polynomials of age and year fixed effects), along with their corresponding covariances. The within-firm component includes idiosyncratic variation from worker fixed effects ($\alpha - \bar{\alpha}$) and time-varying controls ($Xb - \bar{X}b$) within firms, residuals (ν), and their covariance.

further ones. First, employed and non-employed workers receive the same number of offers from an identical distribution, making the job search non-directed and effortless. Second, workers value firms identically, up to i.i.d. idiosyncratic preferences, which yields a unique firm ranking—or job ladder—across all workers.

Employers offer contracts $\langle w_{it}, a_j \rangle$ and differ in size g_j , exogenous separation rates δ_j , exogenous reallocation rates ρ_j , and exogenous hiring rates f_j .

A worker i values employment at firm j according to the flow payoff $v_j(w_{it}, a_j)$ and the associated continuation value, discounted at rate β . With probability δ_j , the firm lays off the worker. Conditional on layoff, the worker reallocates to a new job before the end of the next quarter with probability ρ_j ; otherwise, she becomes non-employed. In the absence of a separation, workers receive an offer from firm k with probability λ_E and decides whether to accept it and quit or remain at their current firm. With probability $1 - \lambda_E$, she receives no

offer but may still choose to exit to nonemployment. The following specification synthesizes the associated Bellman equation

$$\begin{aligned}
\underbrace{V_j^e(w_{it}, a_{it})}_{\text{value of being at } j} &= \underbrace{v_j(w_{it}, a_{it})}_{\text{flow payoff}} + \beta \cdot \mathbb{E} \left[\underbrace{\delta_j(1 - \rho_j) \int_{\iota_1} \{V^n + \iota_1\} dI}_{\text{exogenous job destruction}} + \underbrace{\delta_j \rho_j \sum_k \int_{\iota_2} \{V_k^e + \iota_2\} dI \tilde{f}_k}_{\text{exogenous employer-to-employer}} \right. \\
&+ \underbrace{(1 - \delta_j)}_{\text{no exogenous job destruction}} \times \left(\underbrace{\lambda_E \sum_k \int_{\iota_3} \int_{\iota_4} \max \left\{ \overbrace{V_k^e + \iota_3}^{\text{accept}}, \overbrace{V_j^e + \iota_4}^{\text{reject}} \right\} dI dI f_k}_{\text{endogenous employer to employer}} \right. \\
&\quad \left. \left. + \underbrace{(1 - \lambda_E) \int_{\iota_5} \int_{\iota_6} \max \left\{ \overbrace{V^n + \iota_5}^{\text{leave}}, \overbrace{V_j^e + \iota_6}^{\text{stay}} \right\} dI dI f_k}_{\text{endogenous job destruction}} \right) \right]
\end{aligned}$$

The empirical strategy revolves around workers who receive offers from actively recruiting firms. The decision to remain with the incumbent firm or to transition to a new job is what reveals their preferences for employers. Given that the framework assumes that all workers broadly agree on the firm ranking, relative flows across any pair of firms help determine the relative value between the two employers,

$$\frac{\underbrace{M_{jk}}_{\text{relative flows}}}{\underbrace{M_{kj}}_{\text{relative offers}}} = \underbrace{\frac{f_j}{f_k}}_{\text{relative offers}} \underbrace{\frac{g_k(1 - \delta_k)}{g_j(1 - \delta_j)}}_{\text{effective relative firm size}} \underbrace{\frac{\mathbb{P}(j \succ k)}{\mathbb{P}(k \succ j)}}_{\text{relative firm values}}$$

where M_{kj} represents employer-to-employer transitions from firm j to employer k .

Sorkin (2018) demonstrates that flows across every pair of firms are not necessary. As long as one focuses on firms hiring workers from or employing workers transitioning to other employers in the SCC, one can identify relative firm values from the entire worker flow network.

$$\frac{\overbrace{\sum_{j \in J} M_{kj} \exp(V_j^e)}^{\text{value-weighted hires}}}{\underbrace{\sum_{j \in J} M_{jk}}_{\text{quits}}} = \underbrace{\exp(V_k^e)}_{\text{firm value}}$$

Lehmann (2025) builds on Sorkin (2018) by imposing a functional form on workers' indirect utility (not the flow payoff), so that

$$V_{it}^j = V_j^e(w_{it}, a_{it}) = w_{it}^\gamma \cdot a_{it}$$

where γ represents the elasticity of substitution between wages and job amenities.

Next, I specify that a_{it} depends on firm-level job amenities $\chi_{J(it)}$ and idiosyncratic preferences ι_{it} ,

$$\log a_{it} = \chi_{J(it)} + \iota_{it}, \quad \iota \sim N(0, \sigma_\iota^2)$$

while wages are jointly determined by worker and firm characteristics, following the AKM model specification in Section A.2 in the Appendix

$$\log w_{it} = \alpha_i + \psi_{J(it)} + X_{it}b + \nu_{it}, \quad \nu \sim N(0, \sigma_\nu^2)$$

Thus, I can rewrite each worker's indirect utility as

$$\log V_{it}^j = \gamma(\alpha_i + \psi_j + X_{it}b) + \chi_j + \underbrace{\gamma\nu_{it} + \iota_{it}}_{\varepsilon}, \quad \varepsilon_{it} \sim N(0, \underbrace{\gamma^2\sigma_\nu^2 + \sigma_\iota^2}_{\sigma_\varepsilon^2})$$

where ε are idiosyncratic and normally distributed i.i.d. shocks. Notice that ε combines individual deviations around common preferences for firm-specific amenities with temporary fluctuations in wages. The specification runs along the same lines as Lentz et al. (2023), who assume that workers sort themselves across firms according to underlying wage distributions, draw an initial wage from those distributions when they transition to a new job, and receive new realizations that depend on past wages. Moreover, since estimates of firm-specific wage premiums ψ_j can be noisy, epsilon also captures noise from the AKM estimation.

A worker i , who receives an offer from firm k while employed at firm j , compares the expected utilities V_{it}^j and V_{it}^k . Whenever the worker chooses to transition to the new job, it reveals that she ranks compensation in firm k higher than in firm j on the job ladder,

$$\begin{aligned} \mathbb{P}(k \succ j) &= \mathbb{P}(\log V_{it}^k > \log V_{it}^j) \\ &= \mathbb{P}\left(\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j) > (\varepsilon_{1it} - \varepsilon_{2it})\right) \\ &= \Phi\left(\frac{\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j)}{2\sigma_\varepsilon}\right) \end{aligned}$$

Therefore, for a worker that values compensation at two firms equally, $V_{it}^j = V_{it}^k$, any difference in firm-specific pay premiums, $\psi_j - \psi_k$, reveal compensating differentials due to uneven job amenities, $\chi_j - \chi_k$, or idiosyncratic fluctuations in preferences and wages, $\varepsilon_{1it} - \varepsilon_{2it}$.

I am hereby deviating from the original approach in Lehmann (2025) for two reasons. First, instead of using wages directly, I employ Equation (A.1) from the AKM model, which accounts for wage differentials originating in time-varying worker characteristics and temporary wage fluctuations. Second, I do not impose $\sigma_\varepsilon = 1/2$, which undermines my ability to interpret γ estimates but reveals the importance of appropriately scaling job amenities χ_j .

A.3.1 Estimation

I estimate the model by maximum likelihood separately for the periods 2012-2015 and 2016-2019. In particular, I solely use endogenous employer-to-employer transitions among firms in the SCC sample, using the following likelihood function

$$\mathcal{L}(\theta|X) = \prod_{n=1}^N \Phi \left(\frac{\gamma(\psi_k - \psi_j) + (\chi_k - \chi_j)}{2\sigma_\varepsilon} \right)^{\frac{1}{g_j} \frac{1}{f_k}} \quad (\text{A.2})$$

where g_j and f_k weight individual contributions by the size of the predecessor firm j and the number of offers made by the successor firm k . The vector θ of parameters has size $J+1$: one for the elasticity of substitution γ and J for each of the firm-specific job amenities χ_j . To estimate the model, I randomly set $M = 10$ initial values for θ , which I maintain across all jackknife and bootstrap estimations for consistency. For each estimation, I keep the parameter set that maximizes the likelihood function.

Notice, however, that parameters are identified up to a scaling parameter $2\sigma_\varepsilon$, such that

$$\hat{\gamma} = \frac{\gamma}{2\sigma_\varepsilon} \quad \text{and} \quad \hat{\chi}_j = \frac{\chi_j}{2\sigma_\varepsilon} \quad \forall j \in J$$

which requires me to follow a series of steps described in Lehmann (2025) before computing the variance of job amenities across samples.

Incidental parameter bias. Since the estimator is nonlinear, and the number of establishments has a similar order of magnitude as the number of transitions in the sample, $\hat{\gamma}$ suffers from incidental parameter bias. Therefore, I implement the jackknife correction detailed in Lehmann (2025), inspired by the original proposal of Hahn and Newey (2004).

Let N be the number of transitions in the sample. Each establishment uses, on average, $T = 2N/J$ observations to identify the corresponding fixed effect, which could be interpreted, analogously to the panel data literature, as the length of the time series. Similar to the original approach, the procedure randomly splits transitions into T separate groups. However, instead of discarding the transitions in the corresponding group for each jackknife replication, this technique weights down the individual contributions by a factor of 1 to 1000. Otherwise, randomly discarding transitions could break the sample's strongly connected structure that is necessary for identification.

Each jackknife replication yields a new estimate $\hat{\gamma}^{T-1}$ of the elasticity of substitution, reducing the bias from $\mathcal{O}(1/T)$ to $\mathcal{O}(1/T^2)$ by using the following equation

$$\hat{\gamma}_{JK} = T\hat{\gamma}^T - (T-1)\mathbb{E}[\hat{\gamma}^{T-1}]$$

where $\hat{\gamma}^T$ corresponds to the baseline estimate.

Job amenities scaling. As mentioned before, estimates for $\hat{\gamma}_{JK}$ and $\hat{\chi}_j$ are not identified separately from the scaling parameter $2\sigma_\varepsilon$. However, the ratio between the two can be established, irrespective of the scaling parameter. In other words, I can define the following,

$$\hat{\chi}^{BRL} = \frac{\hat{\chi}}{\hat{\gamma}_{JK}} = \frac{\chi}{\gamma_{JK}}$$

corresponding to firm-specific job amenities expressed in the same currency as wages: log-BRL (*reais brasileiros*). Therefore, hereafter, I refer to job amenities in this unit of measure.

Variance correction. As in the AKM framework, firm-specific job amenities are estimated with noise, which leads to an upward bias in the plugin variance estimate. Therefore, I implement the correction discussed in Lehmann (2025).

The first step is to estimate the baseline noise. I fix $\gamma = \hat{\gamma}_{JK}$ and bootstrap the likelihood function $B = 50$ times. To maintain the same network structure in the SCC, instead of randomly sampling transitions, I randomly assign them positive weights ω_i , such that

$$\sum_N \omega_n = N$$

Next, I normalize each estimation so that the average level of job amenities across all establishments equals zero, and define a firm-specific noise term π_j^2 , which corresponds to the variance in firm-specific job amenities across all bootstrap samples.

The second step is to recalculate the plugin estimate of the variance using the following formula

$$\mathbb{V}[\chi^{BRL}] = \frac{\sum_J size_j \left[\left(\frac{J}{J-1} \right) (\hat{\chi}^{BRL} - \bar{\chi}^{BRL})^2 - \pi_j^2 \right]}{\sum_J size_j}$$

where $size_j$ denotes the size of firm j and $\bar{\chi}^{BRL}$ represents the employment weighted average of job amenities (Sorkin, 2018; Lehmann, 2025).

A.3.2 Estimation results

Table A.3 presents key information on the maximum likelihood estimation using Equation (A.2), separately for the periods 2012-2015 and 2016-2019.

The first thing worth noting is the sharp decline in endogenous employer-to-employer transitions in Brazil over time. The immediate consequence is that job amenities in the latest period are more noisily estimated. To account for this, I construct 95% confidence intervals by reporting the 5th and 95th percentiles of all corrected plugin variance estimations from each bootstrap.

The confidence intervals across samples confirm the suspicion that estimates in the later period contain more noise, yet still suggest a significant difference in magnitude.

Table A.3 – *MLE estimation of firm-specific job amenities in log-BRL*

2012-2015				2016-2019			
<i>Baseline estimation</i>							
M	10	Transitions (N)	8,002,800	M	10	Transitions (N)	4,609,276
T	21	Firms (J)	765,419	T	19	Firms (J)	498,859
γ	2.938	Bias: $O(1/T)$	0.0476	γ	3.855	Bias: $O(1/T)$	0.0526
<i>Jackknife correction</i>							
γ_{JK}	5.412	Bias: $O(1/T^2)$	0.0022	γ_{JK}	2.680	Bias: $O(1/T^2)$	0.0027
<i>Variance of job amenities</i>							
B	50	Non-corrected	0.0632	B	50	Non-corrected	0.4046
		Corrected	0.0627			Corrected	0.4052
(0.0401,0.0891)				(0.1344,0.6850)			

Notes: The table presents estimates, corrections, and key information on the maximum likelihood estimation using Equation (A.2), separately for the periods 2012-2015 and 2016-2019. The sample consists solely of quarterly employer-to-employer transitions between establishments in the SCC sample. I use quarterly RAIS employment spells for workers aged 18-54 with open-ended contracts in the urban private sector. I randomly set $M = 10$ initial values for the parameter vector θ , which I keep across all T jackknife and $B = 50$ bootstrap estimations for consistency. $T = 2N/J$ corresponds to the average number of transitions used to identify each individual job amenities fixed effect. For each sample, I report the baseline γ and the jackknife correction γ_{JK} , along with their respective biases. Finally, I report the baseline and corrected plugin estimators for the variance of job amenities in log-BRL.

Appendix to Chapter 3

B.1 Framework: Derivation of profit weights

The following section details the derivation of profit weights from a framework first introduced by Rotemberg (1984). For a more comprehensive description of the assumptions and properties of the model, see Backus et al. (2021).

First, consider the standpoint of shareholders, who aim to maximize cash flow rights across all companies in which they hold shares, which converges into the overall market value of their portfolios.

$$V_{it}(x_{jt}, x_{-jt}; \beta_t) = \sum_{\forall j \in \mathcal{J}} \beta_{jt}^i \pi_{jt}(x_{jt}, x_{-jt}) \quad (\text{B.1})$$

where β_{jt}^i represents the fraction of outstanding shares that investor $i \in \mathcal{S}$ holds in the company $j \in \mathcal{J}$ in year t . Moreover, the vectors x_{jt} and x_{-jt} denote an arbitrary set of managerial decisions taken by the manager at company j and all other companies in $\mathcal{J}$, respectively, and π_{jt} defines the profits obtained by company j given all managerial decisions.

The model's main assumption is that managers do not solely maximize their companies' profits. Instead, they maximize a weighted average of the overall market value of the shareholders' portfolios at the firm that appointed them. In other words, they maximize the following expression

$$Q_{jt}(x_{jt}, x_{-jt}; \gamma_t) = \sum_{\forall i \in \mathcal{S}} \gamma_{jt}^i V_{it}(x_{jt}, x_{-jt}) \quad (\text{B.2})$$

where γ_{jt}^i measures the control that investor i holds on company j in year t . Therefore, when managers account for the consequences that their decisions, x_{jt} , have on the profits of other related companies, they use γ_{jt}^i to address the importance of each specific portfolio.

The framework follows a common claim in the corporate finance literature and in several legal sources, saying that a firm should always answer to its investors (Jensen and Meckling, 1976). Nonetheless, it could also reflect a shift in power dynamics between shareholders and managers since the late 1990s. For instance, Wells (2016) lists a series of events that contributed to the rise of shareholder power and the crystallization of two critical instruments for stockholder activism: jawboning and shareholder proposals.¹

The first one began in 1992, when the SEC revised the proxy solicitation rules. While the original goal was to ease communication between large shareholders, they inadvertently promoted the proliferation of meetings and other less formal, less visible interactions between shareholders and corporate management. The second one became widely used in the mid-90s, leading to the formation of several shareholder coalitions that discuss and draft proposals daily.²

Combining the expressions in Equations (B.1) and (B.2), yield profit weight values for the maximization program of managers

$$\begin{aligned}
 Q_{jt}(x_t; \beta_t, \gamma_t) &= \sum_{\forall i \in \mathcal{S}} \gamma_{jt}^i V_{it}(x_t) \\
 &= \sum_{\forall i \in \mathcal{S}} \gamma_{jt}^i \sum_{\forall k \in \mathcal{J}} \beta_{kt}^i \pi_{kt}(x_t) \\
 &= \sum_{\forall k \in \mathcal{J}} \sum_{\forall i \in \mathcal{S}} \gamma_{jt}^i \beta_{kt}^i \pi_{kt}(x_t) \\
 &\propto \pi_{jt}(x_t) + \sum_{k \in \mathcal{J} - \{j\}} \left(\frac{\sum_{\forall i \in \mathcal{S}} \gamma_{jt}^i \beta_{kt}^i}{\sum_{\forall i \in \mathcal{S}} \gamma_{jt}^i \beta_{jt}^i} \right) \pi_{kt}(x_t) \\
 &\propto \pi_{jt}(x_t) + \sum_{k \in \mathcal{J} - \{j\}} \kappa_{jkt} \pi_{kt}(x_t)
 \end{aligned}$$

where x_t denotes both the endogenous vector of managerial decisions x_{jt} —for executives appointed to company j —and the corresponding set of response functions for companies in

¹. For example, the author mentions the creation of the Institutional Shareholder Services (ISS) in 1985, the issuance of the *Avon Letter* by the US Department of Labor in 1988, the enactment of the Dodd-Frank Act by the US Congress in 2010, and the closer ties between unions and institutional shareholders—fostered by an increasing focus on pension funds since the mid-90s. For a more in-depth explanation on how these changes affected the corporate power dynamics, see Wells (2016).

². For example, James McRitchie, founder of the blog Corporate Governance, acted as the plaintiff in a lawsuit against Meta Platforms Inc. executive officers, including Mark Zuckerberg. The accusation is that board members own an excessive shareholding in the company, thereby making decisions that, though beneficial for Meta, ignore their effects on stockholders' portfolios. This is the exact mechanism proposed in Rotemberg (1984). For the full version of the document, refer to <https://www.documentcloud.org/documents/23117937-james-mcrichtie-v-board-of-directors-meta>.

$\mathcal{J} - \{j\}$: x_{-jt} .³ The interpretation of κ_{jkt} is straightforward. Profit weights determine how the manager appointed by firm j weighs the profits of company k in year t when making an arbitrary set of decisions x_{jt} that imposes externalities on the profits of that company.

Using vectorial notation, I can write profit weights as a ratio of the following cross products

$$\begin{aligned}\kappa_{jkt} &= \frac{\sum_{i \in S} \gamma_{jt}^i \beta_{kt}^i}{\sum_{i \in S} \gamma_{jt}^i \beta_{jt}^i} \\ &= \frac{\langle \gamma_{jt}, \beta_{kt} \rangle}{\langle \gamma_{jt}, \beta_{jt} \rangle} \\ &= \frac{\cos(\gamma_{jt}, \beta_{kt}) \|\gamma_{jt}\| \|\beta_{kt}\|}{\cos(\gamma_{jt}, \beta_{jt}) \|\gamma_{jt}\| \|\beta_{jt}\|}\end{aligned}$$

Furthermore, the change to polar notation yields a decomposition into two distinct components. On the one hand, the degree of cashflow rights concentration across firms, given that the degree of control rights concentration cancels out. On the other hand, a ratio that compares control and cashflow rights across companies. The numerator determines how similar the control rights in company j are to the cashflow rights in company k across all shareholders. The denominator denotes the similarity between control and cashflows rights within the same company j .

Note that using profit weights requires appropriately distinguishing between ownership and control. In this chapter, I assume proportional control, implying that γ_{jt} equals β_{jt} .⁴ As a result, profit weight can be expressed as

$$\kappa_{jkt} = \cos(\beta_{jt}, \beta_{kt}) \sqrt{\frac{IHHI_{kt}}{IHHI_{jt}}}$$

where $IHHI_{jt} = \|\beta_{jt}\|^2$ corresponds to the Herfindahl-Hirschman index (HHI). That is, a measure of investor concentration within the ownership structure of company j in period t .

³. The fourth step in the derivation stems from a simple normalization, so that every manager assigns a profit weight equal to 1 to the company that appointed them.

⁴. The assumption is not harmless, but is standard in the literature. Alternatively, similar studies have proposed incorporating attentiveness into the analysis (Gilje et al., 2020) or representing control rights as a quadratic form of cashflow rights (Antón et al., 2025b). For a discussion of the tradeoffs between cashflow and control rights—incentives versus influence in tunneling—see Newham et al. (2025).

B.2 IV strategy: Predicted shareholdings

Several prior studies have used changes in index components as exogenous shocks in the ownership structure of companies.⁵ The premise is that index funds offer value to investors by tracking a set of sufficiently diversified assets, such as the S&P 500 index. Thus, upon changes in the index constituents, passive funds and other investors replicating the strategy must acquire equity in companies entering the index and sell the stocks of companies leaving it. In this section, I discuss the limitations of a few instruments in the literature and present a detailed description of the empirical approach used in the chapter.

Additions and deletions in the literature. According to Lewellen and Lowry (2021), additions and deletions are inappropriate instruments for addressing the endogeneity of ownership structures. First, which companies make the cut into the index depends on a committee, whose decisions might reflect the companies' most recent performance.⁶ For instance, committee members could be anticipating the trading partnerships of entering companies to improve, or the opposite for companies leaving the index. Second, being added to the index attracts the attention of the media, analysts, and a wider range of investors. As a result, the company's ownership structure can change through channels that are hard to isolate. For example, the authors note that companies recently added to indexes increase their institutional ownership while crowding out blockholders.

Boller and Morton (2020) and Antón et al. (2023) take a different approach, by leveraging S&P 500 additions without considering the outcomes on the added companies themselves. Instead, they focus on competitors, for whom the changes in index constituents are completely unrelated. The proposal stands out in horizontal settings, where managers make unilateral decisions. In these contexts, changes in index constituents would affect firms' choices only through changes in the degree of overlapping ownership, since the ownership structure of competitors remains unaltered. Thus, the exclusion restriction holds by con-

⁵. For further reference, some examples of studies that have used Russel Index reconstitutions as an instrument are Boone and White (2015), Kennedy et al. (2017), Brooks et al. (2018), and Kostovetsky and Manconi (2020). Aghion et al. (2013), on the other hand, use S&P 500 additions.

⁶. According to the S&P 500 US Indices Methodology documentation: "*Constituent selection is at the discretion of the Index Committee and is based on the eligibility criteria. [...] Sector balance, [...] in the relevant market capitalization range, is also considered in the selection of companies for the indices.*" It later adds, "*S&P Dow Jones Indices Index Committees reserve the right to make exceptions when applying the methodology if the need arises. In any scenario where the treatment differs from the general rules stated in this document or supplemental documents, clients will receive sufficient notice, whenever possible.*"

struction. Moreover, since the increase is mechanical, it constitutes an exogenous source of variation.

Unfortunately, the same approach does not apply to vertical settings, where both companies choose to enter into a trading partnership. Since the addition to the index always affects the ownership structure of one of the firms in the customer-supplier dyad, the exclusion restriction no longer holds.

B.2.1 Constructing predicted shareholdings

The identification strategy for this chapter proposes a workaround by constructing analogous versions of all ownership structure measures, using predicted rather than actual shareholdings.

Provided that persistence in the supply chain declines faster than that of ownership structures, estimated portfolios satisfy the exclusion restriction. However, that requires the analysis to focus on particular sets of companies and shareholders. For that reason, I exclude companies in years when they participate in the index's constituent changes, as well as shareholders in years in which they purchase shares in a company for the first time. Then, I construct predicted shareholdings based on over a decade of past holdings of institutional shareholders across publicly listed companies in the US, as well as differences in the market value of annual additions and deletions from the S&P 500 index. Finally, I use predicted shareholdings to construct instruments for the degree of overlapping ownership, the relative shareholder concentration, and profit weights.

Stable shareholders and companies. The empirical approach first requires dealing with additional potential sources of endogeneity. In particular, those that could influence the chain of effects linking additions and deletions in the S&P 500 to the companies' ownership structures. I drop 300,514 out of the 10,238,314 ownership ties from 2000 to 2013, involving companies in years when they participated—directly or indirectly—in an index addition or deletion. This includes not only entrant and exiting firms but also related parent companies, subsidiaries, spin-offs, and so on.

Furthermore, I leverage the fact that indexing⁷ has become a widely used form of passive investing—with most institutional shareholders typically employing a diversified strategy

⁷. I refer to *indexing* as a strategy in which investors replicate the performance of specific market indexes, such as the S&P 500, by closely matching their holdings and weights to those of the index.

that combines active and passive funds—and focus on shareholders with more stable positions within their portfolios. I exclude institutional investors purchasing shares in a company for the first time, which amounts to 3,248,053 additional ownership ties. For the purpose of computing ownership structures, I henceforth consider these investors as retail shareholders. That is, small individual investors who buy and sell company stock in tiny amounts.

S&P 500 index weighting. Every company in the S&P 500 index has an associated weight that depends on the market capitalization of all firms. Therefore, when the S&P 500 changes constituents, index weights tend to shift, driven by differences in market value between additions and deletions. For example, when additions have a higher market capitalization than deletions, the index weights of all other constituents must fall proportionally to make room for the larger weights associated with the entrants. The opposite holds as well.

Because the index lists the 500 most influential companies in the US according to S&P Indices, a typical change in constituents should involve replacing a low-market-capitalization company with a high-market-capitalization one. However, this is seldom the case. Publicly listed companies in the US regularly engage in mergers, acquisitions, and spin-offs, prompting the S&P Indices Committee to respond quickly and appropriately to such operations. As a result, the index does not always delete a company with a relatively low market value compared to the addition, making it challenging to anticipate the net change in the constituent companies' market values.

I manually collect news releases from Standard & Poor's, S&P Indices, and S&P Dow Jones Indices between 1999 and 2013. I select all announcements for additions and deletions related to the S&P 500, S&P MidCap 400, and S&P 600 SmallCap indexes. For the subset of S&P 500 constituents, I identify the exact date of each addition and deletion, the names and tickers of the companies involved, and a detailed description of the announcements. This way, I classify entering and exiting firms by whether the decision involves a change in the company's market capitalization, a merger, an acquisition, a spin-off, a renaming, or a delisting.

Shareholdings and index weights. To minimize endogeneity concerns arising from correlations between index weight determinants and other factors, such as outstanding shares and share prices, I set each firm's market capitalization to the quarter immediately preceding the corresponding addition or deletion. Furthermore, the S&P 500 belongs to the *float-adjusted market-capitalization-weighted indices* segment of S&P Dow Jones, which has two immediate implications. First, stocks with higher market capitalization have a greater impact on the

index's performance than those with lower market capitalization. Second, market capitalization is based solely on floating outstanding shares. That is, it excludes shares held by long-term strategic shareholders, such as insiders, private equity firms, or the government, although I cannot make that distinction.⁸

I compute index weights ω_{jt} as follows

$$\omega_{jt} = \frac{MktVal_{jt}}{\sum_{\forall k} MktVal_{kt}} = \frac{MktVal_{jt}}{IdxVal_t}$$

where $MktVal_{jt}$ denotes the market capitalization of company $j \in \mathcal{J}$ in year t and $IdxVal_t$ represents the aggregate market value of all 500 companies in the index during period t .⁹ By assuming away changes in the market capitalization of company j , which I assume fixed, I obtain the relative change of index weights as a function of fixed quantities: past index aggregate values and differences in the market capitalization of additions and deletions.

Thus, I compute the quarterly aggregate value of the index during the previous year and use the cross-year average to obtain the following expression

$$\begin{aligned} \frac{\omega_{jt} - \omega_{jt-1}}{\omega_{jt-1}} &= \frac{IdxVal_{t-1}}{IdxVal_t} - 1 \\ &= \frac{IdxVal_{t-1}}{IdxVal_{t-1} + \sum_k MktVal_{kt}(\mathbb{1}_{k \in \mathcal{A}_t} - \mathbb{1}_{k \in \mathcal{D}_t})} - 1 \\ W_{jt} &= \begin{cases} \frac{\omega_{jt} - \omega_{jt-1}}{\omega_{jt-1}} & \text{if } j \in \text{S\&P 500} - \{\mathcal{A}_t, \mathcal{D}_t, \mathcal{R}_t\} \\ 0 & \text{if } j \notin \text{S\&P 500} \end{cases} \end{aligned}$$

where $\mathcal{A}_t$ and $\mathcal{D}_t$ additions and deletions in year t , respectively.

As I mentioned before, it is important to exclude other companies related to the added and deleted firms. That group of companies is represented by $\mathcal{R}_t$, which includes parent companies, subsidiaries, spin-offs, and other companies explicitly mentioned in the announcement. In all cases, the explicit mention of a company in the announcements reveals that the Index Committee has considered its relationship with at least one of the additions and deletions when calling for a constituent change.

⁸. For more information about the float adjustment methodology and investable weight factors, refer to <https://www.spglobal.com/spdji/en/documents/index-policies/methodology-sp-float-adjustment.pdf>.

⁹. Notice the total market value of all companies in the S&P 500 index does not coincide with the index's market capitalization, even by including investable weight factors. The difference arises because S&P Dow Jones Indices scales the aggregate index value to avoid abrupt changes in its price over time.

Predicted shareholdings. I estimate Equation (3.5) using OLS for the remaining 6,648,273 ownership ties in the sample. The estimation yields the following coefficients

$$(\alpha, \rho, \tau) = (0.0007, 0.9074, 0.0193)$$

all of which are statistically significant at the 99% confidence level using heteroskedastic-robust standard errors. More importantly, the R^2 is 0.815, suggesting that past shareholdings and index weight changes from additions and deletions to the S&P 500 capture most of the variation in portfolios.

The underlying assumption in the identification strategy is that, as long as the persistence in the supply chain declines faster than that of the ownership structure, the unexplained variation in this estimation captures most endogeneity and simultaneity in ownership ties. As such, I extend the same argument to the analogous versions of the degree of overlapping ownership, the relative shareholder concentration, and the profit weights, which I construct using the predicted shareholdings from the estimated Equation (3.5).

To conclude, I am more inclined to use Equation (3.4) for estimating the 2SLS. As I mentioned above, the key determinant of the identification strategy is appropriately controlling for supply chain persistence. Thus, the inclusion of time-varying company fixed effects can do more to capture this persistence, as long as it is driven by certain customers and suppliers rather than idiosyncratic factors at the customer-supplier dyad level.

B.3 The tropes of partial vertical integration

The literature on industrial organization has shown that vertical integrations can affect upstream and downstream market outcomes (Bolton and Whinston, 1993; Boehm and Sonntag, 2023). Partial integration and vertical overlapping owners can attain similar results, but without companies engaging in mergers and acquisitions. For example, Fee et al. (2006) explore the role of partial vertical integration on the stability of customer-supplier relationships, while Freeman (2025) extrapolates similar results to third-party overlapping shareholders. Therefore, it seems reasonable to consider that overlapping shareholders affect the supply chain network through typical features of vertical integration and vertical control.¹⁰

¹⁰. Vertical integration is a way of organizing a trading relationship in which a company aims to gain control of multiple steps along the supply chain. Instead, vertical control implies transferring decision-making rights of some but not all aspects of the trading partnership.

To begin with, overlapping owners could enable valuable trading partnerships that otherwise would not exist due to contractual frictions, such as incomplete contracts or injunctions from competition policy authorities.¹¹ For example, by easing access to data, overlapping shareholders could help to reduce the cost of getting information about prospective trading partners and alleviate conflicting interests. Since the screening process and transaction costs occur on both sides of the supply chain relationship, they could make it more likely for companies with similar ownership structures to pick each other. Moreover, competition policy authorities often overlook ownership ties across companies, so overlapping shareholders could act as makeshifts for the contracts, mergers, and acquisitions that authorities would deem anti-competitive behavior. Similarly, they could help work around the discouraging costs associated with such financial transactions.

Beyond nudges at the customer-supplier dyad, overlapping shareholders could make systemic risk more evident for individual companies. Therefore, it could be that managers internalize the benefits of engaging with redundant trading partners to reduce the impact of natural disasters (Carvalho et al., 2021; Barrot and Sauvagnat, 2016), bullwhip effects (Croson and Donohue, 2005), wars, or other external shocks (Wang, 2025a). Moreover, they could increase supply reliability for customers (Bolton and Whinston, 1993; Wu and Birge, 2014), alleviate suppliers' cash constraints (Fee et al., 2006), smooth sales dependence on inherently uncertain markets, or provide companies that hold or require essential facilities with a competitive advantage.

Subsection 3.4.2 explores an alternative mechanism, emphasizing the role of tunneling. Nonetheless, transferring value across companies may take many forms and does not necessarily entail cash flows. In particular, overlapping shareholders could seek tighter control over the supply chain, thereby gaining an advantage across multiple markets simultaneously.

This interpretation resembles the typical drawbacks of partial vertical integration in the literature. For example, a higher degree of overlapping ownership could lead to foreclosure in upstream and downstream markets (Levy et al., 2018; Boehm and Sonntag, 2023) or tighter market-entry barriers (Newham et al., 2025). I leave Subsection 3.4.2 for the analysis of such mechanisms, focusing here on the positive aspects discussed in prior studies: holdup and double marginalization.

^{11.} For instance, revenue-sharing contracts have been under scrutiny by competition policy authorities in both the US and Europe. As an example, the Directorate-General for Competition of the European Commission accused Google in 2018 of using these contracts to bundle Google Chrome and the PlayStore to the Android OS.

Since these particular mechanisms fall beyond the scope of the chapter, I offer only suggestive evidence that aligns with prior studies. These investigations leverage better settings and specific sources of variation, demonstrating the validity of the particular channels. Nevertheless, the relative importance of each mechanism remains to be determined.

Holdup. Customers and suppliers often face the opportunity to boost profits through coordination and long-term contracting. However, contractual frictions may preclude potential trading partners from pursuing these opportunities. Alternatively, these companies can opt to merge or acquire each other as a workaround.

However, the costs associated with merging or acquiring each other can be high, so companies would pursue these strategies only when signing a contract is not an option. Nonetheless, contracting may become completely unfeasible under certain circumstances. Either due to uncertainty or information asymmetries, a few situations can arise that prevent companies from agreeing on the contract's content, on who should retain residual control over it, or how to substitute specific clauses that competition policy authorities often deem as anticompetitive.

Holdup constitutes a typical example of contractual frictions between customers and suppliers. The core of the problem lies in the disagreement over how to split profits originating from partnership-specific investments. In particular, because agents cannot truthfully commit without a binding document, one of the trading partners refrains from making the investment, concerned about being held up by the other party.

The literature presumes that overlapping shareholders can help alleviate these informational problems, either by disclosing otherwise private information (Pawliczek et al., 2022; Boone and White, 2015) or by enforcing tacit agreements through other means, including voting (Fichtner et al., 2017; Bubb and Catan, 2022; Antón et al., 2023), interlocking boards (Azar, 2022), putting managers on the payroll of institutional funds (Freeman, 2025), tying compensation to ESG-based outcomes (Bebchuk and Tallarita, 2022; Cohen et al., 2023), or through selective omission (Wells, 2016; Hemphill and Marcel, 2019).¹²

¹². *Interlocking boards* refers to institutional shareholders who have a seat on the board of multiple firms. *ESG-based compensation* ties executive pay to environmental, social, and governance metrics, which can restrict customers or suppliers to companies meeting very specific characteristics. *Selective omission* represents situations in which shareholders encourage actions that simultaneously benefit the company's profits and investors' portfolios, but remain silent otherwise. Wells (2016) provides a few examples that illustrate how institutional investment funds tipped the scales and amplified their voice in corporate decision-making through several means at their disposal, e.g., jawboning, shareholders' proposals, proxy voting, among others.

Along the same lines, prior studies have found empirical support for this intuition. For instance, Freeman (2025) shows that overlapping shareholders lengthen trading partnerships while improving innovation outcomes, and Deng et al. (2025) finds that suppliers invest more intensively in partnership-specific assets. I further examine the possibility by exploring the heterogeneous effects of overlapping ownership across markets where holdup appears to be more prevalent.

The literature broadly considers three dimensions to identify innovation activities related to holdup (Aghion et al., 2013; Freeman, 2025; Deng et al., 2025; Antón et al., 2025a). The innovation input reflects the resources and efforts invested by firms; the innovation output represents the outcomes of the innovation process; and the innovation specificity refers to the degree of customization to a particular trading partner's needs. I measure the prevalence of holdup by classifying companies by their level of R&D intensity. That is, the ratio of annual R&D expenses to total book assets, which is a standard proxy for innovation input in the literature. A caveat of using only innovation input is that the variable captures only the average likelihood that companies may face holdup problems with any given trading partner, unlike input specificity, which provides pairwise measures that would allow a classification of customer-supplier dyads rather than individual companies or market aggregates.

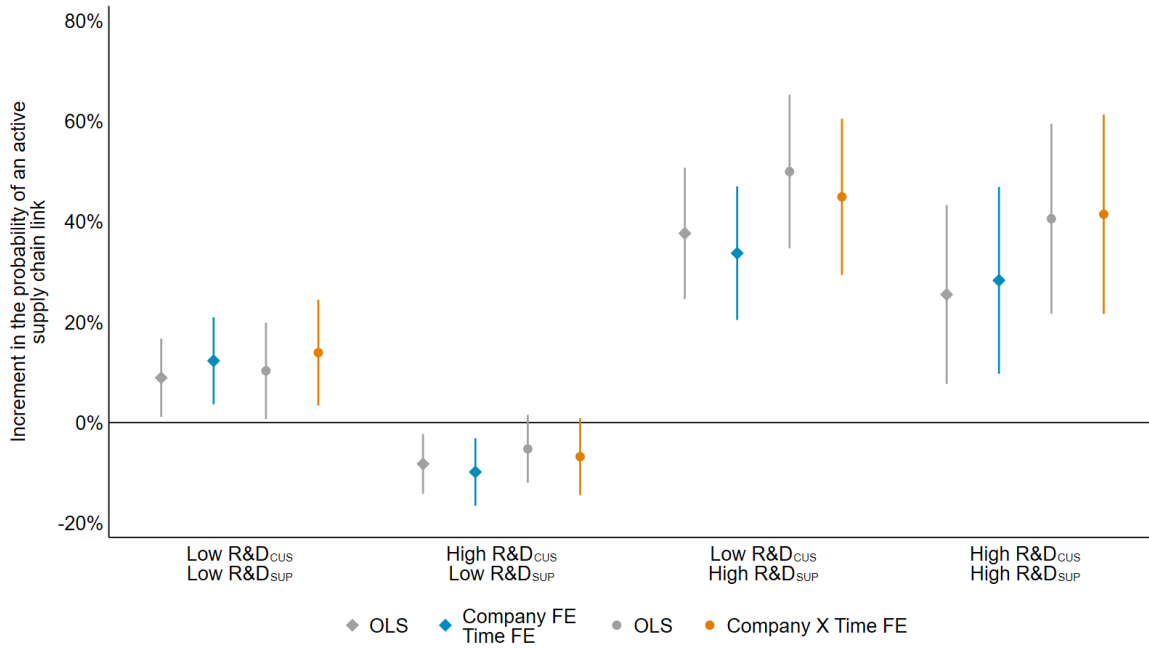
I classify customers and suppliers based on whether they are above or below the average R&D intensity (6.89%) among firms in the sample, i.e., publicly listed US companies between 1999 and 2013, excluding the financial and utility sectors. I then extend Equation (3.3)

$$Link_{sct} = \delta_{st} + \delta_{ct} + OvrOwn_{sct} \times \sum_{j \in \{L, H\}} \sum_{k \in \{L, H\}} \tau^{jk} (R\&D)^{jk} + \varepsilon_{sct} \quad (B.3)$$

where $(R\&D)^{LH}$ indicates that customer c has a relative low R&D intensity, while supplier s has a relatively high intensity. The same interpretation applies to the remaining dummies and the four coefficients τ^{LL} , τ^{LH} , τ^{HL} , and τ^{HH} . Likewise, I report results from extending Equation (3.2) as well.

Figure B.1 shows that an increase in the degree of overlapping ownership has a larger impact when suppliers have an R&D investment above the average, consistent with the results of Freeman (2025) and Deng et al. (2025), who find that overlapping shareholders facilitate innovation only when investments hinge on the supplier side.

When the supplier is the trading partner mostly engaged in R&D, an increase in ownership structure similarity by one standard deviation raises the unconditional probability of an

Figure B.1 – *The role of overlapping owners in alleviating holdup*

Notes: The figure depicts the effect of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different degrees of holdup and fixed-effects specifications. I classify customers and suppliers based on whether their R&D intensity is above or below the sample average (6.89%). I report the effect of increasing the average degree of overlapping ownership (0.0905) by one standard deviation (0.1361). I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). In orange, I report 2SLS estimates from Equation (B.3), and in blue, 2SLS estimates from an analogous extension of Equation (3.2). Estimates for the OLS are reported in gray to the left of the corresponding 2SLS estimates. The 2SLS estimations use a first-order Taylor expansion of the ownership structure similarity based on predicted shareholdings, as described in Section 3.3. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification and customer-supplier dyads with missing information on annual R&D expenses or total book assets. For further details, refer to Table B.3 in the Appendix.

active trading partnership by 33.67% to 44.88%, depending on the specification. The effects seem slightly smaller when both companies invest heavily in R&D, with coefficients ranging from 28.27% to 41.41%. More importantly, notice that coefficients are positive when customers and suppliers have a below-average R&D intensity, ranging from 12.27% to 13.89%, suggesting that holdup cannot fully explain the role of overlapping shareholders. For further details of the estimations, see Table B.3 in the Appendix.

Double marginalization. When customers and suppliers operate in less competitive markets, they apply successive markups to their marginal costs, leading to a problem known as *double marginalization*. Although not a general rule, double marginalization tends to reduce profits for all companies along the supply chain (Hamilton and Mqasqas, 1996), so

firms often avoid it by employing downstream-profit revenue-sharing contracts and nonlinear pricing. In addition, this type of contract is attractive to companies because it allows them to incorporate clauses that address the specificities of supply chain relationships, including product quality, technical specifications, retail services, and so on.

However, when not explicitly forbidden by competition authorities, these contracts are usually hard to enforce. In particular, because customers and suppliers need full information about each other’s actions, which becomes increasingly demanding the more steps there are in the supply chain. Thus, as in the previous case, double marginalization can be a good example of contractual frictions in which overlapping shareholders can play a decisive role in overcoming it.

I proceed likewise as in the previous case, using market concentration as a rough indicator of competition. Although this often requires establishing a definition for product markets, I lean on a particular version of the Herfindahl-Hirschman Index (HHI) based on a text-based network industry classification (TNIC). Hoberg and Phillips (2016) constructs product differentiation distances across all publicly listed companies in the US. Therefore, every other firm in the economy can be a potential competitor for a specific output produced by the firm.

The network approach makes it unnecessary to define juxtaposed market definitions across industries, which is more appropriate in a setting with publicly listed companies in the US for two reasons. On the one hand, most of these companies are multi-product firms, so confining them to a single industry artificially inflates concentration measures within that market (though potentially creating biases in the opposite direction for other product markets). On the other hand, some companies may not have close substitutes within standard market classification—such as SIC or NAICS—, which can also introduce problems into the analysis. For more details on text-based product similarity and its properties, see Hoberg and Phillips (2016), Frésard et al. (2020), and Ederer and Pellegrino (2025).

I classify customers and suppliers based on whether their TNIC-HHI concentration is above or below the median (2,040 out of 10,000, which corresponds to moderate levels of concentration) among all firms in the sample. Namely, publicly listed US companies between 1999 and 2013, excluding the financial and utility sectors. I then extend Equation (3.3) to assess heterogeneous effects of overlapping ownership across markets with different degrees of double marginalization.

$$Link_{sct} = \delta_{st} + \delta_{ct} + OvrOwn_{sct} \times \sum_{j \in \{L,H\}} \sum_{k \in \{L,H\}} \tau^{jk} (TNIC - HHI)^{jk} + \varepsilon_{sct} \quad (\text{B.4})$$

where $(TNIC - HHI)^{LH}$ indicates that customer c faces relatively high competition, while supplier s holds a more dominant position. The same interpretation applies to the remaining dummy variables and the four coefficients τ^{LL} , τ^{LH} , τ^{HL} , and τ^{HH} . Likewise, I report results from extending Equation (3.2), following the same approach as above.

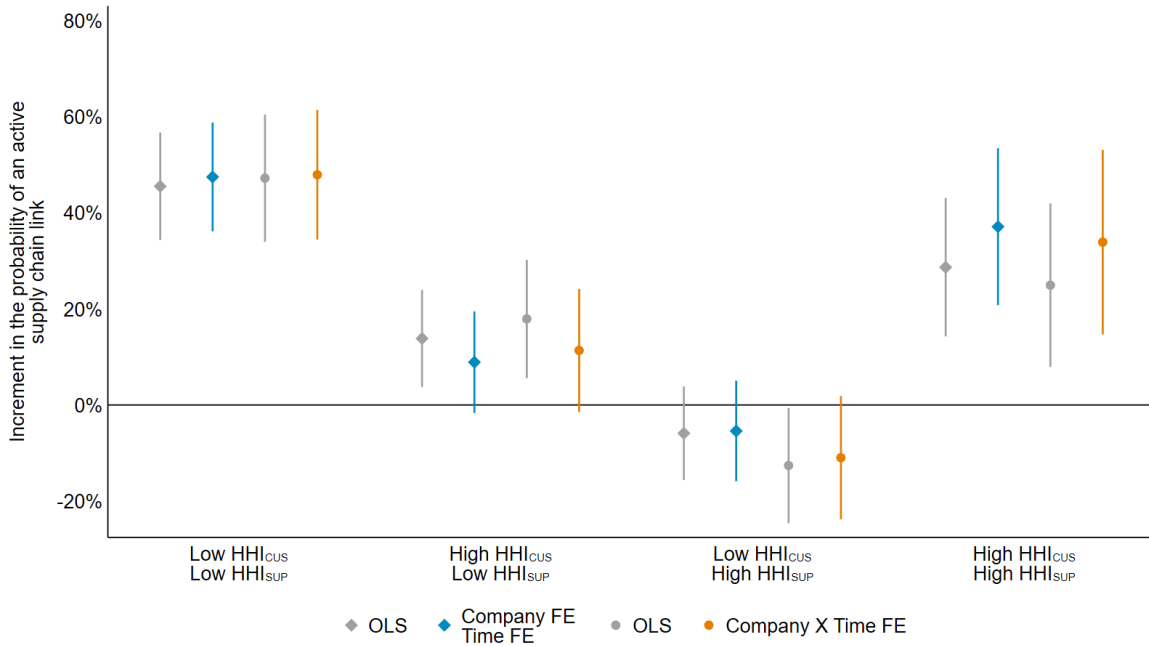
Figure B.2 illustrates several insightful results. First, overlapping shareholders play an extensive role when both the customer and the supplier hold a dominant position—which aligns with double marginalization—showing that an increase in the degree of overlapping ownership increases the likelihood of an active trading partnership between 33.87% and 37.08%, depending on the specification. Second, when upstream and downstream markets exhibit greater competition, coefficients are larger, ranging from 47.43% to 47.88%. Contrary to the hypothesis of double marginalization, the last conclusion suggests that overlapping shareholders play a more significant role when customers and suppliers face more intense competition. For further details of the estimations, see Table B.4 in the Appendix.

All in all, overlapping shareholders can help customers and suppliers overcome informational and contractual difficulties, overcoming problems that often require more cumbersome arrangements. Although the results support this conclusion for situations involving holdup and double marginalization, my findings do not rule out the possibility that overlapping owners play a significant role in other margins. That is, other contractual and informational frictions, as well as systemic risks affecting the supply chain. If anything, the effects of intense competition faced by customers and suppliers corroborate the importance of further exploring additional channels.

Nevertheless, these results add suggestive evidence to the literature, which has found beneficial consequences of having overlapping owners on both sides of the supply chain relationship, with the potential to create value for trading partners, their shareholders, and the economy as a whole (Gao et al., 2022; Riva, 2022; Freeman, 2025; Deng et al., 2025). In particular, my findings propose that overlapping owners, beyond extending the duration of trading partnerships already in place, also facilitate the creation of supply chain relationships that might not have occurred otherwise.

B.4 Robustness

The following section condenses all relevant robustness checks, each corresponding to a methodological decision described in the chapter.

Figure B.2 – *The role of overlapping owners in alleviating double marginalization*

Notes: The figure depicts the effect of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different degrees of double marginalization and fixed-effects specifications. I classify customers and suppliers based on whether their TNIC-HHI is above or below the sample median (2,040 out of 10,000). I report the effect of increasing the average degree of overlapping ownership (0.0905) by one standard deviation (0.1361). I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). In orange, I report 2SLS estimates from Equation (B.4), and in blue, 2SLS estimates from an analogous extension of Equation (3.2). Estimates for the OLS are reported in gray to the left of the corresponding 2SLS estimates. The 2SLS estimations use a first-order Taylor expansion of the ownership structure similarity based on predicted shareholdings, as described in Section 3.3. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification. For further details, refer to Table B.4 in the Appendix.

B.4.1 Alternative overlapping ownership measures

While my preferred measure of overlapping ownership between customers and suppliers relies on deriving profit weights, the literature offers a few other candidate metrics for measuring ownership structures and their similarities.

The following subsection tests the consistency of my findings using two standard measures found in prior studies. In addition, I propose a third variable that combines both profit weights into a single metric, which measures how likely it is for managers to agree on the benefits of engaging in a trading partnership.

Overlapping market value. Antón and Polk (2014) and Freeman (2025) are proponents of the following metric, which represents the fraction of the total market value of the supplier,

V_{st} , and the customer, V_{ct} , that is owned by the set of overlapping shareholders $i \in \mathcal{O}$ in year t . Namely,

$$ovrMktVal_{sct} = \frac{\sum_{i \in \mathcal{O}} V_{sit} + V_{cit}}{V_{st} + V_{ct}}$$

Overlapping-shares product. Hansen and Lott (1996) and Freeman (2025) present the following measure, which is the cross-product of the customer’s and supplier’s shareholdings. That is,

$$ovrShrProd_{sct} = \sum_{i \in \mathcal{S}} \beta_{sit} \times \sum_{i \in \mathcal{S}} \beta_{cit}$$

Manager’s agreeability. Last, I propose a metric that uses the smaller of the two profit weights, rather than the higher one used in Subsection 3.4.2.

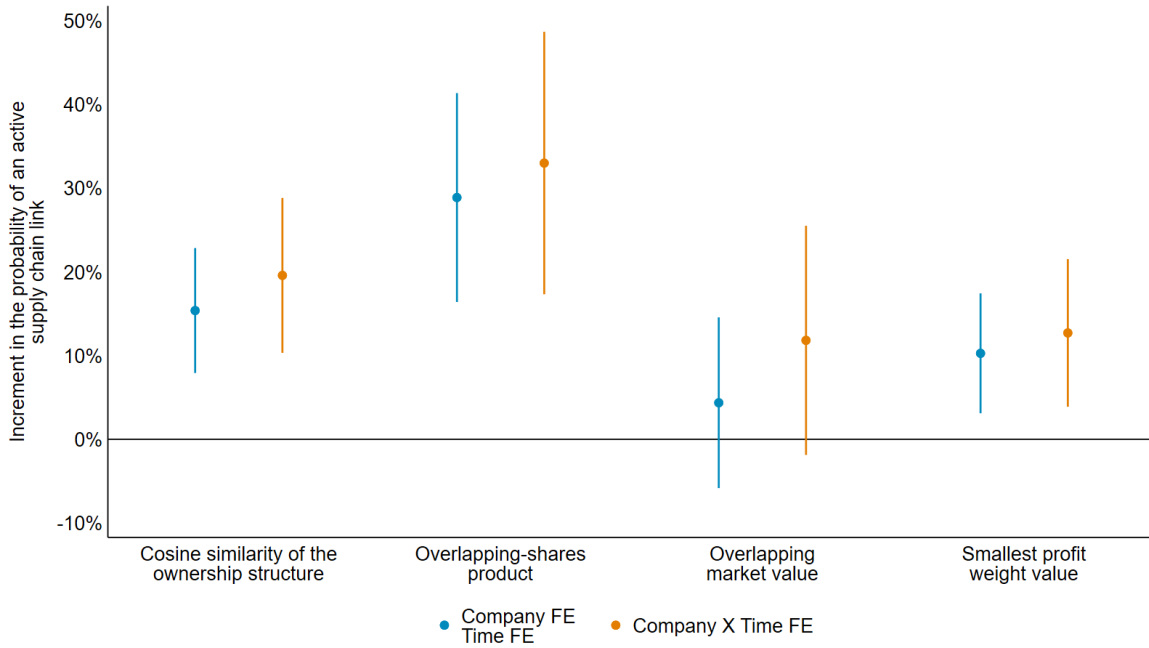
$$minKappa_{sct} = \min(\kappa_{sct}, \kappa_{cst})$$

Intuitively, this metric should capture when both managers internalize the aggregate benefits of engaging in a trading partnership.

Robustness. Figure B.3 shows that coefficients are positive and statistically significant for all alternative metrics except for the overlapping market value, which is consistent with the findings in Freeman (2025). The author consistently reports coefficients for this metric that halve the effects identified by other alternatives. This is not a surprise, since the measure relies on companies’ market capitalization in addition to their ownership structure, which introduces a further source of endogeneity that my instrumental variables approach does not address. As such, these results suffer from measurement error and identification problems, both of which bias the coefficients towards zero. For further details on the estimations, see Table B.5 in the Appendix.

B.4.2 Alternative functional forms in the IV approach

A potential threat to identification arises from the inherent nonlinearities in the ownership measures, which may capture unintended interactions between shifts in index weights and shareholdings. Throughout the chapter, I present results using a linear approximation of the original expression (first-order Taylor expansion). In this subsection, I replicate the baseline analysis using the nonlinear functional form and a quadratic approximation (second-order Taylor expansion).

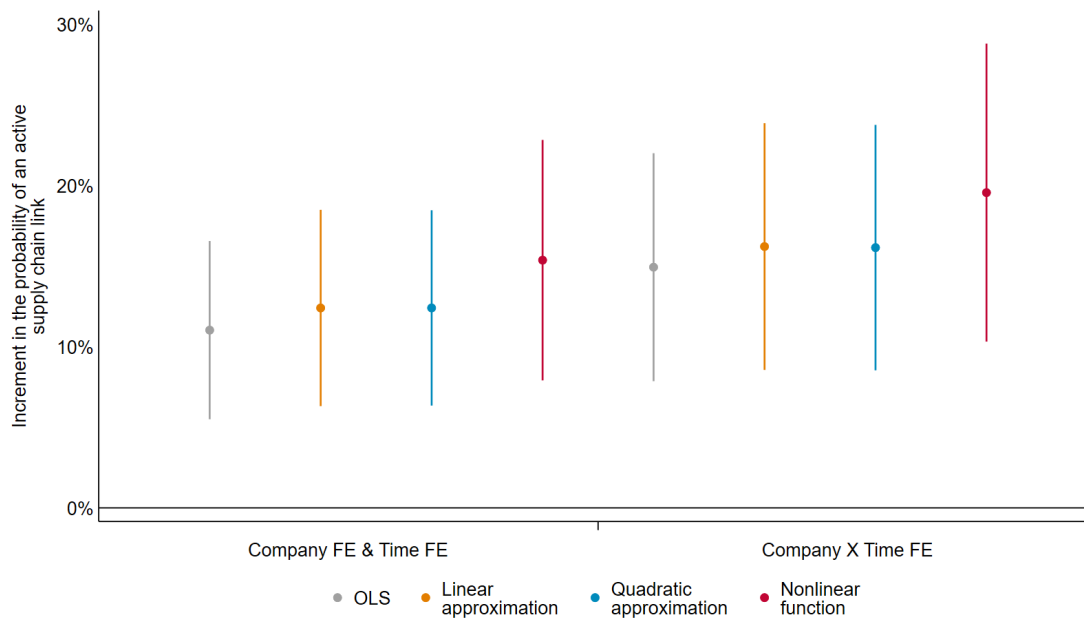
Figure B.3 – *Robustness: Alternative overlapping ownership measures*

Notes: The figure depicts the effect of increasing the degree of overlapping ownership on the likelihood of an active trading partnership, across different fixed-effects specifications and alternative measures of overlapping ownership. For each metric, I report the effect of increasing the sample average by one standard deviation. I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). In blue, I report 2SLS estimates analogous to Equation (3.2) and, in orange, estimates analogous to Equation (3.3). The 2SLS estimations use analogous definitions for each variable based on predicted shareholdings, as described in Section 3.3. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification. For further details, see Table B.5 in the Appendix.

Robustness. Figure B.4 reveals that using the nonlinear functional form or any of the two approximations conveys almost identical quantitative results. In particular, coefficients obtained via approximations of the instrument are slightly smaller than those obtained using the nonlinear version. In conclusion, methodological choices regarding the instrument's functional form do not drive the patterns observed across the empirical exercises. For more details about the estimations, see B.6 in the Appendix.

B.4.3 Alternative sample definitions

In the following subsection, I elaborate on the potential biases introduced by the wording of SFAS N° 131 and present robustness tests using four alternative criteria to define the sample. The first two sample definitions arise from discrepancies encountered in the two datasets that I use to define feasible and active trading partnerships. In contrast, the last

Figure B.4 – *Robustness: Alternative functional forms in the IV approach*

Notes: The figure depicts the effect of increasing the degree of overlapping ownership on the likelihood of an active trading partnership, across different fixed-effects specifications and alternative functional forms for the instrument. I report the effect of increasing the average degree of overlapping ownership (0.0905) by one standard deviation (0.1361). I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). To the left, I report 2SLS estimates from Equation (3.3), and to the right, 2SLS estimates from Equation (3.2). The 2SLS estimations use the nonlinear functional form (red), a linear approximation (orange), or a quadratic approximation (blue) of the ownership structure similarity based on predicted shareholdings, as described in Section 3.3. Estimates for the OLS are reported in gray to the left of the corresponding 2SLS estimates. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification. For further details, see Table B.6 in the Appendix.

two sample definitions rely on alternative approaches for determining the feasibility of each customer-supplier dyad.

Small customers. The wording of SFAS N° 131 can censor relatively small customers, even when these companies buy only from a few suppliers. Since these companies make smaller purchases, they become less likely to surpass the mandated 10% threshold. Furthermore, anticipating the direction of potential biases is not straightforward, as it depends on whether the degree of overlapping ownership among these dyads lies above or below the current sample average.

Since 2SLS regressions estimate local effects—for this particular instrument and setting, that means customers and suppliers with more stable ownership structures over time—this source of bias does not affect the interpretation for the compliers in the sample.

If anything, the censorship could have introduced a downward bias in empirical exercises on double marginalization and holdup, since the missing trading partnerships would often be those in which a small customer requires a higher level of customization for its inputs or faces a monopolist supplier.

Discrepancies between feasible and active trading partnerships. When merging the datasets corresponding to feasible (Frésard et al., 2020) and active trading partnerships (Barrot and Sauvagnat, 2016), a few active supply chain relationships show a null vertical upstream relatedness. On the one hand, the first paper mentions that the procedure loses track of a small fraction of dyads when mapping CUSIP (American Bankers Association) and GVKEY (Compustat) identifiers. On the other hand, the second paper explains that using phonetic string-matching algorithms can lead to incongruent trading partnerships, even after manual review. Therefore, discrepancies arising when merging the datasets could indicate errors in one or both sources.

Throughout the chapter, I lean on a compromise solution. First, I define all customer-supplier dyads in the dataset of active trading partnerships as feasible. Second, for supply chain relationships with a null vertical upstream relatedness, I assign them the average across active trading partnerships in the same year. For the remaining customer-supplier dyads, I retain the corresponding value reported in the dataset of feasible trading partnerships. To assess the importance of the assumption in my findings, I consider two distinct scenarios in which the error arises solely from one dataset.

Should inconsistencies arise from improper mapping between firms' names and GVKEY identifiers, I define the *Feasible-priority sample* by excluding all non-feasible customer-supplier dyads, even if they appear in the dataset of active trading partnerships. Instead, if discrepancies arise in mapping vertical upstream relatedness between CUSIP and GVKEY identifiers, I define the *Active-priority sample* by assigning a weight equal to one to all active trading partnerships in the sample.

Alternative samples under different criteria. Alternatively, I identify a different set of feasible trading partnerships from patterns in the original supply chain network and the text-based network industry classification.

I denote the first definition as the *Network sample*, which requires me to blend all active trading partnerships between 1976 and 2013 into a unique supply chain network, a common practice in the literature on *link analysis*. After that, I proceed as follows. For any arbitrary customer c , I identify all its suppliers in the network and label the set as $\mathcal{S}_0$. Next, I retrieve

all the customers of companies in $\mathcal{S}_0$ and tag them as $\mathcal{C}_1$. This step provides a set of closely related companies to c , even if they operate in different industries. Then, I retrieve all suppliers of companies in $\mathcal{C}_1$ and label them as $\mathcal{S}_1$. Finally, I define all companies in $\mathcal{S}_1$ as feasible suppliers of c . After repeating the procedure for every customer in the dataset of active trading partnerships, I retain annual observations in which companies simultaneously report book assets to the SEC between 1999 and 2013, in industries other than the financial and utility sectors.

The second definition, the *TNIC sample*, uses text-based product similarity to identify all competitors of customers and suppliers in the dataset of active trading partnerships (Hoberg and Phillips, 2016). As a result, every company j is assigned to a year-specific set of closely related companies, $\mathcal{P}_t^j$, in the product space. Then, for any given year t and customer c in the dataset, I consider all companies in $\mathcal{P}_t^c$ as potential customers of every observed supplier of c . Likewise, for any given supplier s , I define all companies in $\mathcal{P}_t^s$ as potential suppliers of every observed customer of s .

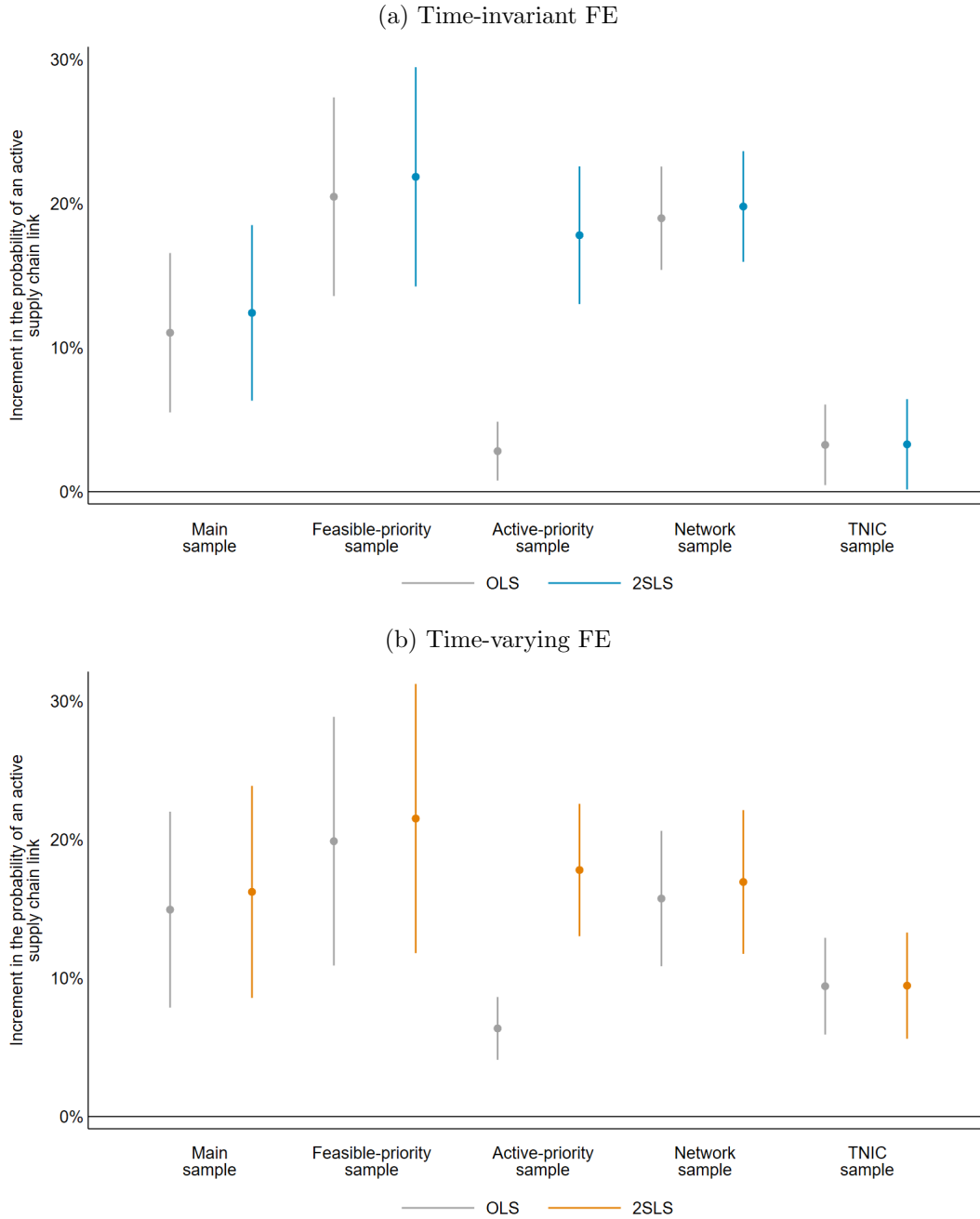
Robustness. Figure B.5 shows that conclusions remain unchanged across all alternative sample definitions. Quantitatively, a few patterns emerge that warrant mention.

First, differences across samples are sharper when using the specification with time-invariant customer and supplier fixed effects. This comes as no surprise, since this specification is more sensitive to supply chain persistence, which can significantly vary with the sample definition. This contributes to another reason for preferring interpretations based on time-varying customer and supplier fixed effects.

Second, coefficients for the *TNIC sample* are statistically significantly smaller. However, when compared to other definitions, the unconditional probability of an active trading partnership in this sample is 10 times higher, making the direct comparison less obvious. In any case, the sign and magnitude of the coefficients still point to similar conclusions as the remainder of the chapter.

Third, the largest differences between OLS and 2SLS estimates occur in the *Active-priority sample*, where active trading partnerships carry much greater weight than the remaining customer-supplier dyads. This feature highlights an important aspect of the workarounds discussed in Section 3.3.

On the one hand, the reweighting breaks the properties discussed in the section, as the approach no longer imposes *ex-ante* feasibility probabilities, leading to larger biases in the OLS estimates. On the other hand, the pattern shows that the IV strategy offers a solution

Figure B.5 – *Robustness: Alternative sample definitions*

Notes: The figure depicts the effect of increasing the degree of overlapping ownership on the likelihood of an active trading partnership, across different fixed-effects specifications and alternative sample definitions. I report the effect of increasing the average degree of overlapping ownership by one standard deviation in the sample. I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample. *Main sample* represents the baseline definition as described in Subsection 3.2.2, while the remaining categories correspond to alternative definitions, as described in Subsection B.4.3. Panels (a) and (b) report estimates from Equations (3.3) and (3.2), respectively. Each panel compares OLS estimates (in gray) with 2SLS estimates (in blue and orange) using a first-order Taylor expansion of the ownership structure similarity based on predicted shareholdings, as described in Section 3.3. All samples consist of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification. For further details, see Tables B.7 and B.8 in the Appendix.

even when feasibility is unobserved and unaccounted for, as it addresses biases arising from omitted variables.

Taken together, and regardless of any disparities in the magnitude of the effects, all the specifications point to the same conclusion about the causal relationship between the degree of overlapping ownership between customers and suppliers and the likelihood of an active trading partnership.

B.5 Additional tables and figures

Table B.1 – *Overlapping ownership increases the likelihood of active trading partnerships*

	Dependent variable: Active trading partnership					
	OLS		FS		2SLS	
	(1)	(2)	(3)	(4)	(5)	(6)
Overlapping ownership	0.1102*** (0.028192)	0.1493*** (0.036067)			0.1240*** (0.031058)	0.1621*** (0.039039)
Predicted overlapping ownership			0.8126*** (0.000130)	0.8176*** (0.000141)		
<i>N</i>	167,902,019	167,899,338	147,818,198	147,815,827	147,818,198	147,815,827
F-statistic			39,156,655	33,591,125		
Time FE	✓		✓		✓	
Company FE	✓		✓		✓	
Time × Company FE		✓		✓		✓

Notes: The table reports OLS and 2SLS estimates of the effect of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different fixed-effects specifications. Columns (1), (3), and (5) estimate Equation (3.3). Columns (2), (4), and (6) estimate Equation (3.2). The first two columns present OLS estimates. I report the effect of increasing the average degree of overlapping ownership (0.0905) by one standard deviation (0.1361). I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). The last two columns present 2SLS estimates based on a first-order Taylor expansion of the ownership structure similarity using predicted shareholdings, as described in Section 3.3. Columns (3) and (4) report the first stage for each specification, including the corresponding Kleibergen-Paap Wald rk F statistic. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification.*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.2 – *More diffuse shareholdings amplify the effects of increasing the degree of overlapping ownership*

	Dependent variable: Active trading partnership									
	OLS (1)	(2)	FS (3)	(4)	2SLS (5)	OLS (6)	(7)	FS (8)	(9)	2SLS (10)
Overlapping ownership	0.0000 (0.000076)				0.0003 (0.000152)	0.0000 (0.000091)				-0.0002 (0.000208)
Highest relative concentration	-0.0000 (0.000003)				0.0000 (0.000018)	-0.0000 (0.000004)				-0.0000* (0.000017)
Highest profit weight value	0.0001** (0.000039)				0.0000 (0.000087)	0.0002*** (0.000045)				0.0003* (0.000122)
Predicted overlapping ownership		0.8379*** (0.000216)	-0.9543*** (0.002049)	1.2541*** (0.000439)			0.8406*** (0.000205)	-0.9976*** (0.002380)	1.2377*** (0.000555)	
Predicted highest relative concentration		0.0001*** (0.000001)	0.0024*** (0.000021)	-0.0001*** (0.000002)			0.0002*** (0.000002)	0.0062*** (0.000046)	0.0007*** (0.000004)	
Predicted highest profit weight value		-0.0105*** (0.000099)	0.0676*** (0.000931)	0.0033*** (0.000116)			-0.0084*** (0.000091)	0.0694*** (0.001113)	0.0080*** (0.000210)	
N	134,355,491	118,984,614	118,984,614	118,984,614	118,984,614	134,354,421	118,983,694	118,983,694	118,983,694	118,983,694
F-statistic		28,800	29,658	28,305	9,615		15,113	16,040	14,660	5,306
Time FE	✓	✓	✓	✓	✓					
Company FE	✓	✓	✓	✓	✓					
Time × Company FE						✓	✓	✓	✓	✓

Notes: The table reports OLS and 2SLS estimates of the effects of increasing ownership structure similarity, relative shareholder concentration, and profit weights between customers and suppliers on the likelihood of an active trading partnership, across different fixed-effects specifications. I drop customer-supplier dyads with a relative concentration above the 90th percentile (5.8918). Columns (6) to (10) estimate Equation (3.4), while Columns (1) to (5) estimate an analogous extension of Equation (3.2). Columns (1) and (6) present OLS estimates. Columns (5) and (10) report 2SLS estimates based on a first-order Taylor expansion of the ownership structure similarity, the relative shareholder concentration, and profit weights based on predicted shareholdings, as described in Section 3.3, including the corresponding Kleibergen-Paap Wald rk F statistic. Columns (2) to (4) and (7) to (9) report the first stage for each specification and report the Sanderson-Windmeijer F Statistic for each endogenous regressor. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification.

*** p < 0.01, ** p < 0.05, * p < 0.1.

Table B.3 – *The role of overlapping owners in alleviating holdup*

	Dependent variable: Active trading partnership											
	OLS			FS			2SLS	OLS			FS	2SLS
<i>Overlapping ownership</i>	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
$R\&D_{CUS}^{Low}$ $R\&D_{SUP}^{Low}$	0.0889* (0.039740)					0.1227** (0.044128)	0.1028* (0.049014)					0.1389** (0.053715)
$R\&D_{CUS}^{High}$ $R\&D_{SUP}^{Low}$	-0.0824** (0.030303)					-0.0984** (0.034250)	-0.0525 (0.034442)					-0.0679 (0.038965)
$R\&D_{CUS}^{Low}$ $R\&D_{SUP}^{High}$	0.3762*** (0.066563)					0.3367*** (0.067756)	0.4991*** (0.077959)					0.4488*** (0.079340)
$R\&D_{CUS}^{High}$ $R\&D_{SUP}^{High}$	0.2547** (0.090724)					0.2827** (0.094780)	0.4053*** (0.096413)					0.4141*** (0.101183)
<i>Predicted overlapping ownership</i>												
$R\&D_{CUS}^{Low}$ $R\&D_{SUP}^{Low}$		0.8338*** (0.000155)	-0.0048*** (0.000019)	-0.0046*** (0.000018)	-0.0019*** (0.000007)			0.8357*** (0.000168)	-0.0039*** (0.000025)	-0.0037*** (0.000025)	-0.0016*** (0.000007)	
$R\&D_{CUS}^{High}$ $R\&D_{SUP}^{Low}$		-0.0184*** (0.000050)	0.8229*** (0.000244)	-0.0042*** (0.000016)	-0.0030*** (0.000024)			-0.0149*** (0.000068)	0.8242*** (0.000252)	-0.0029*** (0.000016)	-0.0025*** (0.000041)	
$R\&D_{CUS}^{Low}$ $R\&D_{SUP}^{High}$		-0.0185*** (0.000050)	-0.0044*** (0.000016)	-0.0044*** (0.000026)	-0.0030*** (0.000024)			-0.0149*** (0.000068)	-0.0031*** (0.000017)	0.8253*** (0.000247)	-0.0024*** (0.000041)	
$R\&D_{CUS}^{High}$ $R\&D_{SUP}^{High}$		-0.0110*** (0.000037)	-0.0114*** (0.000061)	-0.0111*** (0.000059)	0.8151*** (0.000039)			-0.0076*** (0.000037)	-0.0092*** (0.000098)	-0.0092*** (0.000095)	0.8161*** (0.000394)	
N	167,868,548	147,786,912	147,786,912	147,786,912	147,786,912	147,786,912	167,865,873	147,784,546	147,784,546	147,784,546	147,784,546	147,784,546
F-statistic	30,740,905	15,033,197	15,703,394	7,016,266	8,489,255		26,556,748	14,023,141	14,507,691	7,000,641	5,912,669	
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Company FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time × Company FE												

Notes: The table reports OLS and 2SLS estimates of the effects of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different degrees of holdup and fixed-effects specifications. I classify customers and suppliers based on whether their R&D intensity is above or below the sample average (6.89%). I report the effect of increasing the average degree of overlapping ownership (0.0905) by one standard deviation (0.1361). I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). Columns (7) to (12) estimate Equation (B.3), while Columns (1) to (6) estimate an analogous extension of Equation (3.2). Columns (1) and (7) present OLS estimates. Columns (6) and (12) report 2SLS estimates based on a first-order Taylor expansion of the ownership structure similarity based on predicted shareholdings, as described in Section 3.3, including the corresponding Kleibergen-Paap Wald rk F statistic. Columns (2) to (5) and (8) to (11) report the first stage for each specification and report the Sanderson-Windmeijer F Statistic for each endogenous regressor. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification and customer-supplier dyads with missing information on R&D intensity. *** p < 0.01, ** p < 0.05, * p < 0.1.

Table B.4 – *The role of overlapping owners in alleviating double marginalization*

Dependent variable: Active trading partnership											
	OLS	FS		OLS		FS		FS		2SLS	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
											(12)
<i>Overlapping ownership</i>											
HHI_{CUS}^{Low}	0.4548*** (0.056967)					0.4743*** (0.057677)	0.4717*** (0.067517)				0.4788*** (0.068798)
HHI_{CUS}^{High}	0.1384** (0.051474)					0.0890 (0.053876)	0.1790** (0.062796)				0.1135 (0.065265)
HHI_{CUS}^{Low}	-0.0588 (0.049675)					-0.0540 (0.053287)	-0.1258* (0.061063)				-0.1096 (0.065409)
HHI_{CUS}^{High}	0.2864*** (0.073376)					0.3708*** (0.083199)	0.2491** (0.086770)				0.3387*** (0.097887)
<i>Predicted overlapping ownership</i>											
HHI_{CUS}^{Low}		0.8411*** (0.000176)	-0.0070*** (0.000033)	-0.0071*** (0.000033)	-0.0087*** (0.000027)			0.8408*** (0.000181)	-0.0052*** (0.000048)	-0.0053*** (0.000049)	-0.0086*** (0.000022)
HHI_{CUS}^{High}		-0.0070*** (0.000028)	0.8389*** (0.000198)	-0.0066*** (0.000021)	-0.0113*** (0.000042)			-0.0057*** (0.000043)	0.8396*** (0.000207)	-0.0045*** (0.000017)	-0.0109*** (0.000062)
HHI_{CUS}^{Low}		-0.0070*** (0.000028)	-0.0065*** (0.000021)	0.8380*** (0.000198)	-0.0113*** (0.000042)			-0.0055*** (0.000043)	-0.0043*** (0.000017)	0.8387*** (0.000208)	-0.0108*** (0.000062)
HHI_{CUS}^{High}		-0.0055*** (0.000016)	-0.0082*** (0.000031)	-0.0083*** (0.000031)	0.8304*** (0.000229)			-0.0026*** (0.000014)	-0.0070*** (0.000047)	-0.0071*** (0.000047)	0.8319*** (0.000249)
N	160,759,796	142,314,146	142,314,146	142,314,146	142,314,146	142,314,146	160,759,592	142,313,958	142,313,958	142,313,958	142,313,958
F-statistic		27,075,967	21,741,691	21,379,160	15,665,887	8,157,231		25,123,720	19,779,720	19,330,307	13,330,739
Time FE	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓
Company FE	✓	✓	✓	✓	✓	✓		✓	✓	✓	✓
Time × Company FE											

Notes: The table reports OLS and 2SLS estimates of the effects of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different degrees of double marginalization and fixed-effects specifications. I classify customers and suppliers based on whether their TNIC-HHI is above or below the sample median (2,040 out of 10,000). I report the effect of increasing the average degree of overlapping ownership (0.0905) by one standard deviation (0.1361). I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). Columns (7) to (12) estimate Equation (B.4), while Columns (1) to (6) estimate an analogous extension of Equation (3.2). Columns (1) and (7) present OLS estimates. Columns (6) and (12) report 2SLS estimates using a first-order Taylor expansion of the ownership structure similarity based on predicted shareholdings, as described in Section 3.3, including the corresponding Kleibergen-Paap Wald rk F statistic. Columns (2) to (5) and (8) to (11) report the first stage for each specification and report the Sanderson-Windmeijer F Statistic for each endogenous regressor. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification. *** p < 0.01, ** p < 0.05, * p < 0.1.

Table B.5 – Robustness: Alternative overlapping ownership measures

	Dependent variable: Active trading partnership											
	Cosine similarity of ownership structures						Overlapping market value					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Overlapping ownership	OLS	FS	2SLS	OLS	FS	2SLS	OLS	FS	2SLS	OLS	FS	2SLS
Predicted overlapping ownership	0.1102*** (0.028192)	0.6947*** (0.000157)	0.1536*** (0.03808)	0.1493*** (0.036067)	0.6968*** (0.000175)	0.1955*** (0.047149)	-0.0469 (0.033840)	0.9178*** (0.000132)	0.0436 (0.051986)	0.0491 (0.046308)	0.9425*** (0.000148)	0.1180 (0.069769)
<i>N</i>	167,902,019	147,818,198	147,818,198	167,899,338	147,815,827	147,815,827	167,814,861	147,737,778	147,737,778	167,812,193	147,735,417	147,735,417
F-statistic	19,457.211				15,894.432			47,979.123			40,567.465	
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Company FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time × Company FE				✓	✓	✓				✓		
Overlapping-shares product												
	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)
	OLS	FS	2SLS	OLS	FS	2SLS	OLS	FS	2SLS	OLS	FS	2SLS
Overlapping ownership	0.1847*** (0.039561)	0.2884*** (0.063580)	0.2608*** (0.049453)	1.0213*** (0.000482)	0.3296*** (0.079903)	0.0669* (0.026380)	0.7202*** (0.000177)	0.1026** (0.036507)	0.0807* (0.033264)	0.7096*** (0.000197)	0.1269** (0.044953)	
Predicted overlapping ownership												
<i>N</i>	167,902,019	147,818,198	147,818,198	167,899,338	147,815,827	147,815,827	167,902,019	154,379,966	154,379,966	167,899,338	154,377,370	154,377,370
F-statistic	4,296.984				2,948.339			16,482.435			13,009.941	
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Company FE	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time × Company FE				✓	✓	✓				✓		

Notes: The table reports OLS and 2SLS estimates of the effects of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different fixed-effects specifications and alternative measures of overlapping ownership, as described in Subsection B.4.3. For each metric, I report the effect of increasing the sample average by one standard deviation. I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample (0.02%). Columns (1) to (6) use the baseline metric, as described in Section 3.3. Columns (7) to (12) use overlapping market value, Columns (13) to (18) overlapping-share product, and Columns (19) to (24) the smallest profit weight. For each variable, the first three columns report analogous estimates of Equation (3.3), while the last three columns do the same for Equation (3.2). For each specification, the first column reports OLS estimates, the third one reports 2SLS estimates based on analogous definitions of the variable based on predicted shareholdings, as described in Section 3.3, and the second one reports the first stage with the corresponding Kleibergen-Paap Wald rk F statistic. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.6 – *Robustness: Alternative functional forms in the IV approach*

Dependent variable: Active trading partnership							
	OLS	Linear approximation		Quadratic approximation		Nonlinear function	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	OLS	FS	2SLS	FS	2SLS	FS	2SLS
Overlapping ownership	0.1102*** (0.028192)		0.1240*** (0.031058)		0.1240*** (0.030907)		0.1536*** (0.038038)
Predicted overlapping ownership		0.8126*** (0.000130)		0.8130*** (0.000130)		0.6947*** (0.000157)	
<i>N</i>	167,902,019	147,818,198	147,818,198	147,818,198	147,818,198	147,818,198	147,818,198
F-statistic		39,156,655		39,039,206		19,457,211	
Time FE		✓	✓	✓	✓	✓	✓
Company FE		✓	✓	✓	✓	✓	✓
Time × Company FE							
	OLS	Linear approximation		Quadratic approximation		Nonlinear function	
	(8)	(9)	(10)	(11)	(12)	(13)	(14)
	OLS	FS	2SLS	FS	2SLS	FS	2SLS
Overlapping ownership	0.1493*** (0.036067)		0.1621*** (0.039039)		0.1614*** (0.038830)		0.1955*** (0.047149)
Predicted overlapping ownership		0.8176*** (0.000141)		0.8182*** (0.000142)		0.6968*** (0.000175)	
<i>N</i>	167,899,338	147,815,827	147,815,827	147,815,827	147,815,827	147,815,827	147,815,827
F-statistic		33,591,125		33,376,791		15,894,432	
Time FE							
Company FE							
Time × Company FE		✓	✓	✓	✓	✓	✓

Notes: The table reports OLS and 2SLS estimates of the effect of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different fixed-effects specifications and alternative functional forms for the instrument. Columns (1) to (7) report estimates from Equation (3.3), while Columns (8) to (14) report estimates from Equation (3.2). For each specification, the first column reports OLS estimates. The next six columns report three different specification based on the same IV approach. Each specification employs a linear approximation, a quadratic approximation, or the nonlinear functional form. For each specification, the second column reports the 2SLS estimates, while the first column reports the first stage, including the corresponding Kleibergen-Paap Wald rk F statistic. The sample consists of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.7 – *Robustness: Discrepancies between feasible and active trading partnerships*

Dependent variable: Active trading partnership									
	Main sample			Feasible-priority sample			Active-priority sample		
	(1) OLS	(2) FS	(3) 2SLS	(4) OLS	(5) FS	(6) 2SLS	(7) OLS	(8) FS	(9) 2SLS
Overlapping ownership	0.1102*** (0.028192)		0.1240*** (0.031058)	0.2045*** (0.035148)		0.2183*** (0.038798)	0.0281** (0.010395)		0.0969*** (0.020626)
Predicted overlapping ownership		0.8126*** (0.000130)			0.8126*** (0.000130)			0.8149*** (0.000289)	
Observations	167,902,019	147,818,198	147,818,198	167,892,799	147,810,035	147,810,035	167,902,019	147,818,198	147,818,198
F-statistic		39,156,655			39,140,670			7,930,506	
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Company FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time × Company FE									
	Main sample			Feasible-priority sample			Active-priority sample		
	(10) OLS	(11) FS	(12) 2SLS	(13) OLS	(14) FS	(15) 2SLS	(16) OLS	(17) FS	(18) 2SLS
Overlapping ownership	0.1493*** (0.036067)		0.1621*** (0.039039)	0.1987*** (0.045782)		0.2150*** (0.049560)	0.0636*** (0.011557)		0.1779*** (0.024368)
Predicted overlapping ownership		0.8176*** (0.000141)			0.8176*** (0.000141)			0.8206*** (0.000270)	
Observations	167,899,338	147,815,827	147,815,827	167,892,799	147,810,035	147,810,035	167,899,338	147,815,827	147,815,827
F-statistic		33,591,125			33,584,180			9,217,730	
Time FE									
Company FE									
Time × Company FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
$E[Link_{act}]$		0.0002			0.0002			0.0272	
$E[OvrOwn_{act}]$		0.0905			0.0905			0.0916	
$SD[OvrOwn_{act}]$		0.1361			0.1361			0.1375	

Notes: The table reports OLS and 2SLS estimates of the effect of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different fixed-effects specifications and alternative sample definitions. This table focuses on alternative sample definitions arising from discrepancies between datasets used to determine feasible and active trading partnerships, as described in Subsection B.4.3. I report the effect of increasing the average degree of overlapping ownership by one standard deviation in the sample. I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample. Columns (1) to (9) report estimates from Equation (3.3), while Columns (10) to (18) report estimates from Equation (3.2). Columns (1) to (3) and (10) to (12) correspond to the baseline sample described in Subsection 3.2.2. Columns (4) to (6) and (13) to (15) exclude non-feasible customer-supplier dyads among active trading partnerships. Columns (7) to (9) and (16) to (18) set a weight equal to one to all active trading partnerships in the sample, irrespective of their vertical upstream relatedness. For each combination of sample and specification, the first column reports OLS estimates, the third column reports 2SLS estimates using a first-order Taylor expansion of the ownership structure similarity based on predicted shareholdings, as described in Section 3.3, and the second column reports the first stage with the corresponding Kleibergen-Paap Wald rk F statistic. All samples consist of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.8 – *Robustness: Alternative samples under different criteria*

Dependent variable: Active trading partnership									
	Main sample			Network sample			TNIC sample		
	(1) OLS	(2) FS	(3) 2SLS	(4) OLS	(5) FS	(6) 2SLS	(7) OLS	(8) FS	(9) 2SLS
Overlapping ownership	0.1102*** (0.028192)		0.1240*** (0.031058)	0.1896*** (0.018287)		0.1978*** (0.019584)	0.0325* (0.014255)		0.0328* (0.015989)
Predicted overlapping ownership		0.8126*** (0.000130)			0.8657*** (0.000401)			0.8406*** (0.001054)	
Observations	167,902,019	147,818,198	147,818,198	5,991,328	5,319,061	5,319,061	850,933	729,242	729,242
F-statistic		39,156,655			4,657,702			636,216	
Time FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Company FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
Time × Company FE									
	Main sample			Network sample			TNIC sample		
	(10) OLS	(11) FS	(12) 2SLS	(13) OLS	(14) FS	(15) 2SLS	(16) OLS	(17) FS	(18) 2SLS
Overlapping ownership	0.1493*** (0.036067)		0.1621*** (0.039039)	0.1573*** (0.024919)		0.1692*** (0.026461)	0.0940*** (0.017803)		0.0944*** (0.019545)
Predicted overlapping ownership		0.8176*** (0.000141)			0.8802*** (0.000416)			0.8570*** (0.001061)	
Observations	167,899,338	147,815,827	147,815,827	5,989,967	5,317,802	5,317,802	839,906	718,502	718,502
F-statistic		33,591,125			4,486,139			651,975	
Time FE									
Company FE									
Time × Company FE	✓	✓	✓	✓	✓	✓	✓	✓	✓
$\mathbb{E}[Link_{set}]$		0.0002			0.0046			0.0322	
$\mathbb{E}[OvrOwn_{set}]$		0.0905			0.0937			0.1071	
$\mathbb{SD}[OvrOwn_{set}]$		0.1361			0.1503			0.1591	

Notes: The table reports OLS and 2SLS estimates of the effect of increasing the similarity in ownership structures between customers and suppliers on the likelihood of an active trading partnership, across different fixed-effects specifications and alternative sample definitions. This table focuses on alternative sample definitions based on the approaches described in Subsection B.4.3. I report the effect of increasing the average degree of overlapping ownership by one standard deviation in the sample. I normalize variables so that coefficients represent percentage changes with respect to the unconditional probability of an active trading partnership in the sample. Columns (1) to (9) report estimates from Equation (3.3), while Columns (10) to (18) report estimates from Equation (3.2). Columns (1) to (3) and (10) to (12) correspond to the baseline sample described in Subsection 3.2.2. Columns (4) to (6) and (13) to (15) identify feasible trading partnerships from the original supply chain network from 1976 to 2013. Columns (7) to (9) and (16) to (18) identify feasible trading partnerships using a text-based product similarity distance. For each combination of sample and specification, the first column reports OLS estimates, the third column reports 2SLS estimates using a first-order Taylor expansion of the ownership structure similarity based on predicted shareholdings, as described in Section 3.3, and the second column reports the first stage with the corresponding Kleibergen-Paap Wald rk F statistic. All samples consist of feasible customer-supplier dyads for publicly listed companies in the US between 2000 and 2013, excluding the financial and utility sectors. Confidence intervals correspond to cluster robust standard errors at the dyad level with a 95% confidence level. I drop singleton observations for the corresponding fixed-effects specification.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix to Chapter 4

C.1 Model

The framework consists of a partial equilibrium, posting, random search model with *ex-ante* homogeneous workers. Random implies that firms match with an exogenously determined number of workers of a given type α . Posting means that there is no bargaining between workers and firms. Partial equilibrium and posting imply that we take firm productivity heterogeneity and the distribution of workers' outside options as given. For a complete characterization of agents, choices, and timing, see Section 4.4.

Solution. The equilibrium of the framework arises as the optimal solution of three decisions. First, the firm lays off a worker whenever the productivity realization plus the continuation value for the specific employment relationship falls short of the posted wage w . As such, we can define the layoff rate as a function of wages, namely $\delta_\psi(w)$. Second, a worker quits whenever the realized outside option b exceeds the posted wage w , which also establishes the quit rate as a function of wages, that is $\rho(w)$. Third, the firm chooses the wage that maximizes expected profits.

From the firm's standpoint, the layoff and quit rate functions are mechanical objects; that is, when choosing an optimal wage, the firm knows these functions and treats them as given. The argument follows from two elements of the setting. First, the firm has no control over workers' quit decisions beyond setting wages. Second, the firm cannot reliably commit to a layoff policy before observing the productivity shock.

C.1.1 Equilibrium

Definition (Equilibrium). *Let the productivity of any given firm be ψ . Then, the equilibrium is fully characterized by the retention function $\rho(w)$, the layoff function $\delta_\psi(w)$, and the optimal wage w_ψ^* , such that*

(I) *Each worker quits whenever $w < b$, so that the retention function corresponds to*

$$\rho(w) = \mathbb{P}_b(b \leq w) = F_b(w)$$

(II) *The firm lays off a worker whenever their realized markdown is negative, which implies that the layoff function corresponds to*

$$\delta_\psi(w) = \mathbb{P}_\eta(\mu_\psi(w) + \eta \leq 0) = F_\eta[-\mu_\psi(w)]$$

(III) *The firm chooses a wage that maximizes the expected present value of profits, such that*

$$w_\psi^* = \operatorname{argmax}_w V_\psi(w) \quad (\text{C.1})$$

where $V_\psi(w)$ represents the firm's expected present value of profits

$$V_\psi(w) \equiv \overbrace{\rho(w)}^{\text{retention rate}} \cdot \overbrace{[1 - \delta_\psi(w)]}^{\text{layoff rate}} \cdot \left\{ \overbrace{\psi + \alpha - w}^{\text{instant markdown}} + \underbrace{\mathbb{E}_\eta[\eta | \mu_\psi(w) + \eta \geq 0]}_{\text{expected productivity shock for non-laid-off workers}} + \overbrace{\beta V_\psi^*}^{\text{continuation value}} \right\}.$$

and $\mu_\psi(w)$ represents the firm's ex-ante average markdown

$$\mu_\psi(w) \equiv \psi + \alpha - w + \beta V_\psi^* + \mathbb{E}_\eta[\eta] = \overbrace{\psi + \alpha - w}^{\text{instant markdown}} + \underbrace{\beta V_\psi^*}_{\text{continuation value}}$$

under the normalization that $\mathbb{E}_\eta[\eta] = 0$.

The equilibrium must therefore simultaneously satisfy the following conditions

$$\mu_\psi^* \equiv \mu_\psi(w_\psi^*) \quad V_\psi^* \equiv V_\psi(w_\psi^*) \quad \rho_\psi^* \equiv \rho(w_\psi^*) \quad \delta_\psi^* \equiv \delta_\psi(w_\psi^*)$$

Notice that the expression for $V_\psi(w)$ represents the expected present value of profits (henceforth, the *value*) for each worker employed by the firm (each employment relationship), rather than the total value of the firm. Nonetheless, optimizing these two objects is equivalent

because the number of meetings is exogenously determined. Additionally, note that the continuation value in $\mu_\psi(w)$ and $V_\psi(w)$ is βV_ψ^* , and not $\beta V_\psi(w)$. This follows immediately from the firm's inability to commit to maintaining the same wage in subsequent periods.

Without additional assumptions, existence and uniqueness are not guaranteed. We impose weak restrictions on the distributions of productivity and outside-option shocks to discipline the solution and ensure a unique equilibrium.

Assumption C.1. *Let the distribution F_b of workers' outside-options-shocks be a log concave function, and let the distribution F_η of workers' productivity-shocks be a function such that*

$$\frac{\partial \mathbb{E}_\eta [\eta | \eta > x]}{\partial x} \leq 1$$

Log concavity in F_b implies a unimodal distribution, without heavy tails, and with an increasing hazard rate. Conveniently, several of these conditions hold for many standard distributions often used in search models and duration analysis, including normal, exponential, and uniform families. We impose an even weaker condition on F_η . We require a distribution with a non-increasing mean residual life (DMRL). This condition implies that, by construction, that the average excess above the cutoff x declines as x increases. Beyond log concave functional forms, other distributions without heavy tails can also satisfy this property.

C.1.2 Proofs

Lemma C.1. *Let*

$$H_\eta(x) \equiv \mathbb{E}_\eta [\eta | \eta \geq x] - x$$

and

$$H_b(x) \equiv \left(\frac{\partial \ln F_b(x)}{\partial x} \right)^{-1}$$

The optimal wage w_ψ^ satisfies the first order condition*

$$H_\eta(-\mu_\psi^*) = H_b(w_\psi^*)$$

Proof. Immediately follows from differentiating Equation (C.1) with respect to w and equating the expression to zero. $\square$

Lemma C.2. *Under Assumption (C.1),*

$$\frac{\partial H_b(x)}{\partial x} > 0$$

Proof. By Assumption (C.1), $F_b(x)$ is log concave. Consequently, $F'_b(x) > 0$ and $F''_b(x) < 0$ by definition. It suffices to show that

$$H_b(x) \equiv \frac{F_b(x)}{F'_b(x)} \Rightarrow H'_b(x) = 1 - \frac{F_b(x) F''_b(x)}{F'_b(x)^2} > 0$$

□

Lemma C.3. *Under Assumption (C.1)*

$$\frac{\partial H_\eta(x)}{\partial x} < 0$$

Proof. That $H'_\eta(x)$ is negative follows immediately from Assumption (C.1), which states that $F_\eta(x)$ satisfies DMRL. □

Proposition C.1. *Under Assumption (C.1), the optimal wage w_ψ^* increases with firm productivity ψ .*

Proof. Replacing μ_ψ^* in Lemma (C.1) and differentiating with respect to ψ yields

$$\frac{dw_\psi^*}{d\psi} = \frac{H'_\eta(-\mu_\psi^*) \cdot (1 + \beta V_\psi^{*'})}{H'_\eta(-\mu_\psi^*) - H'_b(w_\psi^*)}$$

Since the value of a match $V_\psi^{*'}$ increases with firm productivity ψ , and given that $H'_b(x) > 0$ and $H'_\eta(x) < 0$ as established in Lemmas (C.2) and (C.3), the optimal wage w_ψ^* is increasing in firm productivity ψ . □

Proposition C.2. *Under Assumption (C.1), the optimal average markdown μ_ψ^* increases with firm productivity ψ .*

Proof. Replacing w_ψ^* in Lemma (C.1) and differentiating with respect to ψ , yields

$$\frac{d\mu_\psi^*}{d\psi} = \frac{\partial \mu_\psi^*}{\partial \psi} + \frac{\partial \mu_\psi^*}{\partial w_\psi^*} \cdot \frac{dw_\psi^*}{d\psi} = \frac{H'_b(w_\psi^*) (1 + \beta V_\psi^{*'})}{H'_b(w_\psi^*) - H'_\eta(-\mu_\psi^*)}$$

Since the value of a match $V_\psi^{*'}$ is increasing in firm productivity ψ , and that $H'_b(x) > 0$ and

$H'_\eta(x) < 0$, as established in Lemmas (C.2) and (C.3), the average markdown μ_ψ^* increases with firm productivity ψ . $\square$

Proposition C.3. *Under Assumption (C.1), the optimal quit rate $1 - \rho_\psi^*$ decreases with firm productivity ψ .*

Proof. Since the optimal wage w_ψ^* increases with firm productivity ψ and the retention rate function $\rho(w)$ is increasing in wages w , the optimal quit rate $1 - \rho_\psi^*$ decreases with firm productivity ψ . $\square$

Proposition C.4. *Under Assumption (C.1), the optimal layoff rate δ_ψ^* decreases with firm productivity ψ .*

Proof. Since the optimal average markdown μ_ψ^* increases with firm productivity ψ and the layoff rate function $\delta_\psi(w)$ is decreasing in expected markdowns $\mu_\psi(w)$, the optimal layoff rate δ_ψ^* decreases with firm productivity ψ . $\square$

Proposition C.5. *Under Assumption (C.1), the steady-state firm size s_ψ increases with firm productivity ψ .*

Proof. Firm size evolves according to the following expression

$$s_{t+1} = \rho(w_t) \cdot (1 - \delta_\psi(w_t)) (1 + s_t)$$

In steady-state, $s_t = s_{t+1} = s_\psi$ by definition. Then, solving for s_ψ yields that

$$s_\psi = \frac{\rho_\psi^* \cdot [1 - \delta_\psi^*]}{1 - [\rho_\psi^* \cdot (1 - \delta_\psi^*)]}$$

Moreover, since the optimal retention rate ρ_ψ^* increases in firm productivity ψ and the optimal layoff rate δ_ψ^* decreases in firm productivity ψ , the steady-state firm size s_ψ is increasing in firm productivity ψ . $\square$

C.2 AKM estimation

The following section details the estimation procedure for firm pay premiums and examines the restrictions imposed by the AKM model (Abowd et al., 1999; Card et al., 2013; Song

et al., 2019; Gerard et al., 2021). For a more exhaustive description of the empirical approach and additional evidence addressing concerns about systematic biases in the estimation for Brazil, see Section A.2 in the Appendix.

C.2.1 Clustering and the AKM model

The AKM model identifies firm pay premiums through worker flows across the largest weakly connected set of workers and firms. Weakly means that firms are connected through at least one hire or separation. Among sparsely linked connected sets, estimates of firm pay premiums become noisy. Although individual coefficients remain unbiased, the introduction of noise leads to an overestimation of the variance of firm effects at the expense of the covariance between worker and firm fixed effects.

To improve estimation precision, we first classify firms into 100 groups using the method proposed by Bonhomme et al. (2019), which clusters firms with similar wage distributions. Formally, we partition firms by solving the following weighted k -means problem:

$$\min_{\substack{k(1), \dots, k(J), \\ H_1, \dots, H_K}} \sum_{j=1}^J n_j \int \left(\hat{F}_j(y) - H_{k(j)}(y) \right)^2 d\mu(y),$$

where $\hat{F}_j(y)$ denotes the empirical cumulative distribution function (CDF) of log wages in firm j , n_j its firm size, μ a given measure supported on a finite grid, $\{k(1), \dots, k(J)\}$ a partition of J firms into K clusters, and $\{H_1, \dots, H_K\}$ cluster-specific wage CDFs.

We then estimate firm pay premiums using the AKM model, separately for men and women, and with cluster rather than firm fixed effects. In particular, we assume the following specification for the log hourly wages Y_{it} of worker i in year t given its gender g ,

$$\log Y_{it} = \alpha_i + \psi_{K(it)}^g + X_{it}\beta^g + r_{it}, \quad (\text{C.2})$$

where α_i denotes a worker fixed effect capturing the portable component of individual wages, X_{it} a set of time-varying controls (including year fixed effects and a polynomial of education-specific age trends), and ψ_k a pay premium paid specific to cluster k . The term $K(it)$ indexes the cluster that corresponds to the workplace of worker i in year t , and r_{it} denotes an error component that captures all other residual variation among employment relationships. We estimate the model using OLS and allow all parameters to vary by gender, including pay premiums.

Table C.1 – *Variance decomposition of log hourly wages*

	All	Female	Male
Mean of log-wages	2.332	2.209	2.420
Standard deviation of log-wages	0.664	0.654	0.658
<i>AKM decomposition</i>			
SD of worker effects	0.504	0.510	0.489
SD of cluster effects	0.213	0.200	0.220
SD of covariates	0.098	0.095	0.100
Corr. of worker and cluster effects	0.598	0.607	0.585
<i>Percentage of variance of log wages due to:</i>			
Worker effect	57.5	60.8	55.4
Custer effect	10.2	9.3	11.2
Cov. of worker and cluster effects	29.0	28.9	29.1
Cluster effects + covariance	39.3	38.3	40.2
Moves	12,013,818	4,913,018	7,100,800
Employment spells	74,910,201	31,473,978	43,436,223

Notes: The table decomposes wage variation in Brazil, separately for men and women. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private manufacturing sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1. We construct firm clusters using a k -means algorithm, as described in Appendix C.2. The analysis follows the estimation of Equation (C.2) and decomposes the residualized variation in log hourly wages ($Y - X\hat{b}$) into worker fixed effects (α), cluster fixed effects (ψ), and the covariance between worker and cluster fixed effects. Time-varying controls include year fixed effects and education-specific age-trend polynomials. The last two rows report the number of moves and employment spells (worker-year observations).

Table C.1 presents a standard variance decomposition of Equation (C.2). Results are broadly consistent with Gerard et al. (2021), who analyze the same setting. Notice, however, that our estimates for cluster fixed effects are smaller than the firm effects they report in their paper. This difference highlights an important tradeoff in both methods. On the one hand, estimating firm fixed effects, as we mentioned above, may overestimate pay premiums due to small-sample bias. In contrast, cluster fixed effects may underestimate them by ignoring within-cluster variation (Card et al., 2024, 2025).

C.2.2 Exogenous mobility

The exogenous mobility condition is a relabeling of the traditional omitted-variable bias. In a sense, the restriction requires that, conditional on individual and firm characteristics,

mobility is as good as random. This also implies that wages follow a log-linear structure; otherwise, the error term would capture match-specific variation, rendering the model misspecified. We test these restrictions with the approach proposed by Sorkin (2018).

According to the model in Equation (C.2)

$$\begin{aligned}\log Y_{it} &= \alpha_i + \psi_{K(it)}^g + x_{it}\beta^g + r_{it} \\ \log Y_{it+1} &= \alpha_i + \psi_{K(it+1)}^g + x_{it+1}\beta^g + r_{it+1}\end{aligned}$$

which allows us to take first differences as follows

$$\Delta \log Y_{it} - \Delta x'_{it}\beta^g = \Delta \psi_{K(it)}^g + \Delta r_{it}$$

Next, we take expectations, conditional on moving

$$\mathbb{E}[\Delta \log Y_{it} - \Delta x'_{it}\beta^g | M_{it} = 1] = \Delta \mathbb{E}[\psi_{K(it)}^g | M_{it} = 1] + \mathbb{E}[\Delta r_{it} | M_{it} = 1]$$

where M_{it} indicates whether worker i changed clusters in year t .

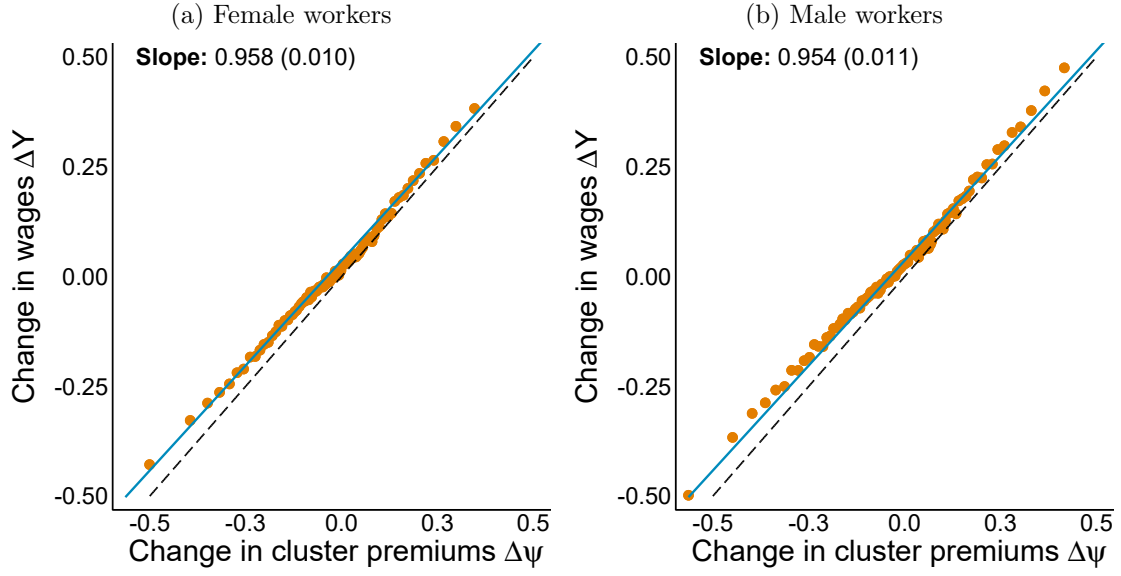
The key assumption on estimating Equation (C.2) using OLS is that the probability of moving is uncorrelated with the error term, that is $\mathbb{E}[\Delta r_{it} | M_{it} = 1] = 0$. Under this assumption

$$\mathbb{E}[\Delta \log Y_{it} - \Delta x'_{it}\beta^g | M_{it} = 1] = \Delta \mathbb{E}[\psi_{K(it)}^g | M_{it} = 1]$$

Figure C.1 plots this relationship separately for women and men, showing that wage changes closely follow changes in cluster pay premiums. Although the linear fit does not exactly match the identity, the slope is above 0.95 in both cases. Thus, we conclude that cluster pay premiums may still contain some noise but do not immediately reveal large biases. This is supportive evidence that our AKM framework fits the data well.

C.3 Institutional setting in Brazil: Additional details

The public pension fund: FGTS. All formally employed workers in the private sector are required to have an account at *Caixa*, a public bank. This account is known as *Fundo de Garantia do Tempo de Serviço* (FGTS). Employers must deposit 8% of each worker's

Figure C.1 – *Wage change corresponds to firm fixed effect change*

Notes: The figure plots changes in log hourly wages against changes in cluster pay premiums among movers. Log hourly wages are residualized with respect to time-varying characteristics using estimates from Equation (C.2) and correspond to hourly rates in the last year at the predecessor firm and in the first year at the successor firm. Panel (a) reports results for female workers, while Panel (b) does the same for male workers. Orange dots correspond to a binscatter regression following the method in Cattaneo et al. (2025). The black-dashed diagonal corresponds to the 45-degree line. The solid blue line plots the best linear fit to the underlying microdata. The corresponding coefficient and standard error (95% confidence) are reported at the upper right corner. Estimates are based on annual RAIS employment spells for workers with urban open-ended contracts in the private manufacturing sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

gross monthly salary into this account. Furthermore, if a worker is laid off, they receive a severance payment amounting to 40% of the total balance accrued in their FGTS account. Workers can access these funds if they are laid off or upon reaching retirement age.

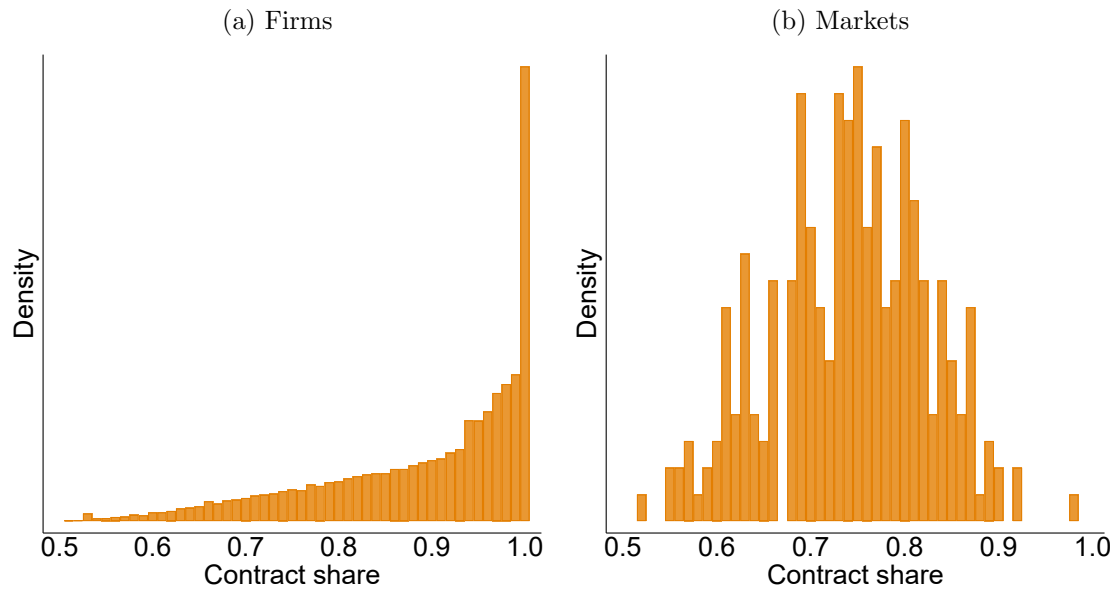
Layoff fine. In the event of a layoff, firms are required to pay a government fine equivalent to 10% of the worker's total FGTS balance. This is in addition to the 40% severance payment that the employer must directly transfer to the worker.

Unemployment insurance. Workers who are laid off are eligible for unemployment insurance, contingent on eligibility, which depends on the length of their formal employment relationships. Workers employed for 6 to 11 months within the last 36 months receive three months of benefits. Workers employed for 12 to 23 months within the last 36 months receive four months of benefits. Workers employed for 24 months or more within the last 36 months receive five months of benefits.

As a reference, the monthly unemployment insurance in 2012 ranged from 1 to 1.76 times the minimum wage, depending on the worker's average salary before being laid off.

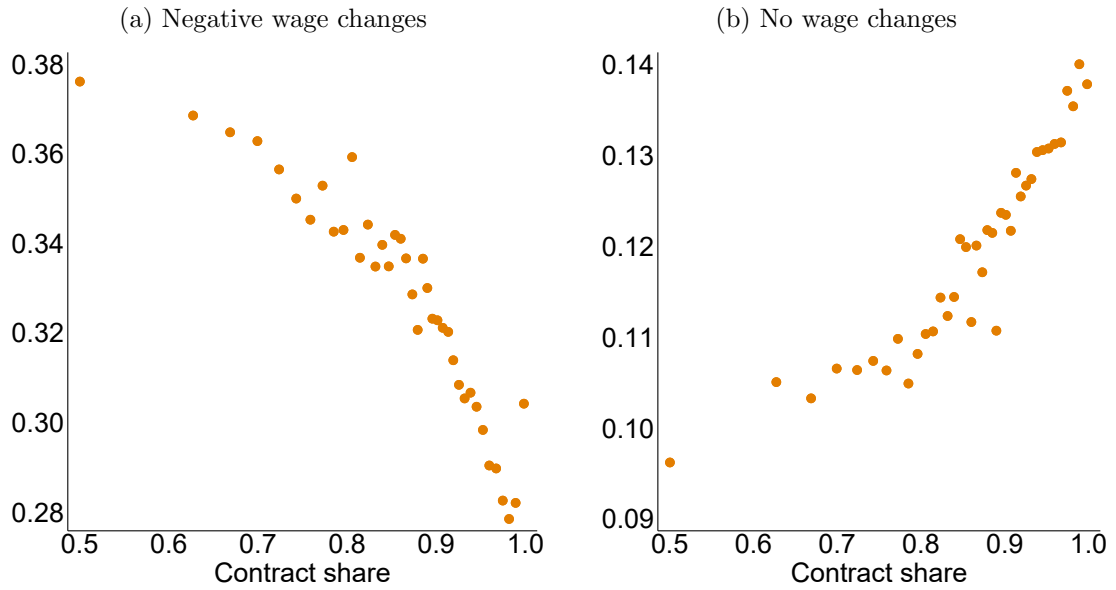
C.4 Additional tables and figures

Figure C.2 – *Distribution of wage rigidity measures across firms and markets*



Notes: The figure depicts the distribution of wage rigidity across firms and markets. Panel (a) uses the firm's contract share (CS) to proxy wage rigidity, defined as the fraction of the total wage bill attributable to contract wages, as in Equation (4.4). Panel (b) uses the average CS across firms within a labor market, plotting the distribution of $\overline{CS}$ across them. A labor market is defined as the combination of state and 2-digit industry codes. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

Figure C.3 – *Worker sorting does not drive the relationship between wage rigidity and the contract share*



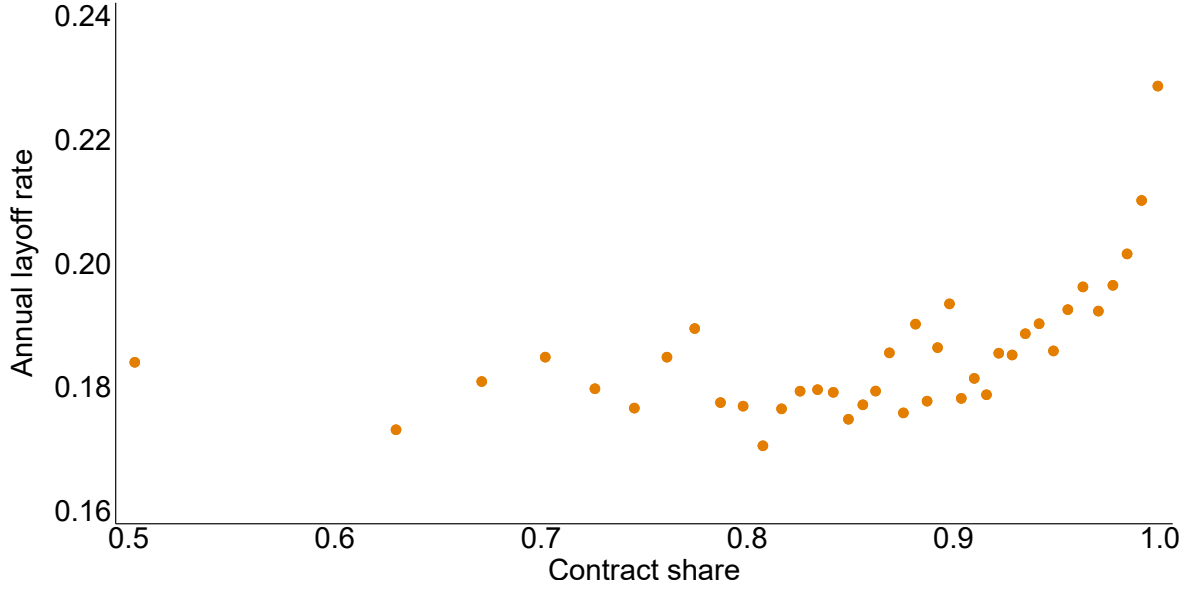
Notes: The figure shows that worker sorting does not drive the relationship between wage rigidity and contract share (CS). CS denotes the fraction of the total wage bill attributable to contract wages, as defined in Equation (4.4). The panels depict the correlation between CS and the share of stayers in a firm that face interannual wage declines (Panel a) or no change (Panel b). We control for worker heterogeneity using individual-level data from RAIS, including tenure and tenure squared, education- and gender-specific quadratic age trends, race fixed effects, occupation fixed effects, and worker fixed effects from the AKM estimation. Orange dots correspond to a firm-level binscatter regression following the method in Cattaneo et al. (2025), weighted by firm size. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

Table C.2 – *Layoffs explain most of the productivity-separation correlation*

	(1)	(2)	(3)	(4)
<i>Panel A - Value added</i>				
β_{Layoff}	-0.048*** (0.0002)	-0.034*** (0.0002)	-0.025*** (0.0002)	-0.024*** (0.0002)
$\beta_{\text{Separation}}$	-0.055*** (0.0002)	-0.039*** (0.0002)	-0.032*** (0.0002)	-0.030*** (0.0002)
$\frac{\beta_{\text{Layoff}}}{\beta_{\text{Separation}}}$	0.871*** (0.0055)	0.852*** (0.0090)	0.776*** (0.0096)	0.804*** (0.0116)
Observations	9,308,341	9,307,701	9,308,341	9,307,701
<i>Panel B - Firm size</i>				
β_{Layoff}	-0.016*** (0.0000)	-0.011*** (0.0000)	-0.012*** (0.0000)	-0.009*** (0.0000)
$\beta_{\text{Separation}}$	-0.017*** (0.0000)	-0.012*** (0.0000)	-0.012*** (0.0000)	-0.010*** (0.0000)
$\frac{\beta_{\text{Layoff}}}{\beta_{\text{Separation}}}$	0.928*** (0.0032)	0.950*** (0.0048)	0.946*** (0.0046)	0.957*** (0.0062)
Observations	49,835,818	49,830,114	49,835,818	49,830,114
<i>Panel C - Firm pay premium</i>				
β_{Layoff}	-0.213*** (0.0002)	-0.183*** (0.0003)	-0.114*** (0.0003)	-0.131*** (0.0003)
$\beta_{\text{Separation}}$	-0.263*** (0.0003)	-0.236*** (0.0003)	-0.163*** (0.0004)	-0.186*** (0.0004)
$\frac{\beta_{\text{Layoff}}}{\beta_{\text{Separation}}}$	0.811*** (0.0018)	0.774*** (0.0025)	0.698*** (0.0034)	0.703*** (0.0033)
Observations	49,835,818	49,830,114	49,835,818	49,830,114
Worker covariates			✓	✓
Worker FE		✓		✓

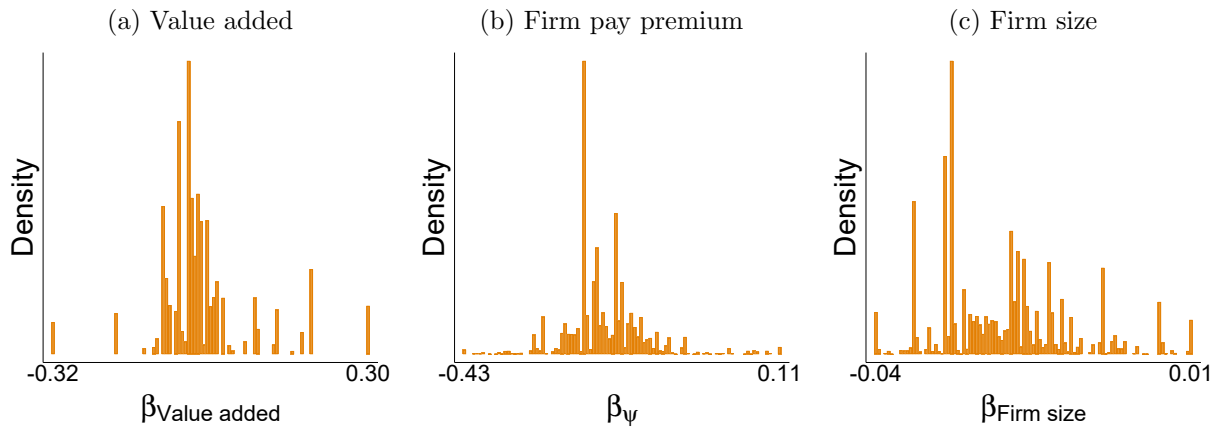
Notes: The table reports the slopes of layoff rates and separation rates, and the corresponding ratio, with respect to firm productivity. We estimate these using OLS at the worker level, separately for three measures of firm productivity and across four specifications. Column (1) corresponds to Equation (4.1), while Columns (2) to (4) correspond to Equation (4.3). Panel A uses value added; Panel B, firm pay premiums; and Panel C, firm size. For more details on firm productivity measures, see Subsection 4.2.1 and Section C.2 in the Appendix. We obtain standard errors of the ratio using the delta method. We control for worker heterogeneity using individual-level data from RAIS, including quadratic tenure, education- and gender-specific quadratic age trends, race and occupation fixed effects, and worker fixed effects from the AKM estimation. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1. Panel A uses value added from the PIA and, as such, it reports aggregates at the industry-state level and is further restricted to manufacturing firms.

Figure C.4 – *Worker sorting does not drive the relationship between layoff rates and the contract share*



Notes: The figure shows that worker sorting does not drive the relationship between layoff rates and contract share (CS). CS denotes the fraction of the total wage bill attributable to contract wages, as defined in Equation (4.4). We control for worker heterogeneity using individual-level data from RAIS, including tenure and tenure squared, education- and gender-specific quadratic age trends, race fixed effects, occupation fixed effects, and worker fixed effects from the AKM estimation. Orange dots correspond to a firm-level binscatter regression following the method in Cattaneo et al. (2025), weighted by firm size. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

Figure C.5 – *Distribution of market-level estimates of the productivity-separation correlation $\hat{\beta}_m$*



Notes: The figure reports the distribution of market-level estimates of the productivity-separation slope $\hat{\beta}_m$ from Equation (4.6) using OLS. We estimate the coefficients separately across three measures of firm productivity. Panel (a) uses value added, Panel (b) uses firm pay premiums, and Panel (c) uses firm size. For more details on firm productivity measures, see Subsection 4.2.1 and Section C.2 in the Appendix. The sample covers the period 2010-2017 and uses annual RAIS employment spells for workers with urban open-ended contracts in the private sector in the Southeast region. We further restrict the sample to firms within the largest connected set, as detailed in Subsection 4.2.1.

